DOCUMENT TYPE: Requirements Specification
DOCUMENT ID: BA-SRS-WEB-ERP-001
EFFECTIVE DATE: October 07, 2026

STRICTLY CONFIDENTIAL
ISO/IEC/IEEE 29148:2018 COMPLIANT

Black Angler

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

PUBLIC WEBSITE + CONTENT MANAGEMENT + CLINIC PRACTICE MANAGEMENT SYSTEM (ERP)
CLIENT: PALAK PARESH GANDHI, AUDIOLOGY & SPEECH-LANGUAGE PATHOLOGY CLINIC, SATARA

Document ID:	BA-SRS-WEB-ERP-001	**Version:**	1.0 (DRAFT for client and technical review)
Prepared by:	Black Angler Pvt Ltd	**Prepared for:**	Ms. Palak Paresh Gandhi
Standard Followed:	ISO/IEC/IEEE 29148:2018 / IEEE 830	**Primary Database:**	MongoDB Atlas (Mumbai region)
Related Documents:	MSA (BA-VC-MSIA-2026-V1.0), SOW, Scope Baseline, CIDR	**Scope Coverage:**	Public Website + CMS + Clinic ERP System

DOCUMENT CONTROL & APPROVALS

REVISION HISTORY

VERSION	DATE	AUTHOR	DESCRIPTION
0.1	07 Oct 2026	Black Angler	Initial Requirements Outline & Architecture Baseline
1.0	07 Oct 2026	Black Angler	First Complete Technical Specification & Data Model Draft

APPROVALS & GOVERNANCE SIGN-OFF

ROLE	NAME	SIGNATURE	DATE
Client (Product Owner)	Ms. Palak Paresh Gandhi	[Pending Sign-off]	07 Oct 2026
Black Angler Project Lead	Shreeharsh M. Shivpuje	[Pending Sign-off]	07 Oct 2026
Black Angler Technical Lead	Lead Software Architect	[Pending Sign-off]	07 Oct 2026
Client Advocate / Privacy Review	Legal & Compliance Advocate	[Pending Sign-off]	07 Oct 2026

How to Read This Document:
- **Sections 1 to 3:** Executive Scope, User Classes, Operating Environment & Architecture.
- **Section 4:** Functional Requirements (FR-xxx) grouped by module with priorities.
- **Sections 5 and 6:** MongoDB Data Model, Collections, Indexing, Security & Infrastructure.
- **Section 7 to 20:** REST API, Interfaces, Security, Business Rules, Reports, Testing, DevOps & Traceability.

1. INTRODUCTION

1.1 PURPOSE

This SRS defines, completely and verifiably, the functional and non-functional requirements of the **Clinic Website** and **Clinic ERP** for Ms. Palak Paresh Gandhi's speech-language pathology, audiology and counselling clinic in Satara. It is the contract between the Client and Black Angler for what will be designed, built, tested and accepted, and is the basis for estimation, development, testing, and acceptance (Scope Baseline items WS-xx, DR-xx and CB-xx map to the requirements here).

1.2 PRODUCT SCOPE

The product consists of two coordinated applications backed by one MongoDB database:

Public Website: a fast, bilingual (English and Marathi), SEO-ready website that informs prospective patients, educates the public, and lets people request appointments, contact the clinic, and submit doctor referrals.
Clinic ERP (private web application, "the ERP"): a role-based system for the practitioner and reception staff to manage appointments, patients, structured clinical records for speech-language therapy, audiology and cognitive behavioural therapy (CBT), follow-ups and recalls, reports, billing, inventory (hearing aids and consumables), expenses, dashboards and compliance logs. It also includes the **CMS** to manage website content.
Automated Clinical Diagnostic & Assessment Report Generator: A specialized, high-precision clinical report generation engine embedded natively within the Clinic ERP. The software automatically compiles clinical data, executes real-time normative formula calculations (PTA1, PTA2, WRS, SNR, %SS, SSI-4, CAPE-V, GRBAS, MPT, s/z ratio, ALD-MM DQ, PHQ-9, GAD-7, FOIS, IDDSI), renders standardized dual audiogram graphs, ear outline graphics, and RCI clinical signature blocks, and produces 10 standardized clinical assessment and diagnostic reports formatted for immediate printing and digital distribution.

IN SCOPE

All items in Section 4.
Single-clinic (single-tenant) deployment, with a data model prepared for a future second branch (optional `branchId`, Section 5.2).
Responsive web UI for desktop, tablet and mobile browsers; installable as a Progressive Web App (PWA).
English and Marathi UI for the public website; English UI for the ERP (Marathi labels for patient-facing documents and consent text; ERP UI localisation is a Could-have).

OUT OF SCOPE (THIS SRS)

ITEM	WHERE SPECIFIED
Google Business Profile management, local SEO operations, citations, review campaigns	Separate GMB/SEO specification
WhatsApp Business automation (chatbot, reminders, templates)	Separate WhatsApp automation specification
Native mobile apps (Android/iOS)	Not planned
Online payment gateway and advance payment	Future (UPI QR display and manual payment recording are in scope)
Patient portal with patient login	Future (data model is prepared, Section 5.6)
Hospital information system, ABDM/ABHA integrations, audiometer device integration	Future
Insurance claims and TPA workflows	Future
Payroll and full accounting/GST return filing	Not in scope (export to Tally/Excel is in scope)

APPOINTMENT REMINDERS (CLARIFICATION)

Automated reminders are delivered through WhatsApp in the separate specification. This SRS therefore defines: (a) the **reminder schedule data** (what must be sent, to whom, when), (b) a **notification outbox** and provider-neutral interface (Section 4.17) that the WhatsApp module will plug into, and (c) **in-app and email** notifications as the baseline channels.

1.3 DEFINITIONS, ACRONYMS AND ABBREVIATIONS

TERM	MEANING
ABR / BERA	Auditory Brainstem Response / Brainstem Evoked Response Audiometry
AVT	Auditory Verbal Therapy
AAC	Augmentative and Alternative Communication
BASLP	Bachelor of Audiology and Speech-Language Pathology
CAPD	Central Auditory Processing Disorder
CAPE-V	Consensus Auditory-Perceptual Evaluation of Voice
CBT	Cognitive Behavioural Therapy
CI	Cochlear implant
CMS	Content Management System
CRUD	Create, Read, Update, Delete

TERM	MEANING
CSRF / XSS	Cross-Site Request Forgery / Cross-Site Scripting
DPDP Act	Digital Personal Data Protection Act, 2023 (India)
Data Fiduciary / Processor	As defined in the DPDP Act: the Client / Black Angler
EMR	Electronic Medical Record
ERP	Enterprise resource planning; here, the clinic practice-management system
FOIS	Functional Oral Intake Scale
GRBAS	Grade, Roughness, Breathiness, Asthenia, Strain (perceptual voice rating)
HNR	Harmonics-to-noise ratio
IDDSI	International Dysphagia Diet Standardisation Initiative (levels 0 to 7)
IST	Indian Standard Time (UTC+05:30)
JWT	JSON Web Token
MPT	Maximum phonation time
OAE	Otoacoustic emissions (TEOAE: transient evoked, DPOAE: distortion product)
PCC	Percentage of consonants correct
PHQ-9 / GAD-7	Patient Health Questionnaire-9 / Generalised Anxiety Disorder-7 scales
PITR	Point-in-time recovery
PTA	Pure tone audiometry (also pure-tone average, disambiguated in context)
RBAC	Role-based access control
RCI	Rehabilitation Council of India
RPO / RTO	Recovery point objective / recovery time objective
SOAP	Subjective, Objective, Assessment, Plan (session note format)
SRT / SDS / WRS	Speech reception threshold / speech discrimination score / word recognition score
SSI-4	Stuttering Severity Instrument, 4th edition
%SS	Percent syllables stuttered
THI / TFI	Tinnitus Handicap Inventory / Tinnitus Functional Index
TRT	Tinnitus Retraining Therapy
UHID	Unique Health ID (the clinic's own patient identifier)
UAT	User acceptance testing
VHI	Voice Handicap Index

1.4 REFERENCES

ISO/IEC/IEEE 29148:2018, Systems and software engineering, Life cycle processes, Requirements engineering.
Digital Personal Data Protection Act, 2023 and rules made under it (as in force).
Information Technology Act, 2000, including section 43A and the reasonable security practices rules, and section 10A (e-contracts).
Rehabilitation Council of India Act, 1992 (professional registration of audiologists and speech-language pathologists).
Consumer Protection Act, 2019 (misleading advertisements).
OWASP Application Security Verification Standard (ASVS) 4.0, Level 2; OWASP Top 10.
WCAG 2.1, Level AA (web accessibility).
Rights of Persons with Disabilities Act, 2016 (accessibility and dignity considerations).
MongoDB Manual (7.x/8.x), Mongoose documentation (8.x).
Master Services & IP Agreement, Scope Baseline and Statement of Work (Black Angler, 2026).

1.5 DOCUMENT CONVENTIONS

Requirement identifiers: `FR-<MODULE>-<NNN>` for functional, `NFR-<AREA>-<NNN>` for non-functional, `BR-<NNN>` for business rules, `DB-<NNN>` for data rules.

Priority (MoSCoW): **M** = Must (needed for go-live), **S** = Should (important, can follow soon after), **C** = Could (nice to have).

Phase: **P1** = launch release, **P2** = within 3 months after launch, **P3** = later.

Verification method: **T** = Test, **D** = Demonstration, **I** = Inspection, **A** = Analysis.

Normative language: "shall" = mandatory, "should" = recommended, "may" = optional.

Dates and time: stored in **UTC** as BSON `Date`; displayed in **IST**. Date-only values (date of birth) are stored as `YYYY-MM-DD` strings.

Currency: INR, stored as integer **paise** (1 INR = 100 paise) to avoid floating-point errors.

1.6 INTENDED AUDIENCE

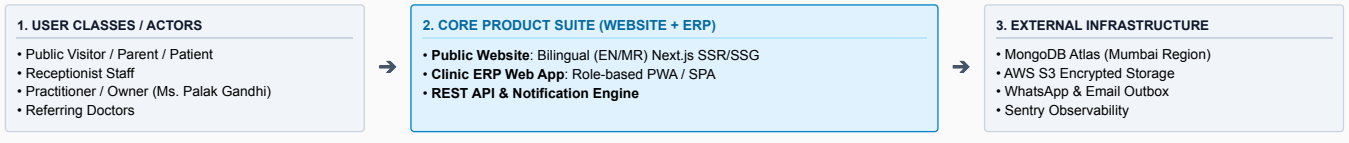
AUDIENCE	USE
Client (practitioner)	Confirm that features match her needs; approve clinical fields.
Black Angler designers and developers	Build the system.
QA	Derive test cases (Section 15).
Client's advocate and CA	Review privacy, consent, billing and tax behaviour.
Future maintainers	Understand design decisions.

2. OVERALL DESCRIPTION

2.1 PRODUCT PERSPECTIVE

The product is a new, self-contained system with no legacy predecessor. It will be accessed over the internet through browsers and uses a small number of external services (email delivery, object storage, maps embedding, optional UPI QR generation) behind adapters.

FIGURE 2.1: SYSTEM PERSPECTIVE & PRODUCT BOUNDARIES



2.2 PRODUCT FUNCTIONS (SUMMARY)

MODULE	SUMMARY
Public Website	Bilingual informational site: home, about, 15 service pages, blog, FAQs, testimonials, contact, map, booking request, doctor referral, self-check tools, legal pages.
CMS	Manage pages, blog posts, FAQs, testimonials, media, menus, SEO metadata and site settings without a developer.
Authentication and RBAC	Secure login, optional 2FA, roles and granular permissions, session control.
Scheduling	Working hours, holidays, slot engine, online booking requests, approvals, reschedule/cancel, waitlist, walk-ins, therapy series, calendar views.

MODULE	SUMMARY
Patients	Registration with UHID, guardian management, duplicate detection, profile, timeline, consent.
Clinical records	Structured case history and assessments for speech-language, audiology and CBT; therapy plans and session notes; hearing aid register; attachments.
Follow-up and recall	Rule-driven recall tasks (therapy review, annual audiometry, hearing aid servicing, newborn re-screen).
Reports and documents	PDF generation: audiology report, therapy summary, referral letter, hearing aid warranty slip, receipt, invoice, certificates.
Automated Clinical Reports	Automated generation of 10 standardized clinical assessment reports (OAE Screening, Pure Tone & Speech Audiological Evaluation, Voice Disorders Proforma, Fluency Disorders Proforma, Comprehensive Speech-Language Battery, CBT & Psychological Screening, Dysphagia, Hearing Aid Fitting, Therapy Progress, Discharge Summary) with single-page A4 layouts, bilateral SVG audiograms, ear indicators, RCI clinical stamps, and bilingual Devanagari output.
Billing	Price list, invoices, receipts, packages, discounts, payments, refunds, daily collection.
Inventory	Hearing aids and accessories with serial tracking, stock movements, vendors, reorder alerts.
Expenses	Expense capture and categorisation.
Dashboards	Daily operations, clinical, financial and marketing-source analytics.
Privacy and audit	Consent records, access logs, data subject requests, retention, break-glass access.

2.3 USER CLASSES AND CHARACTERISTICS

USER CLASS	DESCRIPTION	TECHNICAL SKILL	ACCESS
Public visitor / parent / patient	Satara and nearby residents, many using low-end Android phones, preferring Marathi, often parents of children with speech or hearing concerns, or older adults and their families.	Low	Public website only
Owner / Practitioner (`OWNER`)	Ms. Palak Gandhi. Full clinical and administrative authority.	Low to medium	Full ERP
Receptionist / Front desk (`RECEPTIONIST`)	Manages appointments, registration, billing, enquiries. No clinical details beyond what is needed.	Low to medium	Limited ERP
Clinician (visiting or additional) (`CLINICIAN`)	Optional future audiologist/SLP/psychologist working in the clinic. Access to own patients' clinical records.	Medium	Clinical + scheduling for own calendar
Content editor (`EDITOR`)	Optional person managing blog and website content.	Low	CMS only
System administrator (`ADMIN`)	Black Angler support staff. Infrastructure and configuration; **no access to clinical content** by default (Section 10).	High	Admin console, no patient data
Referring doctor	ENT, paediatrician, neurologist, school staff submitting referrals.	Low	Public referral form (no login)

2.4 OPERATING ENVIRONMENT

ITEM	SPECIFICATION
Client devices	Android 9+ (Chrome 100+), iOS 14+ (Safari 14+), Windows 10/11, macOS 12+, ChromeOS; screens 360 px width and above
Browsers (supported)	Latest two major versions of Chrome, Edge, Firefox, Safari, Samsung Internet
Server runtime	Node.js 20 LTS (or later LTS), Linux containers
Database	MongoDB 7.0+ (Atlas M10 or higher for production), replica set with 3 nodes, region AWS Mumbai (`ap-south-1`)
Object storage	Amazon S3 (`ap-south-1`) with server-side encryption (SSE-KMS)
Network	HTTPS/TLS 1.2+ only; the clinic needs 2 Mbps or better broadband plus a mobile data fallback

2.5 DESIGN AND IMPLEMENTATION CONSTRAINTS

ID	CONSTRAINT
C-01	**MongoDB** is the system of record. No other primary database.
C-02	All patient and personal data shall be stored and processed in **India** (Mumbai region) unless the Client approves otherwise in writing.
C-03	The system shall comply with the DPDP Act, 2023 (consent, purpose limitation, security safeguards, breach notification, data principal rights).
C-04	The public website shall not publish any professional claim or title (RCI number, "Dr.", CBT) unless flagged **verified** in settings (Section 4.5, BR-041).
C-05	Sensitive CBT/mental-health records shall be encrypted at field level and visible only to users with explicit permission.
C-06	The UI shall be usable by non-technical staff and with older adults' needs in mind: large touch targets (min 44x44 px), plain language, Marathi availability.
C-07	Aadhaar numbers shall **not** be collected or stored.
C-08	Money amounts shall be stored in integer paise.
C-09	The code shall be TypeScript end to end, with automated tests and CI.
C-10	Custom code ownership and Background IP boundaries follow the MSA.
C-11	Open-source dependencies shall have licences permitting commercial closed-source use (MIT, Apache-2.0, BSD, ISC); copyleft (GPL/AGPL) libraries are not allowed in the shipped application unless approved.

2.6 ASSUMPTIONS AND DEPENDENCIES

ID	ASSUMPTION / DEPENDENCY
A-01	Clinic is single-location, one primary practitioner, 1 to 2 reception/support staff at launch.
A-02	Expected load: up to 40 appointments/day, 100 patients/month new, 10 concurrent ERP users, 1,000 website visits/day at launch (design headroom 20x).
A-03	The Client provides clinical field lists, sample report formats and approves all clinical content (Questionnaire Q11).
A-04	A domain, hosting accounts and email sender identity are available in the Client's name.
A-05	Reliable broadband in the clinic. Offline-first operation is not required; transient offline tolerance (draft saving) is a Should.
A-06	WhatsApp and GMB specifications will integrate through the hooks in Section 4.17 and the public APIs in Section 7.
A-07	Legal text (privacy notice, consent forms, disclaimers) is approved by the Client's advocate; the SRS provides structures, not legal wording.

2.7 APPORTIONING OF REQUIREMENTS (RELEASE PLAN)

RELEASE	CONTENTS
P1 Launch	Website, CMS, auth/RBAC, settings, service catalogue, scheduling, patients, clinical records (all three service lines), recalls, PDF reports, billing and receipts, dashboards (core), audit, backups.
P2 (up to 3 months)	Inventory with serial tracking, expenses, advanced analytics, packages, waitlist automation, data import tools, PWA refinements.
P3 (later)	Patient portal, online payments, multi-branch, ERP Marathi UI, e-prescription-style home programme sharing, integrations.

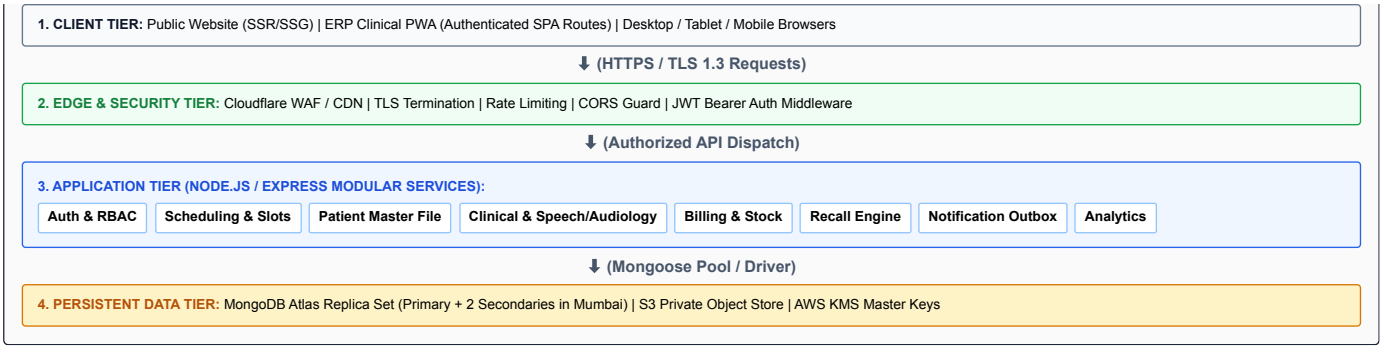
3. SYSTEM ARCHITECTURE

3.1 ARCHITECTURAL GOALS

Simplicity for a small clinic; security of health data; fast public pages on low-end devices; clean modular boundaries so WhatsApp, GMB, payments and a patient portal can be added without rewriting; a data model that is auditable and migration-friendly.

3.2 LOGICAL ARCHITECTURE

FIGURE 3.2: LOGICAL 4-TIER APPLICATION ARCHITECTURE



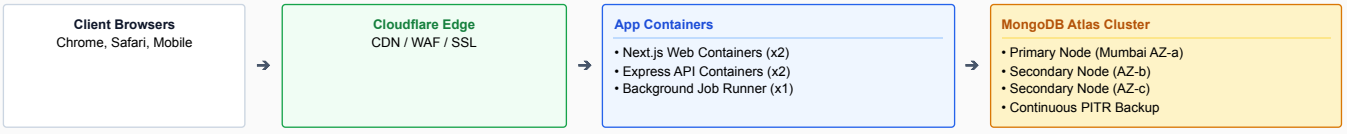
3.3 TECHNOLOGY STACK (BASELINE DECISIONS)

LAYER	CHOICE	RATIONALE
Public website and ERP front end	**Next.js (React, App Router) with TypeScript**; `next-intl` for English/Marathi; Tailwind CSS; shadcn/ui components; TanStack Query; React Hook Form + Zod	SSR/SSG for SEO and speed; one codebase; accessible components
Rich text (CMS)	TipTap editor, sanitised HTML output	Safe, bilingual editing
Charts	Recharts (audiogram drawn with SVG component)	Open licence, light weight
API	**Node.js 20 + Express + TypeScript**; Zod request validation; OpenAPI 3.1 generated docs	Mature, simple to maintain
Database	**MongoDB 7+ on Atlas (Mumbai)** with **Mongoose 8** ODM	Document model fits clinical templates; flexible per-service structures
Authentication	Email/phone + password (Argon2id), JWT access tokens (15 min) + rotating refresh tokens (httpOnly, Secure, SameSite=Strict cookie), TOTP 2FA	Strong, standards-based
Background jobs	`agenda` (MongoDB-backed) or `node-cron` + outbox pattern	No extra infrastructure (no Redis required at this scale)
File storage	S3 with pre-signed URLs, SSE-KMS	Secure uploads of reports and audiogram images
PDF generation	Headless Chromium (Playwright) rendering HTML templates, Noto Sans and Noto Sans Devanagari fonts	Accurate Marathi text rendering
Email	Amazon SES (ap-south-1) or SMTP provider via adapter	Transactional email
Logging and monitoring	Pino structured logs; Sentry (errors); uptime monitor; Atlas alerts	Observability
CI/CD	GitHub Actions: lint, type-check, unit/integration tests, security scan, build, deploy	Quality gates
Containers and hosting	Docker images; AWS (EC2/ECS or Lightsail) in Mumbai behind a CDN/WAF (Cloudflare or CloudFront)	Data residency, performance

Open decisions are listed in Section 18.4.

3.4 DEPLOYMENT ARCHITECTURE

FIGURE 3.4: PRODUCTION CONTAINER & DEPLOYMENT TOPOLOGY



ENVIRONMENT	PURPOSE	DATA
`local`	Developer machines (Docker Compose with MongoDB replica set)	Synthetic
`dev`	Integration testing	Synthetic
`staging`	UAT with the Client (production-like)	Synthetic or anonymised only; **never real patient data**
`production`	Live	Real data

3.5 CODE ORGANISATION (MONOREPO)

```
clinic-platform/
  apps/
    web/                # Next.js: public site + ERP UI (route groups: (public), (erp))
    api/                # Express API
      src/
        config/         # env, logger, db connection
        modules/
          auth/ users/ settings/ services/ scheduling/ patients/
          clinical/ followups/ documents/ billing/ inventory/ expenses/
          cms/ publicapi/ reports/ privacy/ notifications/ search/
        common/         # middleware, errors, utils, crypto, pagination
        jobs/           # scheduled jobs
        main.ts
  packages/
    shared/             # Zod schemas, types, enums, constants used by web and api
    ui/                 # shared UI components
    config/             # eslint, tsconfig, prettier
  infra/                # Dockerfiles, compose, IaC, scripts
  docs/                 # OpenAPI, ADRs, this SRS
```

Each API module contains: `routes.ts` (HTTP), `controller.ts`, `service.ts` (business logic), `repository.ts` (database access), `model.ts` (Mongoose schema), `schema.ts` (Zod DTOs), `permissions.ts`, `tests/`.

3.6 KEY ARCHITECTURAL DECISIONS (ADR SUMMARY)

#	DECISION	REASON
ADR-01	Modular monolith, not microservices	Small team, small clinic, lowest operational burden; modules have clean boundaries for later extraction.
ADR-02	MongoDB with **embedding for owned, bounded data** and **referencing for shared or growing data**	Fast reads of clinical records; avoids unbounded documents (Section 5.3).
ADR-03	**Multikey unique index** (`slotKeys`) to prevent double booking	Database-enforced integrity under concurrency.
ADR-04	Application-level envelope encryption (AES-256-GCM, KMS-wrapped data keys) for CBT and selected sensitive fields	Works on any MongoDB tier; keeps keys outside the database.
ADR-05	Append-only `auditLogs` collection with hash chaining	Tamper evidence for access to patient data.
ADR-06	Money as integer paise	Avoid rounding errors.
ADR-07	Outbox pattern for notifications	Reliable delivery; plug-in channels (email now, WhatsApp later).
ADR-08	Soft delete with `deletedAt` for clinical and financial data; hard delete only through the privacy erasure workflow	Medico-legal retention and auditability.
ADR-09	Public API separate from ERP API (`/public/v1` and `/api/v1`) with different auth, rate limits and response shaping	Prevent accidental exposure of private data.
ADR-10	Server-side rendering and static generation for public pages	SEO and performance on low-end devices.

4. FUNCTIONAL REQUIREMENTS

Each table lists: **ID | Requirement | Priority | Phase**. "The system" means Website + ERP + API as applicable.

4.1 PUBLIC WEBSITE (WEB)

4.1.1 GENERAL

ID	REQUIREMENT	PRI	PH
FR-WEB-001	The website shall be available in **English** and **Marathi**, with a language switcher visible on every page, persisted in a cookie and reflected in the URL (`/en/...`, `/mr/...`).	M	P1
FR-WEB-002	Every public page shall be rendered server-side or statically generated so that the full text content is present in the initial HTML.	M	P1
FR-WEB-003	The site shall be **mobile-first and responsive** from 360 px to 1920 px width, with no horizontal scrolling.	M	P1
FR-WEB-004	The site shall meet Core Web Vitals targets on a mid-range Android phone over 4G: LCP <= 2.5 s, INP <= 200 ms, CLS <= 0.1 (75th percentile in lab testing).	M	P1
FR-WEB-005	The site shall support an **"Opening Soon" mode** (configurable in settings) that shows a countdown/opening date, a waiting-list sign-up and a contact form, and hides online booking until switched off.	M	P1
FR-WEB-006	A persistent header shall provide navigation, a click-to-call button, a language switcher, and a prominent "Book Appointment" call to action.	M	P1
FR-WEB-007	A floating **contact button** shall provide click-to-call and a link to open a chat in the Client's configured messaging URL (the URL is a configurable setting; automation is out of scope).	S	P1
FR-WEB-008	The footer shall show clinic name, address, phone, email, working hours, quick links, social links, legal links, and the website credit line (removable by settings).	M	P1
FR-WEB-009	A **cookie/consent banner** shall appear on first visit; analytics scripts shall load only after consent is given.	M	P1
FR-WEB-010	The site shall display a **medical disclaimer** on service and educational pages stating that content is informational and not a substitute for professional assessment.	M	P1
FR-WEB-011	The site shall show a styled, bilingual **404** and **500** page with navigation and contact information.	M	P1
FR-WEB-012	The site shall be installable as a PWA (manifest, icons, service worker for static assets and an offline fallback page).	C	P2

4.1.2 PAGES AND CONTENT BLOCKS

ID	REQUIREMENT	PRI	PH
FR-WEB-020	**Home page**: hero with clinic name and value statement, primary actions (Book, Call), service category cards (Speech & Language, Audiology, CBT), "Why choose us", about summary, latest articles, FAQs preview, testimonials (if available and consented), location and hours with map, final call to action.	M	P1
FR-WEB-021	**About page**: practitioner profile (photo, qualifications as verified, registration number where verified, languages spoken, approach, experience summary drawn from approved CV content), clinic philosophy, facilities (photo gallery).	M	P1
FR-WEB-022	**Services overview page** listing all services grouped in the three categories with short descriptions, icons and links to detail pages.	M	P1

ID	REQUIREMENT	PRI	PH
FR-WEB-023	**Service detail pages** (one per service in the catalogue, 15 at launch): what it is, who it helps, signs to look for, what happens in the first visit, duration, preparation, FAQs, related articles, booking button, disclaimer. Content is managed in the CMS and linked to the `services` collection.	M	P1
FR-WEB-024	**Book Appointment page**: guided request form (see 4.1.3).	M	P1
FR-WEB-025	**Resources/Blog**: paginated list with category and tag filters, search, and article pages with reading time, author, published date, last reviewed date, related articles, share links, and structured data.	M	P1
FR-WEB-026	**FAQ page** grouped by category with accordion UI and FAQ structured data.	M	P1
FR-WEB-027	**Testimonials page/section** showing only testimonials with `consent.verified = true`; name may be initials only.	S	P1
FR-WEB-028	**Contact page**: address, phone, email, working hours, embedded map (static map image with a link, or a consent-gated interactive map), contact form, directions, parking and accessibility notes.	M	P1
FR-WEB-029	**For Doctors/Referral page** with a referral submission form (see 4.1.5).	M	P1
FR-WEB-030	**Self-check pages** (see 4.1.6).	S	P1
FR-WEB-031	**Legal pages**: Privacy Policy, Terms of Use, Medical Disclaimer, Cookie Policy, Accessibility Statement. Text editable in the CMS.	M	P1
FR-WEB-032	**Gallery** of clinic photographs (lazy-loaded, alt text mandatory).	C	P1
FR-WEB-033	**Awareness/Events page** for screening camps and awareness days (list of upcoming and past events).	C	P2
FR-WEB-034	**Downloadable resources** (parent guides as PDF) with download counts.	C	P2
FR-WEB-035	**Site search** across pages, services and articles in the current language.	S	P2

4.1.3 APPOINTMENT REQUEST (PUBLIC BOOKING)

ID	REQUIREMENT	PRI	PH
FR-WEB-040	The booking form shall be a stepper: (1) choose service or "Not sure, help me choose", (2) patient details, (3) preferred date and time, (4) review and submit.	M	P1
FR-WEB-041	Step 1 shall offer service categories and service names in the selected language, with the option "I need a hearing test/speech assessment, I'm not sure" mapping to the default new-patient assessment service.	M	P1
FR-WEB-042	Step 2 shall capture: booking for (self / child / other family member), patient full name, age or date of birth, sex, parent/guardian name (if minor), mobile number (10-digit Indian mobile, validated), optional email, locality/village, **reason or concern** (free text up to 500 characters with common-concern chips), referral source (dropdown), preferred language for communication, and whether the patient is new or existing.	M	P1
FR-WEB-043	Step 3 shall show **available slots** from the scheduling engine for the chosen service for the next [30] days (window configurable) grouped by day, with morning/afternoon/evening filters, and no slots that violate booking rules (minimum notice, closed days, holidays).	M	P1
FR-WEB-044	The request shall include a consent checkbox for contact and data use, linking to the Privacy Policy; submission shall be impossible without it.	M	P1
FR-WEB-045	On submit, the system shall create an appointment with status `REQUESTED` (or `CONFIRMED` when auto-confirm is enabled and the slot is held), reserve the slot for the configured hold period, generate a human-friendly **booking reference** (for example `BK-7K3QH2`), and show a confirmation screen with the details and clinic contact.	M	P1
FR-WEB-046	A **self-service manage link** shall be generated (token-based, stored as a hash) letting the requester view, reschedule or cancel the request according to policy without logging in.	S	P1
FR-WEB-047	Honeypot field, rate limiting and CAPTCHA fallback (privacy-friendly, such as Cloudflare Turnstile or hCaptcha) shall protect against spam; limit [5] requests per IP per hour and [3] open requests per mobile number.	M	P1
FR-WEB-048	If no slot is suitable, the user shall be able to submit a **callback request** with a preferred time window.	S	P1
FR-WEB-049	A booking confirmation email shall be sent if an email address is given; other channels attach through the notification outbox (Section 4.17).	S	P1
FR-WEB-050	The booking form shall save progress in the browser session so that accidental refresh does not lose entered data (excluding sensitive free text beyond the session).	C	P2

4.1.4 CONTACT FORM AND ENQUIRIES

ID	REQUIREMENT	PRI	PH
FR-WEB-060	The contact form shall capture name, mobile, email (optional), topic (dropdown), message (up to 1,000 characters), consent, and create an `enquiries` record visible in the ERP inbox.	M	P1
FR-WEB-061	The staff shall be notified in-app and by email of new enquiries.	M	P1
FR-WEB-062	The auto-acknowledgement shall tell the sender the expected response time and not request clinical details.	S	P1

4.1.5 DOCTOR REFERRAL FORM

ID	REQUIREMENT	PRI	PH
FR-WEB-070	The referral form shall capture referrer name, designation, hospital/clinic/school, registration number (optional), phone, email, patient name, age/sex,	M	P1

ID	REQUIREMENT	PRI	PH
	guardian contact, reason for referral, service requested, urgency, and attachments (PDF/JPG up to 5 MB each, max 3).		
FR-WEB-071	Submissions shall create a `referrals` record, flagged for staff review, and may be converted to a patient and appointment in the ERP.	M	P1
FR-WEB-072	Attachments shall be virus-scanned and stored privately; they are never publicly accessible.	M	P1

4.1.6 SELF-CHECK TOOLS (SCREENING CHECKLISTS)

ID	REQUIREMENT	PRI	PH
FR-WEB-080	The site shall provide at least three checklists: **Child speech and language milestones**, **Hearing concern checklist**, **Voice health checklist**, each with 8 to 15 yes/no/sometimes questions, defined in the CMS.	S	P1
FR-WEB-081	Results shall show a **non-diagnostic** guidance message (for example "consider an evaluation") with the disclaimer, never a medical diagnosis.	M	P1
FR-WEB-082	Responses may be stored anonymously (no personal data) for aggregate statistics; contact details are stored only if the user chooses "Contact me" and consents.	S	P1
FR-WEB-083	Checklist definitions shall be versioned; results shall record the version used.	S	P1

4.1.7 SEO AND TECHNICAL WEBSITE REQUIREMENTS (ON-SITE ONLY)

ID	REQUIREMENT	PRI	PH
FR-WEB-090	Every page shall have editable `title`, `meta description`, canonical URL, Open Graph and Twitter card tags, with per-language values.	M	P1
FR-WEB-091	The site shall generate `sitemap.xml` (with language alternates through `hreflang`) and `robots.txt`; ERP and admin routes shall be excluded and `noindex`.	M	P1
FR-WEB-092	JSON-LD structured data: `MedicalBusiness` / `LocalBusiness` (name, address, phone, hours), `Physician/Person` for the practitioner (only verified claims), `MedicalWebPage`, `FAQPage`, `Article/BlogPosting`, `BreadcrumbList`.	M	P1
FR-WEB-093	Clean, readable, language-specific URL slugs (for example `/en/services/hearing-aid-fitting`, `/mr/...`); automatic 301 redirects when a slug changes (redirect table in the CMS).	M	P1
FR-WEB-094	Images shall be served in modern formats (AVIF/WebP) with responsive sizes, `loading="lazy"`, explicit dimensions and mandatory alt text.	M	P1
FR-WEB-095	Fonts: Noto Sans and Noto Sans Devanagari (self-hosted subsets), `font-display: swap`.	M	P1
FR-WEB-096	The site shall expose consistent **NAP** (name, address, phone) values from the single settings source on all pages and in structured data.	M	P1
FR-WEB-097	HTTP security headers: HSTS, CSP, X-Content-Type-Options, Referrer-Policy, Permissions-Policy, frame-ancestors none.	M	P1
FR-WEB-098	Analytics: a privacy-respecting analytics tool (for example Plausible/Umami self-hosted, or GA4 with consent) with events for booking started/submitted, call click, referral submitted.	S	P1
FR-WEB-099	Accessibility: WCAG 2.1 AA, keyboard navigation, visible focus, sufficient contrast, labels, skip link, `lang` attributes, reduced-motion support.	M	P1

4.2 CONTENT MANAGEMENT SYSTEM (CMS)

ID	REQUIREMENT	PRI	PH
FR-CMS-001	Authorised users shall create, edit, preview, schedule, publish, unpublish and archive **pages**, **posts**, **FAQs**, **testimonials**, **events**, **service descriptions** and **checklist definitions**.	M	P1
FR-CMS-002	Each content item shall have **per-language fields** (`en`, `mr`) with a status per language (draft, in review, published); a missing translation falls back to English with a banner.	M	P1
FR-CMS-003	Content statuses: `DRAFT`, `IN_REVIEW`, `SCHEDULED`, `PUBLISHED`, `ARCHIVED`. Publishing requires the permission `cms.publish`.	M	P1
FR-CMS-004	A **rich text editor** (headings, lists, links, tables, images, callouts, video embed by URL allow-list) with sanitised output.	M	P1
FR-CMS-005	**Media library**: upload images (JPG/PNG/WebP/SVG-sanitised, max 10 MB) and PDFs (max 20 MB); automatic resizing and format conversion; required alt text per language; usage tracking; delete protection if referenced.	M	P1
FR-CMS-006	**Revision history**: each save creates a revision (who, when, diff summary); restore any previous revision.	S	P1
FR-CMS-007	**Preview** of unpublished content on a signed, expiring URL.	S	P1
FR-CMS-008	**Medical review flag**: content tagged as clinical must carry `reviewedBy` (the practitioner) and `lastReviewedAt`; unreviewed clinical content cannot be published.	M	P1
FR-CMS-009	**Navigation menus** (header, footer) manageable with drag-and-drop ordering and per-language labels.	S	P1
FR-CMS-010	**SEO fields** per item (title, description, canonical, OG image, noindex) with length hints and preview.	M	P1
FR-CMS-011	**Redirect manager** for old to new URLs.	S	P1
FR-CMS-012	**Site settings**: clinic identity, NAP, hours, social links, opening-soon mode, analytics IDs, default OG image, footer text, announcement banner.	M	P1
FR-CMS-013	**Announcement banner** with start and end date (for holiday closures, camps).	S	P1

ID	REQUIREMENT	PRI	PH
FR-CMS-014	**Blocks/sections** approach: pages are built from predefined, validated blocks (hero, text, image+text, cards, FAQ, CTA, gallery, testimonial slider, steps, map, video) to keep design consistent.	S	P1
FR-CMS-015	**Enquiry inbox**: list, filter, assign, mark responded, add internal notes; export CSV.	M	P1
FR-CMS-016	**Referral inbox** with conversion to patient/appointment.	M	P1
FR-CMS-017	**Newsletter/waiting-list subscribers** list with consent timestamp and unsubscribe; export CSV.	S	P1
FR-CMS-018	**Content calendar** view of scheduled posts.	C	P2
FR-CMS-019	**Translation helper**: side-by-side English and Marathi editing; a "needs translation" filter.	S	P1
FR-CMS-020	All CMS actions shall be written to the audit log.	M	P1

4.3 AUTHENTICATION AND SESSION MANAGEMENT (AUTH)

ID	REQUIREMENT	PRI	PH
FR-AUTH-001	Users shall log in with **email or mobile number** and a password.	M	P1
FR-AUTH-002	Passwords shall be hashed with **Argon2id** (memory >= 19 MiB, iterations >= 2, parallelism 1) with a per-user salt and server-side pepper from the secrets manager.	M	P1
FR-AUTH-003	Password policy: minimum 12 characters, no top-10,000 common passwords, no reuse of the last 5, compromised-password check (k-anonymity range lookup) when online.	M	P1
FR-AUTH-004	**Two-factor authentication** (TOTP, RFC 6238) shall be mandatory for `OWNER` and `ADMIN` and optional for others; 10 single-use recovery codes are generated at enrolment.	M	P1
FR-AUTH-005	Access tokens (JWT, 15 minutes) and refresh tokens (opaque, 7 days idle / 30 days absolute) shall be issued; refresh tokens rotate on use with **reuse detection** (reuse revokes the whole token family).	M	P1
FR-AUTH-006	Tokens shall be delivered in **httpOnly, Secure, SameSite=Strict** cookies; a CSRF double-submit token is required for state-changing requests.	M	P1
FR-AUTH-007	**Account lockout**: after 5 failed attempts within 15 minutes the account is locked for 15 minutes, with exponential back-off, and the owner is notified of repeated lockouts.	M	P1
FR-AUTH-008	**Password reset** by emailed single-use link (valid 30 minutes) after identity verification; reset invalidates all sessions.	M	P1
FR-AUTH-009	**Session management**: list active sessions with device, browser, IP (truncated), last used; revoke any or all.	S	P1
FR-AUTH-010	**Idle timeout**: ERP locks after [15] minutes of inactivity (configurable) and requires re-authentication; unsaved drafts are preserved locally and encrypted.	M	P1
FR-AUTH-011	**Login notifications** for new device sign-in to the user's email.	S	P1
FR-AUTH-012	**IP/time restrictions** (optional setting): allow-list for the `OWNER` role.	C	P2
FR-AUTH-013	**Re-authentication (step-up)** is required for high-risk actions: export of patient data, user management, deleting records, viewing CBT notes after a lock, changing 2FA.	M	P1
FR-AUTH-014	All authentication events (success, failure, lockout, reset, 2FA changes) shall be logged to the audit log with IP and user agent.	M	P1
FR-AUTH-015	The public API shall not use these sessions; it uses CSRF-protected, rate-limited, anonymous endpoints (Section 7.3).	M	P1

4.4 USERS, ROLES AND PERMISSIONS (USR)

ID	REQUIREMENT	PRI	PH
FR-USR-001	`OWNER` shall be able to create, edit, deactivate and delete (soft) users and assign roles.	M	P1
FR-USR-002	Roles: `OWNER`, `RECEPTIONIST`, `CLINICIAN`, `EDITOR`, `ADMIN`. Custom roles with a custom permission set shall be possible (P2).	M	P1
FR-USR-003	Access shall be controlled by **permissions** of the form `resource.action` (for example `patient.read`, `clinical.write`, `cbt.read`, `billing.refund`). A role is a named set of permissions; user-level overrides are allowed with an audit entry.	M	P1
FR-USR-004	The default permission matrix shall be as in Appendix 20.3.	M	P1
FR-USR-005	**Clinical data scoping**: `CLINICIAN` sees only patients with whom they have an appointment or an explicit assignment; `OWNER` sees all.	S	P2
FR-USR-006	**CBT permission** (`cbt.read`, `cbt.write`) shall be separate and granted by default only to `OWNER`.	M	P1
FR-USR-007	User profile: name, email, mobile, photo, designation, qualifications, **RCI registration number** (with verified flag), signature image (for reports), languages, working-hours template, calendar colour.	M	P1
FR-USR-008	Deactivated users cannot log in but their historical records and audit entries remain attributed.	M	P1
FR-USR-009	The system shall prevent removal or demotion of the last active `OWNER`.	M	P1
FR-USR-010	Admin (Black Angler) accounts shall have **no clinical read permissions**; support access to patient data requires a time-boxed **break-glass** grant by `OWNER` (Section 4.16).	M	P1

4.5 CLINIC SETTINGS AND CONFIGURATION (SET)

ID	REQUIREMENT	PRI	PH
FR-SET-001	**Clinic profile**: legal/trade name, tagline, address (structured), phone numbers, email, GSTIN (optional), PAN (optional, masked in UI), registration numbers (clinical establishment, shop act), logo, letterhead, signature, stamp image.	M	P1
FR-SET-002	**Credential verification flags**: `rciVerified`, `cbtQualificationVerified`, `titlePermitted` (for "Dr."), each requiring an uploaded proof and the owner's confirmation; controls what the website may display (BR-041).	M	P1
FR-SET-003	**Working schedule**: weekly hours per provider with multiple sessions per day, breaks, and slot granularity (5, 10, 15, 20 or 30 minutes; default 15).	M	P1
FR-SET-004	**Holidays and closures**: date ranges with reason, full or partial day, applying to all or specific providers; import of the Maharashtra holiday list as a template.	M	P1
FR-SET-005	**Booking policy**: minimum notice (default 2 hours), maximum advance (default 30 days), hold time for requested slots (default 24 hours), auto-confirm on/off (global and per service), cancellation window (default 4 hours), reschedule limit (default 2), same-day booking on/off, daily cap per service.	M	P1
FR-SET-006	**Numbering schemes**: UHID prefix and format, invoice and receipt numbering with financial-year reset (April to March), booking reference format.	M	P1
FR-SET-007	**Tax settings**: default tax treatment per item type (exempt, 0%, 5%, 12%, 18%, 28%), HSN/SAC codes, display of GST on invoices, rounding. Rates are configuration, not hard-coded.	M	P1
FR-SET-008	**Payment settings**: accepted modes, UPI ID and QR for display, bank details for invoices.	M	P1
FR-SET-009	**Notification settings**: email sender identity, in-app notification preferences, quiet hours; channel adapters configuration for future modules.	S	P1
FR-SET-010	**Retention settings** (read-only defaults with owner override within legal limits): clinical records 7 years from last visit (adults), until age 21 or 7 years (children, whichever is later); billing 8 years; audit logs 7 years.	M	P1
FR-SET-011	**Document templates**: header/footer, disclaimers, sign-off text for reports; bilingual.	M	P1
FR-SET-012	**Feature toggles**: inventory, expenses, packages, CBT module, online booking, waitlist, referral inbox.	S	P1
FR-SET-013	All setting changes shall be audited with before/after values.	M	P1
FR-SET-014	**Lookup lists** (referral sources, cancellation reasons, expense categories, document types, languages, relations, tags) editable by `OWNER`.	S	P1

4.6 SERVICE CATALOGUE (SVC)

ID	REQUIREMENT	PRI	PH
FR-SVC-001	The system shall maintain a catalogue of services with: code, category (`SPEECH_LANGUAGE`, `AUDIOLOGY`, `CBT`, `OTHER`), names (en, mr), descriptions, default duration, buffer time, whether it is bookable online, new/follow-up variants, price, tax treatment, HSN/SAC, required clinical template, active flag, display order, and **launch status** (`AVAILABLE`, `COMING_SOON`, `HIDDEN`).	M	P1
FR-SVC-002	The catalogue shall be seeded with the 15 services in Appendix 20.1 and be editable.	M	P1
FR-SVC-003	Each service shall be linked to a **clinical template key** (for example `AUDIO_PTA`, `SL_FLUENCY`) that determines which record form opens.	M	P1
FR-SVC-004	**Service variants**: for example "Initial assessment" (60 min), "Therapy session" (45 min), "Review" (30 min); each variant has its own duration and price.	M	P1
FR-SVC-005	A service with `COMING_SOON` status is displayed on the website but cannot be booked.	M	P1
FR-SVC-006	**Provider eligibility**: which providers can deliver which services.	S	P1
FR-SVC-007	**Packages**: bundles (for example 10 therapy sessions) with price, validity days and included services; sold via billing and consumed by appointments.	S	P2
FR-SVC-008	The CBT service shall be bookable only if `cbtQualificationVerified` is true (BR-042).	M	P1

4.7 SCHEDULING AND APPOINTMENTS (APT)

4.7.1 APPOINTMENT LIFECYCLE

```
stateDiagram-v2
    [*] --> REQUESTED: online request
    [*] --> CONFIRMED: staff books / auto-confirm
    REQUESTED --> CONFIRMED: staff approves
    REQUESTED --> REJECTED: staff declines
    REQUESTED --> EXPIRED: hold time elapsed
    CONFIRMED --> CHECKED_IN: patient arrives
    CONFIRMED --> RESCHEDULED: moved to a new slot
    CONFIRMED --> CANCELLED: cancelled
    CONFIRMED --> NO_SHOW: not arrived
    CHECKED_IN --> IN_PROGRESS: session starts
    IN_PROGRESS --> COMPLETED: session ends
    COMPLETED --> [*]
    CANCELLED --> [*]
    NO_SHOW --> [*]
    RESCHEDULED --> [*]
    REJECTED --> [*]
    EXPIRED --> [*]
```

`RESCHEDULED` closes the old appointment and creates a new `CONFIRMED` one linked through `rescheduledFromId`.

4.7.2 REQUIREMENTS

ID	REQUIREMENT	PRI	PH
FR-APT-001	The system shall compute **available slots** for a provider, service variant and date range from: weekly schedule, holidays, existing active appointments (including buffers), blocked time, booking policy, and service duration, at the configured granularity.	M	P1
FR-APT-002	**Double booking shall be impossible**: each active appointment occupies a set of `slotKeys` (one per granularity unit); a **partial unique multikey index** on `(providerId, slotKeys)` for active statuses rejects overlaps atomically (Section 5.5.12).	M	P1
FR-APT-003	Staff shall create an appointment by selecting patient (search or create), service variant, provider, date and time, mode (clinic/online/home), and notes.	M	P1
FR-APT-004	**Calendar views**: day, week, month and agenda list; per provider, colour-coded by service category and status; drag-and-drop rescheduling with validation; "now" indicator; print view.	M	P1
FR-APT-005	**Online requests** appear in a "Pending requests" queue with age indicators; staff can approve (optionally adjust slot), decline with reason, or convert after phoning the patient.	M	P1
FR-APT-006	A held slot for a `REQUESTED` appointment shall expire after the hold period (default 24 h) via a scheduled job, releasing the slot and notifying staff.	M	P1
FR-APT-007	**New vs existing patient**: online requests from unknown people create a **prospect** (embedded in the appointment); upon approval the staff either links an existing patient (duplicate match assistant) or creates a new one in one step.	M	P1
FR-APT-008	**Check-in**: front desk marks arrival; a **token number** is generated per day per provider; the dashboard shows a live waiting queue.	M	P1
FR-APT-009	**Start/complete session**: clinician starts the session (opens the clinical record linked to the appointment) and completes it, which creates or updates the **visit** and may trigger billing and recall suggestions.	M	P1
FR-APT-010	**Cancellation**: with reason (patient request, doctor unavailable, illness, other), who cancelled, and a flag whether it was within the policy window; late cancellations and no-shows are counted on the patient profile.	M	P1
FR-APT-011	**Reschedule** keeps history; limit per policy; staff may override with a reason.	M	P1
FR-APT-012	**No-show** marking after the appointment end plus a grace period (default 15 minutes), manual or automatic via job (configurable).	M	P1
FR-APT-013	**Walk-in** registration creates a `CONFIRMED` + `CHECKED_IN` appointment in the next available slot, or an off-slot "overbook" with permission `appointment.overbook`.	M	P1
FR-APT-014	**Recurring therapy series**: create N sessions (for example 2 per week for 8 weeks) at a fixed weekday/time; conflicts are listed and resolved before saving; the series can be edited or cancelled as a group or single instances.	S	P1
FR-APT-015	**Blocked time** (meetings, equipment calibration, school visits) as non-patient events with recurrence.	M	P1
FR-APT-016	**Waitlist**: patients can be added for a service and preferred times; when a slot frees up the system suggests matching waitlist entries (manual offer, automated offers in the future WhatsApp module).	S	P2
FR-APT-017	**Home visit and online appointment** modes: address for home visits; meeting link and instructions for online sessions; travel time buffer for home visits.	S	P1
FR-APT-018	**Group sessions** (for example parent training, group therapy) with capacity and enrolled patient list.	C	P3
FR-APT-019	**Reminder schedule**: for each confirmed appointment, the system shall create `notifications` outbox items (Section 4.17) according to settings (default 24 h and 2 h before; channel list per patient preference). Sending is performed by the available channel adapters (in-app and email now; WhatsApp later).	M	P1
FR-APT-020	**Patient appointment history** on the profile with status, service, provider, and filters.	M	P1
FR-APT-021	**Conflict warnings**: patient already has an appointment at the same time; provider overtime; duplicate booking of the same service the same day.	S	P1
FR-APT-022	**Appointment notes** and **internal flags** (needs interpreter/language, wheelchair access, child needs extra time, hearing-impaired communication preference).	S	P1
FR-APT-023	**Auto-assignment of provider** if the clinic has multiple providers and the patient has no preference (round-robin by load).	C	P3
FR-APT-024	**Time zone**: all calculations use `Asia/Kolkata`; daylight saving is not applicable; stored in UTC.	M	P1
FR-APT-025	The calendar shall refresh near-real-time (poll every 30 s or server-sent events) so that receptionist and practitioner see the same state.	S	P1
FR-APT-026	**Export** of the day's schedule as a printable PDF.	S	P1
FR-APT-027	Every appointment state change shall record `statusHistory` (status, by, at, reason).	M	P1

4.7.3 SLOT GENERATION ALGORITHM (NORMATIVE)

Inputs: `providerId`, `serviceVariant` (duration `d`, buffer `b`), date range, granularity `g`.
For each date, take the provider's **weekly sessions** (for example 10:00 to 13:00, 14:00 to 18:00), remove **holiday/closure** ranges and **blocked time**.
Generate candidate start times at multiples of `g` within sessions such that `start + d + b <= session end`.
Convert each candidate into `slotKeys = [start, start+g, ..., start+d+b-g]` (UTC ISO strings).
Discard a candidate if any slotKey exists in an active appointment for that provider, if `start < now + minimumNotice`, if `start > now + maxAdvance`, or if the per-service daily cap is reached.
Return remaining candidates grouped by date.
On booking, the API inserts the appointment including `slotKeys` inside a transaction; the unique index guarantees that a concurrent booking of any overlapping slot fails with a duplicate-key error mapped to HTTP 409 `SLOT_TAKEN`.

4.8 PATIENT MANAGEMENT (PAT)

ID	REQUIREMENT	PRI	PH
FR-PAT-001	**Registration form** capturing: salutation, first, middle, last name; sex; date of birth (or approximate age with flag); preferred language(s); primary and alternate mobile; email; address (line, locality/village, taluka, district, state, PIN); occupation; education; marital status; referral source; guardian(s) for minors or dependent adults; medical alerts; photo (optional); communication consent.	M	P1
FR-PAT-002	A **UHID** shall be generated automatically using the configured scheme (default `{PREFIX}-{YY}-{000000}` such as `PGC-27-000001`), unique, immutable, sequential via the counters collection.	M	P1
FR-PAT-003	**Duplicate detection**: on create, the system compares normalised name (phonetic and transliteration-aware for Marathi/English), DOB and mobile, and shows probable matches with scores before saving; the user may open the existing record.	M	P1
FR-PAT-004	**Guardian management**: multiple guardians with relationship, phone, email, `isPrimaryContact`, `isLegalGuardian`, `canConsent`, pick-up authorisation. For patients under 18, at least one guardian with `canConsent` is required.	M	P1
FR-PAT-005	**Family linking**: siblings and family groups for shared contact details (one mobile used for several patients is common in India) with an explicit `sharedContact` flag so that duplicate detection does not treat them as duplicates.	S	P1
FR-PAT-006	**Medical alerts/flags**: hearing aid user, cochlear implant, dysphagia risk (aspiration), seizure disorder, allergies, latex allergy, communication needs (sign language, large print), behaviour notes. Alerts show as badges on the profile, appointment card and clinical screens.	M	P1
FR-PAT-007	**Patient profile** with tabs: Overview, Appointments, Clinical (Speech-Language, Audiology, CBT*), Hearing Aids, Plans and Goals, Documents, Billing, Recalls, Communication log, Consents, Audit (owner only).* (visible with CBT permission).	M	P1
FR-PAT-008	**Timeline**: chronological stream of visits, assessments, notes, documents, payments, and recalls.	S	P1
FR-PAT-009	**Search**: by UHID, name (partial, case-insensitive, Marathi/English), mobile, guardian name, DOB; results in under 1 second for up to 100,000 patients.	M	P1
FR-PAT-010	**Edit** with field-level audit; change of DOB or sex requires a reason.	M	P1
FR-PAT-011	**Merge duplicate records** (permission `patient.merge`): choose master, merge appointments, visits, invoices and documents; merged record keeps a redirect mapping; irreversible without audit-backed restore.	S	P2
FR-PAT-012	**Archive/deactivate** patient (for example deceased or moved); archived patients are excluded from default searches but retained per retention policy.	M	P1
FR-PAT-013	**Deceased** status with date; stops recalls.	S	P1
FR-PAT-014	**Tags** (for example "Stroke", "AVT", "School referral", "VIP") for filtering.	S	P1
FR-PAT-015	**Quick registration** (name, age, mobile, concern) for walk-ins and phone requests, completed later.	M	P1
FR-PAT-016	**Age computation** shall display age in years/months/days appropriate to context (for children under 5 show months).	M	P1
FR-PAT-017	**Patient ID card/label** printable (UHID barcode/QR).	C	P2
FR-PAT-018	**Patient communication log**: manual entries (calls, messages) and system notifications.	S	P1
FR-PAT-019	**Consent status** visible on the profile (treatment consent, data use, communication consent, photo/video consent, research/case-study consent).	M	P1
FR-PAT-020	**Referral tracking**: referring doctor/hospital recorded and used for reports (for example referrals by source) and for automatic referral-letter reply drafting.	S	P1
FR-PAT-021	**Unique constraints**: UHID unique; mobile numbers need not be unique.	M	P1

4.9 CLINICAL RECORDS (CLN)

4.9.1 CLINICAL DATA ARCHITECTURE

Encounter-centred model. Every clinical contact creates a **visit** (encounter). Assessments, session notes, plans and documents hang off the visit and the patient.
Templates. Each assessment type has a typed schema (Section 5.5.20 to 5.5.34). The form UI is generated from a template definition so fields can be added without code changes (configurable fields stored in `clinicalTemplates`).
Versioning. Clinical documents are versioned. Finalising (`FINAL`) makes the document read-only; edits after finalisation create a new version with a reason (**addendum**), retaining the old one.
Signature. Finalisation records the clinician, timestamp, and optionally an e-signature image on the PDF.

4.9.2 GENERAL CLINICAL REQUIREMENTS

ID	REQUIREMENT	PRI	PH
FR-CLN-001	The system shall create a **visit** for each appointment or walk-in with: patient, provider, date/time, service(s), visit type (new, follow-up, review, screening, camp), chief complaint, and links to the documents created.	M	P1
FR-CLN-002	**Case history** (first visit): demographic summary, chief complaint, history of presenting complaint, **prenatal, natal and postnatal history**, developmental milestones, medical and surgical history, medications, hearing history, speech-language history, feeding and swallowing history, family history, educational and social history, behavioural observations, previous evaluations and therapy, languages used at home and school. Pre-filled from earlier data where available.	M	P1
FR-CLN-003	Document status workflow: `DRAFT` (autosaved every 30 seconds) then `FINAL` (signed) then `AMENDED` (new version). Only the author or `OWNER` can finalise.	M	P1
FR-CLN-004	**Autosave and conflict handling**: optimistic concurrency using a `version` field; a conflicting save	M	P1

ID	REQUIREMENT	PRI	PH
	prompts a merge/overwrite choice.		
FR-CLN-005	**Attachments** to any clinical document: audiogram image, scan, previous report, video clip (max 100 MB; S3 pre-signed upload), audio recordings of speech samples (max 50 MB) with patient consent flag.	M	P1
FR-CLN-006	**Unit and range validation** on every numeric field (Section 13), with clear inline messages and a "confirm unusual value" override that is logged.	M	P1
FR-CLN-007	**Computed fields** (PTA averages, degree of loss, PCC, %SS, THI category, GRBAS total, PHQ-9 total, IDDSI mapping) shall be computed by the server and displayed read-only; the formula versions are recorded.	M	P1
FR-CLN-008	**Clinical templates** can be copied as a "prefill from last visit" to speed up follow-up documentation.	S	P1
FR-CLN-009	**Standard phrase library** (macros) per clinician for impressions and recommendations, in English and Marathi.	S	P2
FR-CLN-010	**Amendment** of a final document requires a reason and creates a visible "Amended" marker with history.	M	P1
FR-CLN-011	**Clinical access is logged**: every open/read of a clinical document writes an access audit entry (who, what, when, purpose if break-glass).	M	P1
FR-CLN-012	**Problem list/diagnoses**: structured list with description, ICD-10 code (optional lookup subset for hearing/speech/voice/swallowing/mental-health codes), onset, status (active/resolved).	S	P1
FR-CLN-013	**Outcome measures** captured at intervals and shown as trend charts.	S	P1
FR-CLN-014	**Impression and recommendations** fields with bilingual output option for patient-facing reports.	M	P1
FR-CLN-015	**Referral out**: record referral to ENT, neurologist, psychiatrist, paediatrician with letter generation.	S	P1
FR-CLN-016	**Clinical alerts**: for example "No response at limits in right ear: medical referral advised" shown as rule-based prompts (never automatic diagnoses).	C	P2

4.9.3 SPEECH AND LANGUAGE THERAPY RECORDS

Assessment forms

ID	REQUIREMENT	PRI	PH
FR-CLN-100	**Oral peripheral mechanism examination**: face, lips (symmetry, mobility, strength), tongue (size, protrusion, lateralisation, elevation, tongue-tie flag), jaw, hard and soft palate (cleft, submucous cleft signs, rise), uvula, teeth/occlusion, dentition, breathing pattern, drooling, with normal/abnormal/not tested and notes; **diadochokinetic (DDK) rates** (/pa/, /ta/, /ka/, /pataka/) in syllables per second.	M	P1
FR-CLN-101	**Articulation and phonology assessment**: test name and version (configurable list, including locally used and custom checklists), language of assessment, per-target record: phoneme, word position (initial/medial/final), level (word/phrase/sentence/conversation), response status (`CORRECT`, `SUBSTITUTION`, `OMISSION`, `DISTORTION`, `ADDITION`), substituted sound, **stimulability** (yes/no), notes; computed **PCC**, **PCC-R**, **PVC**, and **speech intelligibility %** (conversation, context-free); **phonological processes** checklist (fronting, backing, stopping, gliding, cluster reduction, final consonant deletion, initial consonant deletion, weak syllable deletion, assimilation, affrication, deaffrication, voicing errors) with occurrence and age-appropriateness.	M	P1
FR-CLN-102	**Language assessment**: tests used (configurable: for example REELS, LPT, PLS-5, CELF, locally adapted Marathi/Hindi tools), subtests, raw score, standard score, percentile, age equivalent, **receptive language age** and **expressive language age** (months), language delay in months (computed), MLU, vocabulary (receptive/expressive counts), syntax and morphology notes, narrative skills, **pragmatics/social communication** (eye contact, turn-taking, topic maintenance, joint attention, requesting, commenting), bilingual/multilingual profile.	M	P1
FR-CLN-103	**Fluency assessment**: speech samples (conversation, reading, monologue) with counts: part-word repetitions, whole-word repetitions, phrase repetitions, prolongations, blocks, interjections, revisions; total syllables; **%SS** computed; **SSI-4** (frequency, duration, physical concomitants, total score and severity); speech rate (syllables per minute) and articulation rate; secondary behaviours (eye blink, head movement, tension); avoidance and situation fears; **OASES** score; onset age, duration, family history, variability, previous therapy; severity grade.	M	P1
FR-CLN-104	**Voice assessment**: perceptual **GRBAS** (each 0 to 3), **CAPE-V** (visual-analogue 0 to 100 for overall severity, roughness, breathiness, strain, pitch, loudness, plus resonance, notes), **acoustic** measures (mean F0 in Hz, F0 range, jitter %, shimmer %, HNR dB, intensity dB SPL), **aerodynamic/phonation**: MPT for /a/, /s/, /z/ (best of three), **s/z ratio**; **VHI-30** and **VHI-10** (item scores and totals with severity); vocal hygiene and demand history (teacher, singer, call-centre worker), laryngeal findings from the ENT report (text and attachment), reflux symptoms.	M	P1
FR-CLN-105	**Aphasia assessment**: onset date, etiology (ischaemic stroke, haemorrhagic stroke, TBI, tumour, other), lesion side and site (text), premorbid languages and handedness; type (global, Broca, Wernicke, conduction, anomic, transcortical motor/sensory/mixed, other); severity; test used (for example WAB-R and adapted language versions) with **Aphasia Quotient** and subtest scores (spontaneous speech, auditory comprehension, repetition, naming); modality profile (listening, speaking, reading, writing, calculation, gesture) on a 0 to 10 scale; apraxia of speech and dysarthria co-occurrence; cognition screen; functional communication (communication needs checklist, ASHA FACS if used); caregiver burden notes.	M	P1

ID	REQUIREMENT	PRI	PH
FR-CLN-106	**Dysarthria assessment**: etiology; type (flaccid, spastic, ataxic, hypokinetic, hyperkinetic, mixed, unilateral upper motor neuron); **Frenchay Dysarthria Assessment** domains (reflex, respiration, lips, palate, laryngeal, tongue, intelligibility) each graded a to e with total; speech subsystem analysis (respiration, phonation, resonance, articulation, prosody) with severity 0 to 4; **intelligibility %** (words, sentences); speaking rate; MPT; AMR/SMR; AAC need.	M	P1
FR-CLN-107	**Dysphagia (swallowing) assessment (clinical)**: history (weight loss, choking, coughing during meals, recurrent chest infections, tube feeding), oral stage (lip seal, mastication, bolus control, residue), pharyngeal signs, **trials** (water swallow test volumes 3 ml, 90 ml; textures per **IDDSI** levels 0 to 7 for drinks and foods), observed signs (cough, wet voice, throat clearing, delayed swallow, SpO2 drop with baseline), positioning, **FOIS** (1 to 7), recommended diet and fluid levels (IDDSI), strategies (chin tuck, supraglottic swallow, effortful swallow), supervision level, **instrumental referral** (VFSS/FEES) with result attachment, risk rating (low, moderate, high), caregiver training given.	M	P1
FR-CLN-108	**Augmentative and alternative communication (AAC)** record: device/system, access method, vocabulary set, training given, outcomes.	S	P2
FR-CLN-109	**Developmental screening** (for infants and toddlers): milestones checklist for hearing, speech, language, motor, social; red flags and referral advice.	S	P1

Therapy plan, goals and sessions

ID	REQUIREMENT	PRI	PH
FR-CLN-120	A **therapy plan** per patient and problem with: diagnosis/impression, long-term goals, short-term objectives (SMART), baseline, target, mastery criterion (for example 80% accuracy over 3 consecutive sessions), planned frequency and duration, start and review dates, status (active, on hold, achieved, discontinued), assigned clinician.	M	P1
FR-CLN-121	**Goal tracking**: each goal has status (not started, in progress, achieved, modified, discontinued), **progress data points** per session (trials, correct, % accuracy, cueing level `INDEPENDENT/VERBAL/VISUAL/MODEL/HAND_OVER_HAND`), with a trend line and "mastery reached" auto-detection.	M	P1
FR-CLN-122	**Session note (SOAP)**: Subjective (parent/patient report), Objective (activities, materials, data, observed behaviours), Assessment (performance, response), Plan (next steps, home programme); structured fields for activities, targets, trial data, reinforcement used, patient participation, attendance, session duration.	M	P1
FR-CLN-123	**Home practice assignments**: description, frequency, materials, due date; compliance tracked at the next visit (completed, partial, not done). The assignment text can be printed or sent through future channels.	M	P1
FR-CLN-124	**Progress summary** auto-draft from goal data over a chosen period, editable.	S	P2
FR-CLN-125	**Parent/caregiver training log**: topics, strategies taught, understanding.	S	P1
FR-CLN-126	**Discharge summary**: reason for discharge, final status, recommendations, follow-up advice.	M	P1
FR-CLN-127	Therapy materials library (optional): tagged activity sheets and files linked to goals.	C	P3

4.9.4 AUDIOLOGY RECORDS

ID	REQUIREMENT	PRI	PH
FR-CLN-200	**Audiology case history**: onset, duration and progression of hearing loss, laterality, noise exposure, ototoxic medication, ear discharge/pain/vertigo, tinnitus, family history, previous hearing evaluations, hearing aids used, **newborn risk factors** (family history, NICU > 5 days, hyperbilirubinaemia needing exchange transfusion, ototoxic drugs, craniofacial anomalies, infections such as CMV/rubella, mechanical ventilation, syndromes, birth weight, gestational age), and a short **ear examination (otoscopy)**: pinna, canal, tympanic membrane, wax, discharge (per ear).	M	P1
FR-CLN-201	**Pure tone audiometry (PTA)** per ear (right, left): **air conduction** thresholds in dB HL at 250, 500, 1000, 2000, 4000, 8000 Hz (optional interoctaves 125, 750, 1500, 3000, 6000 Hz); **bone conduction** thresholds at 250 (optional), 500, 1000, 2000, 4000 Hz; per-point flags **masked**, **no response (NR)** with the limit of the audiometer, **unmasked/masked symbol** selection; **transducer** (supra-aural, insert, bone vibrator, free field); **audiometer** (make/model, calibration date); **reliability** (good/fair/poor); **test conditions** (booth, noise level); **client response** type (hand-raise, button, play audiometry, VRA, CORA); **tester**. The UI shall draw the **audiogram** with standard symbols (O right air unmasked, triangle right masked, X left air unmasked, square left masked, [and] bone conduction) and allow clicking on the graph or typing values.	M	P1
FR-CLN-202	**Computed PTA results** per ear: **PTA(500, 1000, 2000)**, **PTA(500, 1000, 2000, 4000)**, **high-frequency average (4000, 6000, 8000)**, **air-bone gap** per frequency, **degree of hearing loss** from the configured scale (default: normal -10 to 15, slight 16 to 25, mild 26 to 40, moderate 41 to 55, moderately severe 56 to 70, severe 71 to 90, profound 91 and above dB HL), **type** (normal, conductive, sensorineural, mixed; suggested by rules, confirmed by clinician), **configuration** (flat, gently sloping, sloping, steeply sloping, rising, cookie-bite, notched, corner, ski-slope, other).	M	P1
FR-CLN-203	**Speech audiometry** per ear (unaided and aided optionally): **SRT** (dB HL), **SDT** (speech detection threshold), **speech recognition/discrimination score (SDS/WRS) %** with presentation level (dB HL or dB SL) and material (language: English/Marathi/Hindi; list name; monosyllabic/bisyllabic), **MCL**, **UCL/LDL**,	M	P1

ID	REQUIREMENT	PRI	PH
	dynamic range (computed), masking level, test mode (live voice, recorded), **PI-PB rollover** (optional), speech in noise (SNR-50, noise type) optional.		
FR-CLN-204	**Immittance audiometry**: tympanometry per ear (probe tone 226 Hz; 1000 Hz for infants): **type** (A, As, Ad, B, C), peak pressure (daPa), static admittance/compliance (mL or mmho), ear canal volume (mL), tympanometric width/gradient (daPa); **acoustic reflexes** ipsilateral and contralateral at 500, 1000, 2000, 4000 Hz (threshold dB HL or "absent"/"not tested"), reflex decay result.	M	P1
FR-CLN-205	**Otoacoustic emissions (OAE)** per ear: type (TEOAE/DPOAE), protocol/equipment, stimulus level, result (`PASS`, `REFER`, `INCONCLUSIVE`), **SNR** per band (1, 1.5, 2, 3, 4, 5 kHz for TEOAE; DP-gram levels at F2 for DPOAE), noise level, probe fit, test duration; **screening stage** (1st screen, 2nd screen, diagnostic), **screener**, **overall result** (both ears pass, refer unilateral/bilateral). For newborn screening: birth details, **risk indicators**, hospital/birth place, **referral pathway** (rescreen date, ENT/diagnostic ABR), parent counselling documented.	M	P1
FR-CLN-206	**ABR/BERA**: stimulus (click, tone-burst), polarity, rate (per second), intensity series, **wave I, III, V absolute latencies (ms)**, **interpeak latencies (I-III, III-V, I-V)**, amplitude (V and I/V ratio), **wave V threshold (dB nHL)**, sedation/sleep state, electrode montage, interpretation, and attachment of waveform images.	S	P1
FR-CLN-207	**ASSR and electrophysiology notes** (optional).	C	P3
FR-CLN-208	**Tinnitus evaluation**: side/laterality (R, L, bilateral, head), onset, duration, nature (ringing, buzzing, hissing, pulsatile), constancy, associated symptoms (hearing loss, vertigo, hyperacusis, sleep disturbance, anxiety, depression), **pitch match** (Hz), **loudness match** (dB HL/dB SL), **minimum masking level (MML)**, **loudness discomfort levels**, residual inhibition test, **THI (25 items)** with item scoring (0/2/4) and total with grade (slight 0-16, mild 18-36, moderate 38-56, severe 58-76, catastrophic 78-100), **TFI** optional, **VAS** annoyance/loudness (0 to 10), **TRT category** (0 to 4), counselling topics covered, sound therapy device and settings, follow-up schedule.	M	P1
FR-CLN-209	**CAPD evaluation**: referral reason/behaviours (difficulty in noise, following instructions, etc.), prerequisite peripheral hearing normal check, test battery with **per-test, per-ear scores and norms**: speech in noise, dichotic digits, dichotic CV, filtered speech, **gaps-in-noise** or random gap detection, **duration pattern** and **frequency pattern** tests, **masking level difference**, (configurable test library); computed deviation from norms; **diagnosis** (CAPD subprofile), questionnaires (for example CHAPS), management plan: environmental modification, auditory training goals, FM/DM system recommendation.	M	P1
FR-CLN-210	**Hearing aid trial, fitting and programming** (see 4.9.5 for the register): selection rationale, candidacy, counselling, earmould impression details, **trial** (device, dates, duration, outcome), **fitting verification** (real-ear measurement targets and achieved values per frequency; aided thresholds; functional gain), **programming record** (number of channels, program memories, gain settings per band, compression, noise reduction, feedback manager settings, directionality, volume control, telecoil), patient comfort, **fitting outcome measures** (APHAB, IOI-HA, GHABP), **orientation checklist** (insertion/removal, battery change, cleaning, use in noise, phone use, care), **warranty and servicing terms**.	M	P1
FR-CLN-211	**Auditory Verbal Therapy (AVT)** record: child profile (age at identification, age at amplification/implantation, device type and wearing time), **Ling six-sound test** (/m, ah, oo, ee, sh, s/ detection and identification at distances), **auditory development level** per goal (detection, discrimination, identification, comprehension) tracked with **CAP**, **SIR**, **LittlEARS** or **MAIS/IT-MAIS** scores, language age, session plan following the AVT principles (guidance, coaching of parents, auditory hierarchy), session data, **home practice** and **parent engagement** ratings.	M	P1
FR-CLN-212	**Cochlear implant (CI) record**: ear, manufacturer, model, processor, surgery date and centre, activation date, MAP/programming sessions (date, comments, electrode impedances optional), aided thresholds with implant, speech perception scores, therapy referrals.	S	P1
FR-CLN-213	**Newborn hearing screening register**: list of screened infants with result status, rescreen due, and referral completion tracking, supporting camp/hospital screening mode with fast data entry (batch mode, tablet-friendly) and summary export.	M	P1
FR-CLN-214	**Vestibular/balance** basic bedside notes (optional template).	C	P3
FR-CLN-215	**Audiogram printing and PDF** with symbols, legend, patient identity, audiometer details, results table and interpretation.	M	P1
FR-CLN-216	**Comparison view** of audiograms across dates (overlay with colour per date) and **trend** of PTA and WRS.	S	P1

4.9.5 HEARING AID REGISTER

ID	REQUIREMENT	PRI	PH
FR-CLN-230	Register each device issued, loaned or trialled to a patient: ear, **make, model, style** (BTE, RIC/RITE, ITE, ITC, CIC, IIC, body-worn, bone conduction, CROS/BiCROS), technology level, **serial number(s)**, battery size or rechargeable, **earmould/dome** type, **date of trial, date of fitting**, **warranty expiry**, **loss-and-damage cover** expiry, purchase invoice link, price, status (`TRIAL`, `DISPENSED`, `RETURNED`, `UPGRADED`, `LOST`, `DAMAGED`, `UNDER_REPAIR`).	M	P1
FR-CLN-231	**Servicing and repair log**: date, type (cleaning, tubing change, repair, reprogramming, re-fit, loan device given), fault description, outcome, cost, next service due date.	M	P1

ID	REQUIREMENT	PRI	PH
FR-CLN-232	**Reprogramming history**: each fitting session as a dated entry with settings summary and the audiogram used.	M	P1
FR-CLN-233	**Service due and warranty alerts** feed the recall engine (Section 4.10).	M	P1
FR-CLN-234	Link to **inventory** serial item when the device is sold from stock (Section 4.13).	S	P2
FR-CLN-235	**Warranty certificate / fitting slip PDF** with device details and clinic terms.	M	P1

4.9.6 COGNITIVE BEHAVIOURAL THERAPY (CBT) RECORDS (RESTRICTED)

ID	REQUIREMENT	PRI	PH
FR-CLN-300	CBT records are available only if `cbtQualificationVerified` and the user has `cbt.read` / `cbt.write`. All CBT fields shall be **field-level encrypted** (ADR-04), excluded from search indexes, and excluded from general exports.	M	P1
FR-CLN-301	**CBT intake**: presenting problems (list), onset, duration, triggers, severity, functional impact (work/school, relationships, sleep, appetite), history of previous treatment, current medication (name only as reported), substance use, medical conditions, family history, social support, strengths and values, treatment goals in the patient's words.	M	P1
FR-CLN-302	**Risk assessment** at intake and each session: suicidal ideation, plan, intent, means; self-harm; harm to others; protective factors; **risk level** (none, low, moderate, high); **safety plan** record (warning signs, coping steps, contacts, professional and crisis contacts); **mandatory referral prompt** to a psychiatrist/psychologist/emergency services at moderate or high risk; the system displays a banner but never makes autonomous clinical decisions.	M	P1
FR-CLN-303	**Standardised scales** with item-level entry and computed totals/severity: **PHQ-9** (0 to 27: minimal 0-4, mild 5-9, moderate 10-14, moderately severe 15-19, severe 20-27; item 9 positive triggers a risk prompt), **GAD-7** (0 to 21: minimal 0-4, mild 5-9, moderate 10-14, severe 15-21); configurable others: PCL-5, **PSS-10**, ISI (insomnia), SPIN, OCI-R, CORE-10, BDI-II (licensed instruments are included only if the Client holds the licence).	M	P1
FR-CLN-304	**Case formulation** (cognitive model: situation, thoughts, emotions, physical sensations, behaviours; predisposing, precipitating, perpetuating, protective factors) as structured text fields.	S	P1
FR-CLN-305	**Treatment plan**: problem list, goals, planned number of sessions, interventions (cognitive restructuring, behavioural activation, exposure, relaxation, problem solving, graded tasks), review dates.	M	P1
FR-CLN-306	**Session record**: agenda items, mood check (0 to 10), homework review, content, techniques used, patient insights, new homework, session rating (SRS/ORS optional), risk check, next session plan, duration.	M	P1
FR-CLN-307	**Thought record** entries: date/situation, automatic thoughts with belief % (0 to 100), emotions with intensity %, evidence for/against, balanced thought, outcome re-rating; cognitive-distortion tags (all-or-nothing, catastrophising, mind reading, overgeneralisation, personalisation, emotional reasoning, should statements, labelling, mental filter, discounting positives).	S	P1
FR-CLN-308	**Behavioural experiments and activity schedule** records.	S	P2
FR-CLN-309	**Outcome tracking**: scale scores plotted over time with reliable-change indicator.	S	P1
FR-CLN-310	**Consent specific to psychological therapy** (confidentiality limits, risk disclosure, record keeping) captured before the first CBT session; blocked otherwise.	M	P1
FR-CLN-311	**Break-glass**: access to a CBT record by someone other than the author requires an emergency justification and is audited and notified to the `OWNER`.	M	P1
FR-CLN-312	**Disclaimer logic**: CBT features must be hidden entirely when `cbtQualificationVerified = false`.	M	P1

4.10 FOLLOW-UP AND RECALL MANAGEMENT (REC)

ID	REQUIREMENT	PRI	PH
FR-REC-001	A **recall** is a task to contact a patient or schedule a future service, with: patient, type, due date, source (rule or manual), related record, priority, status (`PENDING`, `CONTACTED`, `SCHEDULED`, `COMPLETED`, `DISMISSED`, `OVERDUE`), assigned to, notes.	M	P1
FR-REC-002	**Recall rules** are configurable and shall include at least: therapy **review after N sessions** (default 12), **re-evaluation** every N months by service (default speech-language 6 months), **annual audiometry** for hearing aid users, **hearing aid servicing** every 6 months, **warranty expiry** reminder 30 days prior, **newborn re-screen** per protocol dates, **tinnitus/TRT review** at 1, 3, 6 months, **missed-appointment follow-up** after a no-show, **post-assessment follow-up** call, **CBT review** at session 6 and 12, **recall after discharge** at 3 months.	M	P1
FR-REC-003	Recalls shall be generated automatically when a **visit is finalised** or an event occurs, and by a **nightly job** to catch time-based rules.	M	P1
FR-REC-004	The **recall worklist** shows overdue, due today, due this week and upcoming, filterable by type and assignee, with one-click actions: mark contacted (with outcome), schedule appointment (prefilled), dismiss with reason, snooze.	M	P1
FR-REC-005	Recall notifications go through the outbox (Section 4.17) to the available channels.	M	P1
FR-REC-006	A recall is auto-closed when a matching appointment is completed.	M	P1

ID	REQUIREMENT	PRI	PH
FR-REC-007	Patients flagged deceased or archived shall not generate recalls.	M	P1
FR-REC-008	**Recall effectiveness report**: due vs contacted vs returned.	S	P2
FR-REC-009	**Missed-session policy**: after N consecutive no-shows (default 3) the system flags the therapy plan "at risk of discontinuation" for clinician review.	S	P1
FR-REC-010	Per-patient **do-not-contact** and channel preferences are honoured.	M	P1

4.11 REPORTS AND DOCUMENTS (DOC)

ID	REQUIREMENT	PRI	PH
FR-DOC-001	The system shall generate **PDF documents** from HTML templates with the clinic letterhead, logo, practitioner details (RCI number if verified), digital signature image, date, document number and page numbers.	M	P1
FR-DOC-002	**Master Clinical Reports Suite**: The software shall natively generate the complete suite of **10 specialized clinical assessment reports** plus practice documents, featuring automated metric calculations, bilateral graphs, ear graphics, and RCI clinical signature blocks.	M	P1
FR-DOC-002A	**Otoacoustic Emission (OAE) Screening Report (`RP-CLN-OAE`)**: Generates standardized OAE screening reports for TEOAE and DPOAE. Displays patient metadata, OAE No., Date of Birth, tested frequencies (1000, 1500, 2000, 3000, 4000 Hz) with SNR (dB) and reproducibility (%). Includes prominent visual ear graphics for Right Ear (Red) and Left Ear (Blue) with binary `Pass [ ]` / `Refer [ ]` checkboxes. Auto-evaluates pass criteria (SNR ≥ 6 dB and Reproducibility > 70% across ≥ 3 frequencies). Features structured clinical sections for `IMPRESSION` and `RECOMMENDATION`, plus RCI Audiologist signature block.	M	P1
FR-DOC-002B	**Audiological Evaluation Report (`RP-CLN-AUD`)**: Generates comprehensive bilateral pure tone and speech audiometry reports. Displays Reg. No., PTA No., Audiometer model, patient response consistency (`Consistent` / `Fairly Consistent` / `Inconsistent`). Renders **Two Separate Audiogram Graphs Side-by-Side** (Right Ear in Red with circles `O` and Left Ear in Blue with crosses `X`) across 125 Hz to 8000 Hz and -10 dB to 120 dB HL with standard unmasked/masked/bone-conduction symbol legend. Includes a full-width measurement grid containing `PTA1 (500, 1k, 2k Hz)`, `PTA2 (500, 1k, 2k, 4k Hz)`, `SDT (dB HL)`, `SRT (dB HL)`, `WRS % at dB HL`, `MCL (dB HL)`, `UCL (dB HL)`, `Rinne (512 Hz)`, and `Weber (512 Hz)`. Below the audiograms, renders a full-width Tympanogram & Middle Ear Immittance table (Type A/As/Ad/B/C, Peak Pressure daPa, Static Compliance ml, Ear Canal Volume ml, Clinical Norms). Features clinical interpretation, recommendations, and stamped RCI signature footer.	M	P1
FR-DOC-002C	**Proforma for Voice Disorders (`RP-CLN-VOI`)**: Generates multidimensional clinical voice assessment reports and diagnostic proformas. Form captures vocal abuse/misuse history, phonotrauma, and hydration. Evaluates perceptual voice quality using **CAPE-V** visual-analogue ratings (0-100 mm scale for Overall Severity, Roughness, Breathiness, Strain, Pitch, Loudness; Consistent vs Intermittent) and **GRBAS Scale** (Grade, Roughness, Breathiness, Asthenia, Strain each graded 0 to 3). Calculates Maximum Phonation Time (**MPT** in seconds for sustained /a/, /i/, /u/, best of 3 trials) and **s/z Ratio** (with clinical pathology flag for ratio > 1.4). Records Fundamental Frequency (F_0), Jitter, Shimmer, and **HNR** (Harmonics-to-Noise Ratio in dB). Incorporates **VHI-30 / VHI-10** (Voice Handicap Index) total score and severity classification with personalized vocal hygiene recommendations.	M	P1
FR-DOC-002D	**Fluency Disorders Assessment Report (`RP-CLN-FLU`)**: Generates stuttering and cluttering assessment reports and clinical proformas. Implements the **SSI-4 (Stuttering Severity Instrument, 4th Edition)** protocol: auto-calculates percentage of syllables stuttered (**%SS**) across reading and conversational speech samples, measures average duration of 3 longest stuttering blocks in seconds, and rates physical concomitants (secondary struggle behaviors: facial grimaces, head movements, distracting noises, extremity movements on 0-20 scale). Auto-sums total SSI-4 score, maps percentile rank, and assigns clinical severity (Very Mild, Mild, Moderate, Severe, Very Severe). Calculates Speech Rate in Syllables Per Minute (SPM) and Words Per Minute (WPM). Records covert avoidance behaviors, situational anxiety ratings, and OASES impact scores.	M	P1
FR-DOC-002E	**Comprehensive Speech & Language Assessment Report (`RP-CLN-SLA`)**: Generates complete pediatric and adult speech-language evaluation reports. Compiles Oral Peripheral Mechanism Examination (OPME) and DDK rates (/pa/, /ta/, /ka/, /pataka/). Integrates standardized test batteries: **ALD-MM** (Assessment of Language Development - Mukesh Mohan) calculating Receptive Language Age (RLA), Expressive Language Age (ELA), and Developmental Quotient (**DQ %**); **LPT** (Linguistic Profile Test) analyzing phonological, semantic, syntactic, and morphological mastery; **CAPC-M** (Clinical Assessment of Pragmatics & Communication) rating eye contact, turn-taking, and communicative intent; **DALI** (Dyslexia Assessment for Indian Children) evaluating phonological awareness and rapid naming; and **PCC** (Percentage of Consonants Correct) phonology error profiling. Evaluates **AAC** readiness and **AVT** Erber hierarchy tracking with Ling 6-Sound detection distances. Formulates SMART therapy goals.	M	P1
FR-DOC-002F	**Psychological & Cognitive Behavioural Therapy (CBT) Report (`RP-CLN-CBT`)**: Generates psychometric screening and CBT clinical formulation reports secured	M	P1

ID	REQUIREMENT	PRI	PH
	under **AES-256 field-level encryption**. Compiles standardized scales with item-by-item scoring and auto-summed totals: **PHQ-9** (Depression scale 0-27; Item #9 positive triggers mandatory clinical safety prompt), **GAD-7** (Anxiety scale 0-21), and **PSS-10** (Perceived Stress Scale 0-40). Synthesizes a structured **Clinical Risk Assessment & Safety Plan Matrix** evaluating suicide/self-harm ideation, intent, plans, and protective factors with a 5-step Stanley-Brown Safety Plan (warning signs, coping strategies, crisis contacts). Formulates Beck's Cognitive Model ABC analysis (Activating Event, Automatic Thoughts, Emotional/Behavioral Consequences) and cognitive distortion tagging. Restricted by `cbtQualificationVerified` flag.		
FR-DOC-002G	**Swallowing & Dysphagia Evaluation Report (`RP-CLN-DYS`)**: Generates clinical bedside swallowing evaluation (CSE) reports. Captures oral-pharyngeal examination, cough reflex, and wet-voice signs during water swallow tests (3 ml, 5 ml, 90 ml continuous). Maps functional dietary levels using the **FOIS** (Functional Oral Intake Scale, Levels 1 to 7) and **IDDSI Framework** (International Dysphagia Diet Standardisation Initiative, Levels 0 to 7 for drinks and foods). Specifies compensatory swallowing maneuvers (Mendelsohn, supraglottic swallow, effortful swallow), postural adjustments (chin tuck, head rotation), and aspiration prevention guidelines.	M	P1
FR-DOC-002H	**Hearing Aid Fitting, Trial & Verification Certificate (`RP-CLN-HAF`)**: Generates amplification dispensing and acoustic verification certificates. Records ear laterality (Right, Left, Bilateral), make, model, style (RIC, BTE, CIC, ITE), technology level, and unique hardware serial numbers. Captures Real-Ear Measurement (**REM**) / Coupler Gain target curve matching (NAL-NL2 / DSL v5), Aided Sound Field thresholds vs Unaided thresholds, and functional gain. Details device warranty duration, trial period conditions, loss-and-damage terms, battery care instructions, and recall maintenance schedules.	M	P1
FR-DOC-002I	**Periodic Clinical Therapy Progress Summary (`RP-CLN-PRG`)**: Generates periodic progress review reports synthesized from SOAP session notes and goal tracking data. Compares baseline scores against current performance metrics (Goal Mastery %, Ling 6-sound detection distance, MLU increase, %SS reduction). Displays session attendance rate (% sessions attended vs scheduled), home program adherence, parent coaching milestones, and updated SMART objectives for the next treatment phase.	M	P1
FR-DOC-002J	**Clinical Discharge Summary & Home Maintenance Plan (`RP-CLN-DSC`)**: Generates formal clinical discharge reports upon completion of therapy. Documents initial vs discharge performance benchmarks, treatment duration and total attended sessions, discharge criteria met, individualized home maintenance strategies, and scheduled annual audiological / speech re-evaluation recall date.	M	P1
FR-DOC-003	Documents shall be produced in **English, Marathi, or bilingual** (patient-facing parts), with Devanagari fonts embedded.	M	P1
FR-DOC-004	Each generated document is stored as a `documents` record with file pointer, type, version, hash (SHA-256), generated by, linked visit/patient, and can be **regenerated** from data or **frozen** (final version retained).	M	P1
FR-DOC-005	Final documents shall display a **verification code/QR** linking to a verification page that confirms document authenticity (document number, issue date, clinic) without revealing medical content.	S	P2
FR-DOC-006	**Share/download**: download PDF, print, email (via outbox) with a link that expires in 7 days; no clinical PDFs are attached to emails by default (link with PIN).	M	P1
FR-DOC-007	**Document upload**: patients' external documents (previous reports, scans, certificates, consent scans) with type, date, description and tags.	M	P1
FR-DOC-008	**Digital consent capture** on touch devices: render consent text (bilingual), capture signature on canvas, store signature image, signer identity (patient/guardian), witness, timestamp, IP/device, consent text version hash, and produce a signed PDF.	M	P1
FR-DOC-009	**Template management** (HTML with placeholders and conditionals) editable by `OWNER`/Admin with preview and versioning.	S	P2
FR-DOC-010	Document access is permission-controlled and audited per document.	M	P1
FR-DOC-011	**Watermark** "DRAFT" for non-final documents; "COPY" for reprints.	S	P1
FR-DOC-012	Bulk export of a patient's documents as a ZIP (permission `patient.export`) for transfer requests.	S	P2

4.12 BILLING AND PAYMENTS (BIL)

ID	REQUIREMENT	PRI	PH
FR-BIL-001	**Price list** per service variant and product, with effective dates and optional concession categories (senior citizen, BPL, staff, camp).	M	P1
FR-BIL-002	**Invoice creation** from an appointment/visit or manually: line items (service, product, package, other), description, quantity, unit price, discount (percentage or amount, with reason and approver), tax treatment and rate, HSN/SAC, line total; header discount; round-off; totals in paise.	M	P1
FR-BIL-003	**Invoice numbering** sequential per financial year (for example `INV/2026-27/00001`), gap-free for issued invoices, with a separate series for credit notes (`CN/...`) and receipts (`RCT/...`).	M	P1
FR-BIL-004	**Invoice statuses**: `DRAFT`, `ISSUED`, `PARTIALLY_PAID`, `PAID`, `CANCELLED`, `REFUNDED`, `CREDIT_NOTED`.	M	P1

ID	REQUIREMENT	PRI	PH
	Issued invoices are immutable; corrections use cancellation with reason or credit note.		
FR-BIL-005	**Payments**: record one or more payments per invoice with mode (cash, UPI, card, bank transfer, cheque, other), reference/UTR/cheque number, amount, received date/time, received by; **advance payments** and **adjustment** against future invoices; overpayment handling as advance.	M	P1
FR-BIL-006	**Receipt** PDF/print for every payment (thermal 80 mm and A4 layouts).	M	P1
FR-BIL-007	**UPI QR**: dynamic display of the clinic UPI QR with the amount (UPI deep-link format); payment confirmation is manual.	S	P1
FR-BIL-008	**Refunds**: partial or full with reason, mode and approval permission `billing.refund`; produces a refund record and credit note.	M	P1
FR-BIL-009	**Patient ledger**: running balance, advances, dues; **dues list** with ageing.	M	P1
FR-BIL-010	**Packages and prepaid sessions** (Section 4.6): purchase creates a patient package; appointment completion consumes a session; balances displayed; expiry handled.	S	P2
FR-BIL-011	**Discount rules and approvals**: discounts above a configurable threshold require `OWNER` approval.	S	P1
FR-BIL-012	**Product sales** (hearing aids, accessories, batteries) with batch/serial selection when inventory is enabled; warranty details recorded.	S	P2
FR-BIL-013	**GST handling**: configurable per item (exempt/taxable at rate). When the clinic has a GSTIN, invoices show GSTIN, place of supply, taxable value, CGST/SGST or IGST split, HSN/SAC, and tax in words. When not registered, the invoice shows "Bill of Supply" or a plain bill per settings.	M	P1
FR-BIL-014	**Daily collection report** (cash/UPI/card totals, by user), **day-end closing** with cash counting and variance note.	M	P1
FR-BIL-015	**Billing without clinical access**: the receptionist sees patient, service, amount, and payment but not clinical notes.	M	P1
FR-BIL-016	**Estimate/quotation** for hearing aids or packages, convertible to invoice.	S	P2
FR-BIL-017	**Invoice and receipt in English or Marathi** (labels) per patient language preference.	S	P2
FR-BIL-018	**Export** of invoices and payments to CSV/Excel and Tally-compatible XML/CSV for the Client's CA.	S	P2
FR-BIL-019	Cancelling an invoice requires a reason, and any payments must be refunded or reallocated first.	M	P1
FR-BIL-020	All monetary mutations are audited, with before/after state.	M	P1

4.13 INVENTORY AND HEARING-AID STOCK (INV)

ID	REQUIREMENT	PRI	PH
FR-INV-001	**Product catalogue**: type (hearing aid, accessory, battery, earmould/dome, consumable, equipment), SKU, brand, model, description, HSN, tax rate, unit of measure, MRP, selling price, purchase price, reorder level, **serial-tracked** flag, warranty months, active flag, photo.	S	P2
FR-INV-002	**Vendors/suppliers**: name, contact, GSTIN, address, payment terms, notes.	S	P2
FR-INV-003	**Purchase entry** (goods received): vendor, invoice number/date, items with quantity, unit cost, serial numbers (for serial-tracked items), tax, attachment of vendor invoice; updates stock and creates stock movements.	S	P2
FR-INV-004	**Stock ledger**: every movement (`PURCHASE`, `SALE`, `TRIAL_OUT`, `TRIAL_IN`, `RETURN_TO_VENDOR`, `ADJUSTMENT`, `DAMAGED`, `TRANSFER`) recorded immutably with quantity, serial, reference document, user, date.	S	P2
FR-INV-005	**Serial item lifecycle**: `IN_STOCK`, `RESERVED`, `ON_TRIAL`, `SOLD`, `RETURNED`, `DAMAGED`, `WITH_VENDOR` with linked patient and invoice.	S	P2
FR-INV-006	**Low-stock alerts** (dashboard and notification) at reorder level; **stock valuation** report (FIFO or weighted average, configurable).	S	P2
FR-INV-007	**Trial devices**: issue to patient with expected return date and overdue flags.	S	P2
FR-INV-008	**Equipment register** (audiometer, tympanometer, OAE, video-laryngoscope): purchase date, serial, **calibration due date** with alerts, service history, warranty.	S	P2
FR-INV-009	**Stock count/adjustment** workflow with reason.	C	P2
FR-INV-010	**Barcode/QR scanning** of serial numbers using device camera.	C	P3

4.14 EXPENSES (EXP)

ID	REQUIREMENT	PRI	PH
FR-EXP-001	Record expenses with date, category (rent, utilities, salaries, consumables, equipment, calibration, marketing, professional fees, software, travel, repairs, other), vendor, amount, tax, payment mode, reference, attachment (bill photo), notes, and status.	S	P2
FR-EXP-002	**Recurring expenses** (for example rent) auto-created monthly as drafts.	C	P2
FR-EXP-003	**Monthly summary** by category and **profit and loss (management view)** combining invoices and expenses.	S	P2
FR-EXP-004	Expense edit/delete permissions restricted to `OWNER`; all changes audited.	S	P2
FR-EXP-005	**Export** to CSV/Excel.	S	P2

4.15 DASHBOARDS AND ANALYTICS (DSH)

ID	REQUIREMENT	PRI	PH
FR-DSH-001	**Today dashboard** (reception): today's appointments by status, queue, pending requests, new enquiries, new referrals, recalls due, dues to collect, collection so far.	M	P1
FR-DSH-002	**Practitioner dashboard**: today's patient list with alerts, pending documents to finalise, follow-ups due, hearing aid servicing due, upcoming appointments.	M	P1
FR-DSH-003	**Operational analytics**: appointments by status, no-show and cancellation rates, utilisation (booked minutes / available minutes), average wait time, new vs follow-up, by service and provider, by weekday/hour heatmap.	S	P1
FR-DSH-004	**Clinical analytics (aggregate, no identifiers)**: patients by service line, age band, diagnosis category, hearing loss degree distribution, therapy goal attainment rate, average number of sessions to discharge, hearing aid fittings per month, OAE screening referral rate.	S	P2
FR-DSH-005	**Financial analytics**: revenue by service/day/month, collection by mode, dues ageing, average revenue per visit, discounts given, expense totals.	S	P2
FR-DSH-006	**Source analytics**: referral source and how patients found the clinic (website, Google, doctor referral, word of mouth, camp), booking channel, conversion from request to attended.	S	P2
FR-DSH-007	**Website analytics integration**: counts of booking-form starts and submissions, enquiries, referrals (first-party events).	S	P1
FR-DSH-008	Dashboards shall render within 2 seconds for datasets up to 100,000 patients using indexed queries and pre-aggregated daily summaries.	M	P1
FR-DSH-009	Date range filters, comparison to previous period, export of charts and tables (CSV, PDF).	S	P2

4.16 PRIVACY, CONSENT, AUDIT AND COMPLIANCE (PRV)

ID	REQUIREMENT	PRI	PH
FR-PRV-001	**Consent management**: consent types (`TREATMENT`, `DATA_PROCESSING`, `COMMUNICATION`, `PHOTO_VIDEO`, `CASE_STUDY_RESEARCH`, `TELEHEALTH`, `CBT_THERAPY`) with versioned consent texts (en/mr), grantor (patient/guardian), method (digital signature, paper scan, verbal with witness), timestamp, expiry, and withdrawal.	M	P1
FR-PRV-002	**Withdrawal of consent**: recorded with timestamp; downstream effects (for example stop communications, restrict processing) applied automatically; legal retention obligations still honoured.	M	P1
FR-PRV-003	**Privacy notice** presented at registration and on the website, with the purpose, data categories, retention, rights and the **grievance officer** contact.	M	P1
FR-PRV-004	**Data principal requests** (access, correction, erasure, grievance, nomination): create a request record, track due dates (response within the statutory/configured period), verify identity, generate a data export (JSON/PDF), perform correction, and perform **erasure or anonymisation** where legally permitted, with a record of what was retained and why.	M	P1
FR-PRV-005	**Audit log** shall capture: authentication events, CRUD on all business entities (who, what, when, from where, before/after for sensitive fields), **read access to patient and clinical records**, document downloads/exports, permission changes, settings changes, break-glass use.	M	P1
FR-PRV-006	Audit logs are **append-only**, hash-chained (each entry stores the SHA-256 of the previous entry), written by the application with no update/delete code path, and verified by a nightly integrity job.	M	P1
FR-PRV-007	The `OWNER` shall have an **audit viewer** with filters (user, patient, action, date) and export; a patient-specific "who accessed my record" report.	M	P1
FR-PRV-008	**Break-glass (emergency access)**: select reason, enter justification, time-limited access (default 60 minutes), notification to `OWNER`, automatic review task.	M	P1
FR-PRV-009	**Breach management**: incident register (detection, scope, categories of data, affected persons, containment, notification to the Data Protection Board/individuals as required, remediation).	S	P1
FR-PRV-010	**Retention enforcement job**: identifies records past retention for review; deletion only after `OWNER` approval with an immutable deletion certificate.	S	P2
FR-PRV-011	**Minimisation**: public forms collect only needed data; no Aadhaar; no payment card data stored.	M	P1
FR-PRV-012	**Children's data**: verifiable parental/guardian consent captured for under-18s; no behavioural tracking or targeted advertising of children (analytics are aggregate).	M	P1
FR-PRV-013	**Data localisation**: all data stores in India; third-party processors listed in the register with purpose and location.	M	P1
FR-PRV-014	**Export restrictions**: bulk export of patient data requires step-up authentication and is logged with purpose.	M	P1
FR-PRV-015	**Masking**: mobile numbers and email are partially masked on lists for roles without `patient.contact.full` where possible.	S	P1

4.17 NOTIFICATION ABSTRACTION AND INTEGRATION HOOKS (NTF)

(WhatsApp and GMB logic is out of scope; the hooks below make it plug-in ready.)

ID	REQUIREMENT	PRI	PH
FR-NTF-001	A **notification outbox** (`notifications` collection) shall hold every outbound message intent with: type (for example `APPOINTMENT_CONFIRMATION`, `APPOINTMENT_REMINDER_24H`, `APPOINTMENT_REMINDER_2H`, `RESCHEDULE`, `CANCELLATION`, `RECALL`, `PAYMENT_RECEIPT`,	M	P1

ID	REQUIREMENT	PRI	PH
	DOCUMENT_READY, REFERRAL_ACK, ENQUIRY_ACK), recipient (patient/guardian/user), channel preference list, payload (template key + variables, language), scheduled time, status (PENDING, PROCESSING, SENT, DELIVERED, FAILED, CANCELLED, SUPPRESSED), attempts, provider message ID, error.		
FR-NTF-002	**Channel adapters** implement the interface `send(notification) -> result`. Delivered at launch: IN_APP, EMAIL. Reserved: WHATSAPP, SMS.	M	P1
FR-NTF-003	A **dispatcher job** polls the outbox every minute, uses optimistic locking (PENDING to PROCESSING via `findOneAndUpdate`), retries failures with exponential back-off up to 5 attempts, and sets FAILED.	M	P1
FR-NTF-004	**Suppression rules**: respect consent and do-not-contact flags, quiet hours (default 21:00 to 08:00), and avoid duplicates for the same appointment and type.	M	P1
FR-NTF-005	**Template registry** (`messageTemplates`) with keys, languages (en/mr), channel-specific bodies, variables, and approval status (for future WhatsApp template approvals).	S	P1
FR-NTF-006	**Webhook endpoints** (secured with HMAC signature) to receive delivery receipts and inbound events from future channels; handlers update `notifications` and the communication log.	S	P2
FR-NTF-007	**Domain events** (internal event bus) such as `appointment.confirmed`, `appointment.cancelled`, `visit.finalised`, `invoice.paid`, `recall.created`, `patient.created`; subscribers can create notifications; the event list is documented in Appendix 20.6.	M	P1
FR-NTF-008	**In-app notification centre** (bell icon): list, unread counts, mark read, deep links.	M	P1
FR-NTF-009	**Public integration API** (key-authenticated, read-limited) offering: availability, create booking request, booking status. Used by the future WhatsApp module and website. Keys are scoped, rotated and audited.	S	P1
FR-NTF-010	The ERP shall expose **no clinical data** through any integration API.	M	P1

4.18 DATA IMPORT AND EXPORT (IMP)

ID	REQUIREMENT	PRI	PH
FR-IMP-001	**Patient import** from CSV/Excel with a downloadable template, column mapping, validation report (row-level errors), duplicate detection, dry-run mode, and rollback of the batch. Max 5,000 rows per batch.	S	P2
FR-IMP-002	**Appointment history import** (optional) for migrating from paper or spreadsheets.	C	P2
FR-IMP-003	**Export** of lists (patients, appointments, invoices, expenses, enquiries, recalls) to CSV/Excel with permission checks, column selection, and audit.	M	P1
FR-IMP-004	**Full backup export** (encrypted archive) on demand for the Client, containing database dump and files, only for OWNER after step-up authentication.	S	P2
FR-IMP-005	Imported data is tagged with `source = IMPORT` and batch ID.	S	P2

4.19 GLOBAL SEARCH (SRC)

ID	REQUIREMENT	PRI	PH
FR-SRC-001	A **global search box** in the ERP (shortcut `/`) shall find patients (UHID, name, mobile), appointments (booking reference), invoices (number), documents, and recalls, showing only entities the user may access.	M	P1
FR-SRC-002	Search shall handle Marathi (Devanagari) and English and transliteration variants (for example "Palak"/"पलक") through normalised search keys stored on the patient (`searchKeys`) and, where Atlas Search is available, Atlas Search indexes.	S	P1
FR-SRC-003	Clinical free text and CBT notes shall **not** be indexed for global search.	M	P1
FR-SRC-004	Results within 500 ms (p95) for 100,000 patients.	M	P1

5. DATA REQUIREMENTS: COMPLETE MONGODB DESIGN

5.1 DATABASE DESIGN PRINCIPLES

ID	PRINCIPLE
DB-001	**One MongoDB database per environment**: clinic_dev, clinic_staging, clinic_prod. All collections live in that database.
DB-002	**Replica set only** (needed for transactions and change streams). Local development also runs a single-node replica set.
DB-003	**Schema-on-write**: Mongoose schemas plus MongoDB `$jsonSchema` validators on critical collections (Section 5.9) so that bad data cannot enter even through scripts.
DB-004	**Embed** data that is owned by a parent, read together with it, and bounded in size (for example guardians in a patient, items in an invoice). **Reference** data that is shared, independently queried, or can grow without bound (for example visits of a patient).
DB-005	**No document may exceed 1 MB** in practice (hard limit is 16 MB); arrays that could exceed 200 elements shall be moved to their own collection.
DB-006	**Money** = integer paise; **quantities** = number; **percentages** = number with the fixed scale stated; **dates** = BSON `Date` in UTC, **date-only values** (date of birth, invoice date) = string `YYYY-MM-DD` in IST calendar; **phone numbers** = E.164 string (`+91XXXXXXXXXX`).
DB-007	**Enums** are stored as upper-case strings (CONFIRMED) and defined once in `packages/shared`.
DB-008	**Soft delete** with `deletedAt` and `deletedBy` on clinical, financial and master data; all default queries add `{ deletedAt: null }` through a Mongoose query helper.
DB-009	**Optimistic concurrency** through Mongoose `optimisticConcurrency: true` (uses `__v`) on all mutable business documents.
DB-010	**Every collection has `createdAt, updatedAt, createdBy, updatedBy`** (Mongoose `timestamps`).
DB-011	**Sensitive data** (CBT content, optionally selected free-text clinical notes) is encrypted before it reaches the database (Section 5.10).
DB-012	**Index discipline**: every query pattern in Section 7 is backed by an index; `autoIndex` is disabled in production, and indexes are created by migrations (Section 17).
DB-013	**Transactions** are used only where multi-document atomicity is required (booking, invoice issue, stock movement, patient merge).
DB-014	**Audit trail** is separate and append-only; business collections do not store audit history except `statusHistory` arrays where explicitly stated.

ID	PRINCIPLE
DB-015	**Personal data minimisation**: Aadhaar and card numbers are never stored.
DB-016	**Pre-aggregated statistics** (`dailyStats`) are used for dashboards rather than scanning clinical collections.
DB-017	**Naming**: collections are plural camelCase (`audiologyEvaluations`); fields camelCase; references end with `Id` (single) or `Ids` (array).
DB-018	**Branch readiness**: every master and transactional document may carry an optional `branchId` (default null = the single clinic).

5.2 COMMON FIELDS

All collections (except where noted) include:

FIELD	TYPE	NOTES
`_id`	ObjectId	Primary key.
`createdAt`, `updatedAt`	Date	Mongoose timestamps (UTC).
`createdBy`, `updatedBy`	ObjectId → `users`	Null for system actions or anonymous public submissions.
`deletedAt`, `deletedBy`	Date, ObjectId	Soft delete markers (null when active).
`__v`	Number	Version key for optimistic concurrency.
`branchId`	ObjectId, optional	Future multi-branch support.
`source`	Enum, optional	`ERP`, `WEBSITE`, `IMPORT`, `API`, `SYSTEM`.

In the field tables below these are **not repeated**. `Req` column: **Y** required, **N** optional, **C** conditional.

5.3 EMBEDDING VS REFERENCING DECISIONS

RELATIONSHIP	CHOICE	REASON
Patient to guardians	Embed (array, max 5)	Bounded, always read with the patient
Patient to medical alerts	Embed (array)	Bounded, displayed everywhere
Patient to visits	Reference (`visits.patientId`)	Unbounded, queried independently
Visit to clinical documents	Reference (`visitId` in each document)	Different permissions and lifecycles
Invoice to line items	Embed (array, max 100)	Immutable after issue, read together
Invoice to payments	Reference (`payments.invoiceId`)	Multiple, independent audit and refunds
Audiology evaluation to individual tests (PTA, speech, immittance, OAE, ABR, tinnitus, CAPD)	Embed as sub-objects	One evaluation = one clinical document; read and printed together
Therapy plan to goals	Embed (array, max 30)	Bounded; versioned with plan
Goal progress data points	Reference (`sessionNotes.goalData`)	Unbounded; derived into goal summary
Appointment to status history	Embed (array)	Small and bounded (< 20)
Hearing aid to service events	Embed (array, max 100)	Bounded device lifecycle
Content (pages/posts) to revisions	Reference (`contentRevisions`)	Unbounded
Notification to attempts	Embed (array, max 10)	Bounded

5.4 ENTITY-RELATIONSHIP OVERVIEW

```
erDiagram
    USERS ||--o{ APPOINTMENTS : "provider"
    USERS ||--o{ AUDIT_LOGS : "actor"
    ROLES ||--o{ USERS : "assigned"
    SERVICES ||--o{ APPOINTMENTS : "booked for"
    PATIENTS ||--o{ APPOINTMENTS : "has"
    PATIENTS ||--o{ VISITS : "has"
    APPOINTMENTS ||--o| VISITS : "results in"
    VISITS ||--o{ CASE_HISTORIES : "records"
    VISITS ||--o{ SL_ASSESSMENTS : "records"
    VISITS ||--o{ AUDIOLOGY_EVALUATIONS : "records"
    VISITS ||--o{ SESSION_NOTES : "records"
    VISITS ||--o{ CBT_RECORDS : "records"
    PATIENTS ||--o{ THERAPY_PLANS : "has"
    THERAPY_PLANS ||--o{ SESSION_NOTES : "tracked by"
    PATIENTS ||--o{ HEARING_AIDS : "uses"
    STOCK_ITEMS ||--o| HEARING_AIDS : "dispensed as"
    PATIENTS ||--o{ RECALLS : "needs"
    PATIENTS ||--o{ CONSENTS : "grants"
    PATIENTS ||--o{ DOCUMENTS : "has"
    VISITS ||--o{ INVOICES : "billed in"
    INVOICES ||--o{ PAYMENTS : "settled by"
    PATIENTS ||--o{ PATIENT_PACKAGES : "buys"
    PRODUCTS ||--o{ STOCK_ITEMS : "instances"
    PRODUCTS ||--o{ STOCK_MOVEMENTS : "moves"
    VENDORS ||--o{ PURCHASES : "supplies"
    PURCHASES ||--o{ STOCK_MOVEMENTS : "creates"
    APPOINTMENTS ||--o{ NOTIFICATIONS : "triggers"
    RECALLS ||--o{ NOTIFICATIONS : "triggers"
    PAGES ||--o{ CONTENT_REVISIONS : "versions"
    POSTS ||--o{ CONTENT_REVISIONS : "versions"
    ENQUIRIES }o--o| PATIENTS : "may convert to"
    REFERRALS_IN }o--o| PATIENTS : "may convert to"
```

5.5 COLLECTION SPECIFICATIONS

Conventions: →X = reference to collection X. `Enum[...]` = allowed values. `{...}` = embedded object. `[...]` = array. **Idx** = indexes. **Rel** = relationships and rules.

5.5.1 USERS

FIELD	TYPE	REQ	NOTES
email	String (lower-case)	Y	Unique among non-deleted
mobile	String E.164	N	Unique if present
passwordHash	String	Y	Argon2id (includes salt and params)
passwordChangedAt	Date	Y	
passwordHistory	[String]	N	Last 5 hashes
name	{first, middle, last, display}	Y	
photoMediaId	→media	N	
designation	String	N	For example "Audiologist & Speech-Language Pathologist"
qualifications	[{degree, institution, year}]	N	
rci	{number, verified: Boolean, verifiedAt, proofDocumentId}	N	Controls website display
signatureMediaId	→media	N	Image for reports
languages	[String]	N	`en`, `mr`, `hi`, `gu`
roleIds	[→roles]	Y	At least one
extraPermissions	[String]	N	Overrides (audited)
deniedPermissions	[String]	N	Overrides (audited)
status	Enum[`ACTIVE`,`INVITED`,`SUSPENDED`,`DEACTIVATED`]	Y	
twoFactor	{enabled, secretEnc, recoveryCodeHashes[], enrolledAt}	N	`secretEnc` encrypted
lockout	{failedCount, lastFailedAt, lockedUntil}	Y	
lastLoginAt, lastLoginIp	Date, String	N	
calendar	{color, bookable: Boolean, serviceIds[], defaultSlotMin}	N	Provider settings
preferences	{locale, timezone, notify: {email, inApp}}	N	
invite	{tokenHash, expiresAt, invitedBy}	N	
passwordReset	{tokenHash, expiresAt, usedAt}	N	

Idx: `{email:1}` unique partial `{deletedAt: null}`; `{mobile:1}` unique sparse partial; `{status:1}`; `{roleIds:1}`; `{ "calendar.bookable":1 }`. **Rel:** `roleIds`→`roles`. Last active `OWNER` cannot be removed (BR-030).

5.5.2 REFRESHTOKENS (SESSIONS)

FIELD	TYPE	REQ	NOTES
userId	→users	Y	
familyId	String (UUID)	Y	Groups rotated tokens (reuse detection)
tokenHash	String	Y	SHA-256 of the opaque token
issuedAt, expiresAt, absoluteExpiresAt	Date	Y	
usedAt	Date	N	Set at rotation
revokedAt, revokedReason	Date, String	N	
device	{userAgent, os, browser, ipTruncated, label}	N	
lastSeenAt	Date	N	

Idx: `{tokenHash:1}` unique; `{userId:1, revokedAt:1}`; `{familyId:1}`; **TTL** `{ absoluteExpiresAt:1 }` with `expireAfterSeconds: 604800` (7 days after expiry).

5.5.3 ROLES

FIELD	TYPE	REQ	NOTES
key	String	Y	`OWNER`, `RECEPTIONIST`, `CLINICIAN`, `EDITOR`, `ADMIN`, or custom
name	{en, mr}	Y	
permissions	[String]	Y	`resource.action` list (Appendix 20.3)
system	Boolean	Y	System roles cannot be deleted
description	String	N	

Idx: `{key:1}` unique.

5.5.4 AUDITLOGS (APPEND-ONLY)

FIELD	TYPE	REQ	NOTES
ts	Date	Y	Event time (server)
seq	Number	Y	Monotonic sequence (from `counters`)
actor	{userId, role, name, type: Enum[`USER`,`SYSTEM`,`PUBLIC`,`API_KEY`]}	Y	
action	String	Y	`auth.login.success`, `patient.read`, `clinical.finalise`, ...
entity	{type, id, patientId}	N	`patientId` enables "who accessed this patient" queries
outcome	Enum[`SUCCESS`,`DENIED`,`FAILURE`]	Y	
changes	[{path, before, after}]	N	Sensitive values masked or hashed
context	{ip, userAgent, requestId, sessionId, purpose, breakGlassId}	N	
prevHash	String	Y	Hash of previous entry
hash	String	Y	SHA-256(`prevHash` + canonical entry JSON)

Idx: `{ts:-1}`; `{ "entity.patientId":1, ts:-1}`; `{ "actor.userId":1, ts:-1}`; `{action:1, ts:-1}`; `{seq:1}` unique. **Rules:** application database user has **insert + find only** on this collection (no update/delete). Nightly job verifies the chain and alerts on mismatch.

5.5.5 COUNTERS

FIELD	TYPE	NOTES
_id	String	For example `uhid:2027`, `invoice:2026-27`, `receipt:2026-27`, `creditnote:2026-27`, `audit:seq`, `token:2026-11-10:PROVIDERID`
seq	Number	Last issued number
updatedAt	Date	

Usage: `findOneAndUpdate({_id}, {$inc:{seq:1}}, {upsert:true, returnDocument:'after'})`; gap-free numbers (invoices) are incremented **inside** the issuing transaction so a rollback does not consume a number.

5.5.6 `clinicSettings` (SINGLETON, `_id: "clinic"`)

FIELD	TYPE	REQ	NOTES
`identity`	{legalName, tradeName, tagline{en,mr}, logoMediaId, letterheadMediaId, stampMediaId}	Y	
`contact`	{phones[{label, number, isPrimary, whatsapp: Boolean}], email, website}	Y	Single source for NAP
`address`	{line1, line2, locality, taluka, district, state, pin, country, landmark{en,mr}, geo{type:"Point", coordinates:[lng,lat]}, mapsUrl, parkingNote, accessibilityNote}	Y	
`registrations`	{gstin, pan, clinicalEstablishmentNo, shopActNo, udyamNo}	N	`pan` stored encrypted
`credentials`	{rciVerified, cbtQualificationVerified, titlePermitted, proofDocumentIds[], verifiedBy, verifiedAt}	Y	Gate flags (BR-041, BR-042)
`hours`	{weekly: [{day:0-6, sessions:[{start:"10:00", end:"13:00"}]}], granularityMin, timezone:"Asia/Kolkata"}	Y	Default clinic hours
`bookingPolicy`	{minNoticeMin, maxAdvanceDays, holdHours, autoConfirm, cancelWindowHours, rescheduleLimit, sameDay, dailyCapByService{}, graceNoShowMin}	Y	
`numbering`	{uhidPrefix, uhidFormat, invoicePrefix, receiptPrefix, bookingRefLength}	Y	
`tax`	{defaults{service, product}, showGst, roundingMode}	Y	
`payment`	{modes[], upiId, upiName, bankDetails{accountName, number(enc), ifsc, bank}}	N	
`notifications`	{quietHours{start, end}, reminderOffsetsMin[1440,120], senderEmail, senderName}	Y	
`retention`	{clinicalYears, minorUntilAge, billingYears, auditYears}	Y	
`features`	{inventory, expenses, packages, cbt, onlineBooking, waitlist, referralInbox}	Y	
`website`	{openingSoon:Boolean, openingDate, banner, footerCredit:Boolean, social{}, analytics{}}	Y	
`grievanceOfficer`	{name, email, phone}	Y	DPDP requirement

5.5.7 `lookups`

FIELD	TYPE	NOTES
`type`	String	`REFERRAL_SOURCE`, `CANCEL_REASON`, `EXPENSE_CATEGORY`, `DOC_TYPE`, `RELATION`, `LANGUAGE`, `TAG`, `OCCUPATION`, `DISCOUNT_REASON`, `DX_SUBSET`
`key`	String	Machine key
`label`	{en, mr}	
`order`, `active`	Number, Boolean	
`meta`	Object	Type-specific

Idx: `{type:1, key:1}` unique.

5.5.8 `services`

FIELD	TYPE	REQ	NOTES
`code`	String	Y	`SL-SPEECH`, `AU-PTA` ... unique
`category`	Enum[`SPEECH_LANGUAGE`,`AUDIOLOGY`,`CBT`,`OTHER`]	Y	
`name`, `shortDescription`	{en, mr}	Y	
`slug`	{en, mr}	Y	Website URL
`variants`	[{key, name{en,mr}, durationMin, bufferMin, pricePaise, taxTreatment, hsnSac, bookableOnline, forNewPatient, forFollowUp, active}]	Y	At least one
`templateKey`	String	Y	Clinical template (for example `AU_PTA`)
`launchStatus`	Enum[`AVAILABLE`,`COMING_SOON`,`HIDDEN`]	Y	
`requiresVerifiedCredential`	Enum[`NONE`,`RCI`,`CBT`]	Y	
`providerIds`	[→users]	N	Eligible providers
`onlineModeAllowed`, `homeVisitAllowed`	Boolean	N	
`displayOrder`	Number	Y	
`cmsPageId`	→pages	N	Detail page content
`icon`	String	N	

Idx: `{code:1}` unique; `{category:1, displayOrder:1}`; `{ "slug.en":1 }` unique; `{ "slug.mr":1 }` unique.

5.5.9 `packages`

FIELD	TYPE	NOTES
`name`	{en, mr}	
`serviceVariants`	[{serviceId, variantKey, sessions}]	
`pricePaise`	Number	
`validityDays`	Number	
`active`	Boolean	

5.5.10 `providerSchedules`

FIELD	TYPE	REQ	NOTES
`providerId`	→users	Y	
`effectiveFrom`, `effectiveTo`	String (date)	Y/N	Allows schedule changes over time
`weekly`	[{day:0-6, sessions:[{start, end, modes[]}]}]	Y	`start`/`end` as "HH:mm" in IST
`granularityMin`	Number	Y	5,10,15,20,30

Idx: `{providerId:1, effectiveFrom:-1}`.

5.5.11 `blockedTimes` (HOLIDAYS, CLOSURES, BLOCKS)

FIELD	TYPE	REQ	NOTES
`kind`	Enum[`HOLIDAY`,`CLOSURE`,`BLOCK`,`LEAVE`]	Y	

FIELD	TYPE	REQ	NOTES
providerId	→users	N	Null = whole clinic
title	String	Y	
start, end	Date	Y	UTC
allDay	Boolean	Y	
recurrence	{freq:Enum[NONE,DAILY,WEEKLY,MONTHLY,YEARLY], interval, until, byDay[]}	N	
publicNote	{en, mr}	N	Shown on website if flagged
showOnWebsite	Boolean	N	

Idx: {providerId:1, start:1, end:1}; {kind:1, start:1}.

5.5.12 APPOINTMENTS

FIELD	TYPE	REQ	NOTES
bookingRef	String	Y	Public reference, for example BK-7K3QH2 (unique)
patientId	→patients	C	Required once linked
prospect	{name, ageText, dob, sex, guardianName, mobile, email, locality, concern, language, forWhom, isNew}	C	For online requests not yet linked
providerId	→users	Y	
serviceId	→services	Y	
variantKey	String	Y	
type	Enum[NEW,FOLLOW_UP,REVIEW,WALK_IN,SCREENING, CAMP]	Y	
mode	Enum[CLINIC,ONLINE,HOME]	Y	
start, end	Date	Y	UTC; end includes buffer end separately below
durationMin, bufferMin	Number	Y	
slotKeys	[String]	Y	ISO UTC of each granularity unit between start and end + buffer
holdsSlots	Boolean	Y	True while status is REQUESTED, CONFIRMED, CHECKED_IN, IN_PROGRESS
status	Enum[REQUESTED,CONFIRMED,CHECKED_IN,IN_PROGRESS,COMPLETED,CANCELLED,NO_SHOW,RESCHEDULED,REJECTED,EXPIRED]	Y	
statusHistory	[{status, at, by, reason, note}]	Y	
source	Enum[WEBSITE,PHONE,WALK_IN,STAFF,REFERRAL,API]	Y	
reason	String (≤ 500)	N	Patient's stated concern
internalNote	String	N	
flags	[String]	N	WHEELCHAIR, NEEDS_MARATHI, CHILD_EXTRA_TIME, HEARING_IMPAIRED, INTERPRETER
expiresAt	Date	C	For REQUESTED: hold expiry
token	{date, number, issuedAt}	N	Queue token
checkInAt, startedAt, completedAt	Date	N	
cancel	{by:Enum[PATIENT,CLINIC,SYSTEM], reasonKey, note, at, withinPolicy:Boolean}	N	
rescheduledFromId, rescheduledToId	→appointments	N	
seriesId	ObjectId	N	Recurring series
location	{address, geo, travelBufferMin}	C	For HOME
meeting	{url, passcodeEnc, instructions}	C	For ONLINE
visitId	→visits	N	Set at start
invoiceId	→invoices	N	
packageUseId	ObjectId	N	Consumed package session
manage	{tokenHash, expiresAt}	N	Public manage-link
consentRef	{privacyVersion, acceptedAt, ip}	C	For website submissions

Idx: - { providerId:1, slotKeys:1 } **unique**, partialFilterExpression: { holdsSlots: true } (ADR-03; DB-enforced no double booking). - { bookingRef:1 } unique. - { providerId:1, start:1 }; { patientId:1, start:-1 }; { status:1, start:1 }; { status:1, expiresAt:1 } (hold-expiry job); { "prospect.mobile":1 }; { seriesId:1 }; { "manage.tokenHash":1 } sparse.

Rules: holdsSlots is set by the service layer whenever status changes; status transitions validated by a state machine (Section 4.7.1). Booking is done inside a transaction (Section 6.7).

5.5.13 WAITLIST

FIELD	TYPE	NOTES
patientId / prospect	→patients / object	
serviceId, providerId?	refs	
preferredDays	[Number]	
preferredWindows	[{start, end}]	
fromDate, toDate	String (date)	
status	Enum[ACTIVE,OFFERED,BOOKED,EXPIRED,REMOVED]	
offers	[{appointmentSlot, offeredAt, response}]	

Idx: {serviceId:1, status:1, createdAt:1}.

5.5.14 PATIENTS

FIELD	TYPE	REQ	NOTES
uhid	String	Y	Unique, immutable
name	{salutation, first, middle, last, display, displayMr}	Y	displayMr optional Devanagari spelling
sex	Enum[MALE,FEMALE,OTHER,UNDISCLOSED]	Y	
dob	String YYYY-MM-DD	C	Null if approximate
ageApprox	{years, months, recordedOn}	C	When DOB unknown
preferredLanguages	[String]	Y	Order = preference
homeLanguages	[String]	N	Languages spoken at home
contact	{mobile, mobileAlt, email, mobileIsShared: Boolean, preferredChannel: Enum[CALL,WHATSAPP,SMS,EMAIL,NONE], bestTime}	Y	
address	{line1, line2, locality, taluka, district, state, pin, country}	N	
occupation, education, maritalStatus	String / Enum	N	
guardians	[{name, relation, mobile, email, isPrimaryContact, isLegalGuardian, canConsent, canPickUp, occupation, notes}]	C	Required for minors
familyGroupId	ObjectId	N	Siblings/family link

FIELD	TYPE	REQ	NOTES
referral	{sourceKey, doctorName, hospital, referralId→referrals_in, note}	N	
medicalAlerts	[{key, label, severity:Enum[INFO,WARN,CRITICAL], note, addedAt, addedBy}]	N	HEARING_AID_USER, COCHLEAR_IMPLANT, DYSPHAGIA_RISK, SEIZURES, ALLERGY, COMM_NEEDS
abha	{number(enc), address}	N	Optional; never Aadhaar
photoMediaId	→media	N	
tags	[String]	N	
status	Enum[ACTIVE,INACTIVE,DECEASED,ARCHIVED,MERGED]	Y	
deceasedOn	String date	C	
mergedIntoId	→patients	C	When MERGED
stats	{firstVisitAt, lastVisitAt, totalVisits, noShowCount, lateCancelCount, totalBilledPaise, totalPaidPaise}	Y	Denormalised; maintained by service layer
consentSummary	{treatment, dataProcessing, communication, photoVideo, caseStudy, cbt} each {status, at, consentId}	Y	Denormalised cache of consents
doNotContact	Boolean	Y	
searchKeys	[String]	Y	Normalised tokens (name parts in Latin and Devanagari, transliterations, last 10 digits of mobiles, UHID, guardian names)
portal	{enabled:false, userRef:null}	N	Reserved (Section 5.6)

Idx: {uhid:1} unique; {searchKeys:1}; { "contact.mobile":1 }; { status:1, "stats.lastVisitAt":-1 }; { "name.last":1, "name.first":1 }; { dob:1 }; { familyGroupId:1 } sparse; { tags:1 }; { "guardians.mobile":1 }; (optional Atlas Search index patients_search, Section 6.9). **Rel:** merged patients keep mergedIntoId; all lookups follow it once.

5.5.15 CONSENTS AND CONSENTTEMPLATES

consentTemplates | Field | Type | Notes | |---|---|---| | type | Enum[TREATMENT,DATA_PROCESSING,COMMUNICATION,PHOTO_VIDEO,CASE_STUDY_RESEARCH,TELEHEALTH,CBT_THERAPY] | | | version | Number | | | text | {en, mr} | Full legal text (approved by advocate) | | hash | String | SHA-256 of canonical text | | active | Boolean | | | effectiveFrom | Date | |

consents | Field | Type | Req | Notes | |---|---|---|---| | patientId | →patients | Y | | | type, templateId, templateVersion, textHash | | Y | Exact text consented to | | status | Enum[GRANTED,WITHDRAWN,EXPIRED] | Y | | | grantedBy | {kind:Enum[PATIENT,GUARDIAN], name, relation, guardianIndex} | Y | | | method | Enum[DIGITAL_SIGNATURE,PAPER_SCAN,VERBAL_WITNESSED,WEBSITE_CHECKBOX] | Y | | | signatureMediaId | →media | C | | | witness | {name, userId} | C | | | grantedAt, expiresAt, withdrawnAt, withdrawnReason | Date, String | | | | context | {ip, userAgent, deviceLabel} | N | | | documentId | →documents | N | Signed PDF |

Idx: {patientId:1, type:1, status:1}; {templateId:1}.

5.5.16 DOCUMENTS

FIELD	TYPE	REQ	NOTES
patientId	→patients	C	Null for non-patient documents
visitId	→visits	N	
docType	Enum[AUDIOLOGY_REPORT,NEWBORN_SCREEN_REPORT,SL_EVAL_REPORT,VOICE_REPORT,FLUENCY_REPORT,SWALLOW_REPORT,PROGRESS_SUMMARY,DISCHARGE_SUMMARY,HA_WARRANTY_SLIP,REFERRAL_LETTER,ATTENDANCE_CERT,HOME_PROGRAMME,APPOINTMENT_SLIP,INVOICE,RECEIPT,CONSENT_FORM,EXTERNAL_REPORT,SCAN,OTHER]	Y	
origin	Enum[GENERATED,UPLOADED]	Y	
title	String	Y	
number	String	C	Document number (generated)
language	Enum[en,mr,bilingual]	N	
file	{bucket, key, mime, sizeBytes, sha256, pages, encryptedKeyRef}	Y	Private S3 object
status	Enum[DRAFT,FINAL,SUPERSEDED,VOID]	Y	
version	Number	Y	
supersedes	→documents	N	
sourceRef	{collection, id, version}	N	Record used to generate
verification	{code, qrUrl}	N	Public authenticity check
av	{scanned: Boolean, result, scannedAt}	N	Virus scan
tags	[String]	N	
visibility	Enum[CLINICAL,ADMIN,BILLING]	Y	Permission class

Idx: {patientId:1, docType:1, createdAt:-1}; {visitId:1}; { "verification.code":1 } unique sparse; { number:1 } unique sparse.

5.5.17 VISITS

FIELD	TYPE	REQ	NOTES
patientId	→patients	Y	
appointmentId	→appointments	N	
providerId	→users	Y	
serviceIds	[→services]	Y	
visitType	Enum[NEW,FOLLOW_UP,REVIEW,SCREENING,CAMP,HOME,ONLINE]	Y	
start, end	Date	Y	Actual times
chiefComplaint	String	N	
status	Enum[OPEN,CLOSED]	Y	
documentRefs	[{collection, id, kind, status}]	Y	Clinical docs created
summary	String	N	One-line clinician summary
billing	{invoiceId, billed:Boolean}	N	
locked	Boolean	Y	After all documents final

Idx: {patientId:1, start:-1}; {providerId:1, start:-1}; {appointmentId:1} unique sparse; {status:1}.

5.5.18 PROBLEMS (DIAGNOSIS/PROBLEM LIST)

FIELD	TYPE	NOTES
patientId	→patients	
label	String	Clinical description
icd10	String	Optional (for example H90.3, F80.0)
category	Enum[HEARING,SPEECH,LANGUAGE,FLUENCY,VOICE,SWALLOWING,NEURO,MENTAL_HEALTH,OTHER]	CBT-category items are stored encrypted in cbtRecords only
onset	String date	

FIELD	TYPE	NOTES
status	Enum[ACTIVE,RESOLVED,INACTIVE]	
firstRecordedVisitId	→visits	

Idx: {patientId:1, status:1}.

5.5.19 clinicalTemplates

FIELD	TYPE	NOTES
key	String	AU_PTA, SL_FLUENCY, ... unique
title	{en, mr}	
version	Number	
fields	[{path, label{en,mr}, type, unit, min, max, step, required, options[], computed?, section, order, helpText}]	UI form definition
computedRules	[{key, formulaVersion}]	
active	Boolean	

Idx: {key:1, version:-1}.

5.5.20 caseHistories

FIELD	TYPE	REQ	NOTES
patientId, visitId	refs	Y	
status, version, finalisedAt, finalisedBy	common clinical doc fields	Y	See 5.5.20.1
informant	{name, relation, reliability:Enum[GOOD,FAIR,POOR]}	N	
chiefComplaint, hopi	String	Y	
prenatal	{motherAge, complications[], infections[], medications, consanguinity, alcoholTobacco, pregnancyOutcome}	N	
natal	{gestationWeeks, deliveryMode, birthWeightG, birthCry, apgar{1min,5min}, nicuDays, jaundice, phototherapy, exchangeTransfusion, oxygen, ventilationDays, complications}	N	
postnatal	{feedingDifficulty, seizures, infections[], ototoxicDrugs[], headInjury, hospitalisations[]}	N	
milestones	{headControlM, sittingM, crawlingM, walkingM, babblingM, firstWordM, twoWordM, sentencesM, toiletTrainedM, concerns}	N	Months
medicalHistory	{conditions[], surgeries[], medications[], allergies[], immunisation, vision, earInfections{count, side}, tonsil/adenoid}	N	
hearingHistory	{onset, laterality, progression, noiseExposure, familyHL, previousTests[], devicesUsed[]}	N	
speechLanguageHistory	{parentConcern, onsetAge, progress, previousTherapy[], communicationMode, languagesHeard[], dominantLanguage, exposureHours, bilingualNotes}	N	
feedingSwallowing	{breastBottle, weaningAge, texturesAccepted, coughingChoking, drooling, tubeFeeding, mealDurationMin}	N	
familyHistory	{speechLanguageDisorder, hearingLoss, stuttering, neurological, other}	N	
educationSocial	{school, grade, medium, performance, peerRelations, behaviour, attention, sleep, screenTimeHrs}	N	
behaviourObservation	String	N	
previousEvaluations	[{date, by, summary, documentId}]	N	
riskFlags	[String]	N	
prefilledFrom	→caseHistories	N	

Idx: {patientId:1, createdAt:-1}; {visitId:1}.

5.5.20.1 Common clinical document fields

Every clinical document collection (caseHistories, slAssessments, audiologyEvaluations, therapyPlans, sessionNotes, avtRecords, cbtRecords, ...) also has:

FIELD	TYPE	NOTES
patientId, visitId	→patients, →visits	
authorId	→users	
status	Enum[DRAFT,FINAL,AMENDED]	
version	Number	Increments on amendment
previousVersionId	ObjectId	
amendmentReason	String	
finalisedAt, finalisedBy	Date, →users	
draftSavedAt	Date	Autosave marker
attachments	[→media/documents]	
templateKey, templateVersion	String, Number	
computed	Object	Server computed values with formulaVersion
impression, recommendations	{en, mr} or String	
encrypted	Boolean	True for CBT

5.5.21 slAssessments (SPEECH-LANGUAGE ASSESSMENTS)

A single collection with a kind discriminator so that one visit can hold several assessments:

kind ∈ [ORAL_MECHANISM, ARTICULATION, LANGUAGE, FLUENCY, VOICE, APHASIA, DYSARTHRIA, DYSPHAGIA, AAC, DEVELOPMENTAL_SCREEN].

5.5.21.1 oral_mechanism

FIELD	TYPE	NOTES
structures	{face, lips, tongue, jaw, hardPalate, softPalate, uvula, teeth, occlusion, breathing, drooling} each {status:Enum[NORMAL,ABNORMAL,NOT_TESTED], findings, severity}	
tongueTie	Boolean	
velopharyngeal	{nasalEmission, hypernasality, hyponasality, cleftSigns}	
ddk	{pa, ta, ka, pataka} each number syllables/second	0 to 10

5.5.21.2 `ARTICULATION`

FIELD	TYPE	NOTES
`test`	{name, version, language, administeredOn}	Configurable list
`items`	[{phoneme, position:Enum[`INITIAL`,`MEDIAL`,`FINAL`], level:Enum[`WORD`,`PHRASE`,`SENTENCE`,`CONVERSATION`], target, response, status:Enum[`CORRECT`,`SUBSTITUTION`,`OMISSION`,`DISTORTION`,`ADDITION`], substitutedWith, stimulable:Boolean, note}]	Up to ~120 items
`phonologicalProcesses`	[{process, occurrencePct, ageAppropriate:Boolean}]	See FR-CLN-101 list
`computed`	{pcc, pccR, pvc, errorsBySoundClass, intelligibilityPct}	
`intelligibility`	{contextPct, noContextPct, listenerFamiliarity}	
`severity`	Enum[`NONE`,`MILD`,`MODERATE`,`SEVERE`]	Clinician judgement

5.5.21.3 `LANGUAGE`

FIELD	TYPE	NOTES
`tests`	[{name, version, language, subtests[{name, raw, standard, percentile, ageEquivalentMonths}], totalStandard}]	
`receptiveAgeMonths`, `expressiveAgeMonths`, `chronologicalAgeMonths`	Number	
`computed`	{receptiveDelayMonths, expressiveDelayMonths, delayCategory}	
`mlu`	{morphemes, utterances, value}	
`vocabulary`	{receptiveCount, expressiveCount, notes}	
`syntaxMorphology`, `narrative`, `pragmatics{eyeContact, turnTaking, topicMaintenance, jointAttention, requesting, commenting, repair}`	Mixed	Pragmatics ratings 0 to 4
`multilingualProfile`	[{language, exposurePct, proficiency}]	

5.5.21.4 `FLUENCY`

FIELD	TYPE	NOTES
`samples`	[{task:Enum[`CONVERSATION`,`READING`,`MONOLOGUE`,`PHONE`], durationSec, totalSyllables, counts:{partWordRep, wholeWordRep, phraseRep, prolongation, block, interjection, revision}, notes}]	
`computed`	{pctSS:Number, perSample[], dysfluencyPer100Words, speechRateSPM, articulationRateSPM}	
`ssi4`	{frequencyScore, durationScore, physicalConcomitantsScore, total, percentile, severity}	Instrument version recorded
`secondaryBehaviours`	[String]	
`avoidance`	{words, situations, fearLevel0to10}	
`oases`	{sections:{...}, total, impact:Enum[`MILD`,`MILD_MODERATE`,`MODERATE`,`MODERATE_SEVERE`,`SEVERE`]}	
`history`	{onsetAge, durationMonths, familyHistory, variability, previousTherapy, triggers}	
`severity`	Enum[`VERY_MILD`,`MILD`,`MODERATE`,`SEVERE`,`VERY_SEVERE`]	

5.5.21.5 `VOICE`

FIELD	TYPE	NOTES
`grbas`	{g, r, b, a, s} each integer 0..3	`computed.grbasTotal`
`capeV`	{overallSeverity, roughness, breathiness, strain, pitch, loudness, resonance, other} each 0..100	
`acoustic`	{meanF0Hz, f0RangeHz, jitterPct, shimmerPct, hnrDb, intensityDb, softwareName}	
`phonation`	{mptA:[s,s,s], mptS:[...], mptZ:[...], bestA, bestS, bestZ, szRatio}	seconds
`vhi30`	{functional, physical, emotional, items[30], total, severity}	Items 0..4; total 0..120
`vhi10`	{items[10], total}	Total 0..40
`laryngealFindings`	{source, date, text, documentId}	
`demandHistory`	{occupation, vocalLoadHrsPerDay, singing, smoking, reflux, hydration, medications}	
`hygiene`	[String]	

5.5.21.6 `APHASIA`

`{ onsetDate, etiology, lesion:{side, site, imagingDocId}, handedness, premorbidLanguages[], type, severity, test:{name, version, language}, aq, subtests: {spontaneousSpeech, auditoryComprehension, repetition, naming}, modalities:{listening, speaking, reading, writing, calculation, gesture} (0..10), apraxiaOfSpeech:Boolean, dysarthria:Boolean, cognitionScreen:{tool, score}, functionalCommunication:{tool, score, needs[]}, caregiverNotes }.`

5.5.21.7 `DYSARTHRIA`

`{ etiology, type, severity (0..4), frenchay:{reflex, respiration, lips, palate, laryngeal, tongue, intelligibility, totalGrade} each a..e, subsystems:{respiration, phonation, resonance, articulation, prosody} (0..4), intelligibility:{wordsPct, sentencesPct}, speakingRateWPM, mptSeconds, amr, smr, aacNeed:Boolean }.`

5.5.21.8 `DYSPHAGIA`

`{ history:{weightLossKg, chokingEpisodes, coughingMeals, chestInfections, tubeFeeding:{type, since}, onsetDate, etiology}, oral:{lipSeal, mastication, bolusControl, residue, dentition}, pharyngeal:{delayedTrigger, throatClearing, wetVoice, nasalRegurgitation}, trials:[{consistency, iddsiLevel (0..7), volumeMl, position, cough:Boolean, wetVoice:Boolean, delayed:Boolean, spo2Baseline, spo2After, notes}], waterSwallow:{volume3ml, volume90ml, result}, fois (1..7), recommendations:{diet: {foodIddsi, fluidIddsi}, strategies[], supervision, restrictions[]}, instrumental:{referred:Boolean, type:Enum[VFSS,FEES], resultText, documentId}, riskLevel:Enum[LOW,MODERATE,HIGH], caregiverTraining:[...] }.`

Idx (all kinds): `{patientId:1, kind:1, createdAt:-1}`; `{visitId:1}`; `{status:1, authorId:1}`.

5.5.22 `AUDIOLOGYEVALUATIONS`

One document per visit evaluation, with embedded test sections (each optional):

```
audiologyEvaluations {
  patientId, visitId, authorId, status, version, ...common
  otoscopy:   { right:{pinna, canal, tm, wax, discharge, notes}, left:{...} }
  pta:        { ... 5.5.22.1 }
  speech:     { ... 5.5.22.2 }
  immittance: { ... 5.5.22.3 }
  oae:        { ... 5.5.22.4 }
  abr:        { ... 5.5.22.5 }
  tinnitus:   { ... 5.5.22.6 }
  capd:       { ... 5.5.22.7 }
  equipment:  { audiometer:{make, model, serial, calibratedOn}, tympanometer:{...}, oaeDevice:{...} }
  interpretation: { en, mr }, recommendations: { en, mr }, referral: {...}
}
```

5.5.22.1 `pta` (pure tone audiometry)

FIELD	TYPE	NOTES
`conditions`	{transducer:Enum[`SUPRA_AURAL`,`INSERT`,`FREE_FIELD`,`BONE_VIBRATOR`], booth:Boolean, ambientNoiseDbA, responseMode:Enum[`HAND_RAISE`,`BUTTON`,`PLAY`,`VRA`,`CORA`,`OTHER`], reliability:Enum[`GOOD`,`FAIR`,`POOR`], testedBy}	
`right.air`	[{hz, db, masked:Boolean, nr:Boolean, maskLevelDb}]	hz ∈ {125,250,500,750,1000,1500,2000,300

FIELD	TYPE	NOTES
		0,4000,6000,8000}; db −10..120
right.bone	same shape	hz ∈ {250,500,1000,2000,4000}
left.air, left.bone	same	
freeField	{warble:[{hz, db, aided:Boolean}]}	For children/aided testing
computed.right, computed.left	{pta3, pta4, hfa, abGap:{hz:value}, degree, type, configuration}	degree per configured scale; type and configuration have suggested and confirmed
computed.formulaVersion	String	For reproducibility
audiogramImageMediaId	→media	Rendered PNG/SVG snapshot at finalisation

Validation: db multiple of 5 (or 1 if configured), nr points must have a db equal to the audiometer limit; bone > air + 10 triggers a warning.

5.5.22.2 speech

{ right:{srt, sdt, wrs:[{level, levelRef:Enum[HL,SL], pct, list}], mcl, ucl, dynamicRange(computed), masking, aided:{...}}, left:{...}, binaural:{...}, material:{language:Enum[en,mr,hi,other], listName, type:Enum[MONO,BI,SENTENCE], mode:Enum[LIVE,RECORDED]}, piPb:{maxScore, rolloverScore, rolloverIndex(computed)}, sin:{snr50, noiseType} }.

5.5.22.3 immittance

{ probeToneHz:Enum[226,678,1000], right:{tymp:{type:Enum[A,As,Ad,B,C,OTHER], peakPressureDaPa, complianceMl, ecvMl, widthDaPa}, reflexes:{ipsi:{500,1000,2000,4000}, contra:{500,1000,2000,4000}} each {thresholdDb, status:Enum[PRESENT,ABSENT,NOT_TESTED]}, decay:{freq, result}}, left:{...} }.

5.5.22.4 oae

FIELD	TYPE	NOTES
type	Enum[TEOAE,DPOAE]	
stage	Enum[SCREEN_1,SCREEN_2,DIAGNOSTIC]	
device	{make, model, protocol, stimulusDb}	
right, left	{result:Enum[PASS,REFER,INCONCLUSIVE], snrByBand:[{hz, snrDb, noiseDb, present:Boolean}], dpgram:[{f2Hz, dpDb, noiseFloorDb}], probeFit, durationSec, notes}	
overall	Enum[BOTH_PASS,UNILATERAL_REFER,BILATERAL_REFER,INCONCLUSIVE]	Computed
newborn	{birthDate, gestationWeeks, birthWeightG, nicuDays, riskIndicators[], birthPlace, motherName, hospitalRegNo, screeningAgeHours, nextStep:Enum[DISCHARGE,RESCREEN,DIAGNOSTIC_ABR,ENT_REFERRAL], rescreenDue, counselled:Boolean}	riskIndicators list per JCIH-style factors

5.5.22.5 abr

{ stimulus:{type:Enum[CLICK,TONE_BURST], freqHz, polarity:Enum[RAREFACTION,CONDENSATION,ALTERNATING], ratePerSec}, state:Enum[NATURAL_SLEEP,SEDATED,AWAKE], montage, right:{series:[{intensityDbNhl, waveI:{latMs, ampUv}, waveIII:{...}, waveV:{...}}], thresholdDbNhl, interpeak:{I_III, III_V, I_V}, vAmpRatio}, left:{...}, interpretation, waveformMediaIds[] }.

5.5.22.6 tinnitus

{ laterality, onset, durationMonths, nature[], constancy, associated:{hearingLoss, vertigo, hyperacusis, sleepDisturbance, anxiety, depression, headache, jawPain}, pitchMatchHz, loudnessMatchDb, loudnessMatchRef:Enum[HL,SL], mmlDb, ldl:{...}, residualInhibition:{done, result}, thi:{items[25] (0|2|4), total (0..100), grade}, tfi:{total}, vas:{loudness, annoyance, impactOnSleep}, trCategory (0..4), counselling:[...], soundTherapy:{device, settings, hoursPerDay}, followUpPlan }.

5.5.22.7 capd

{ referralBehaviours[], peripheralHearingNormal:Boolean, battery:[{testKey, testName, version, language, ear:Enum[R,L,BINAURAL], condition, rawScore, scorePct, normMean, normSd, zScore(computed), result:Enum[NORMAL,BORDERLINE,ABNORMAL]}], questionnaires:[{name, score}], diagnosis:{capd:Boolean, subprofile[], severity}, plan:{environmental[], auditoryTraining[], remoteMic:Boolean, school letter}, }.

Idx: {patientId:1, createdAt:-1}; {visitId:1}; { "oae.overall":1, createdAt:-1 }; { "oae.newborn.rescreenDue":1 } sparse.

5.5.23 newbornscreenings (SCREENING REGISTER / CAMPS)

FIELD	TYPE	NOTES
patientId	→patients	Created in quick mode
evaluationId	→audiologyEvaluations	
batchId	String	Camp or hospital batch
screeningDate	String date	
result	Enum[PASS,REFER_UNILATERAL,REFER_BILATERAL,INCONCLUSIVE]	
nextStep, rescreenDue, completedFollowUpAt		
hospital, motherName, contact		

Idx: {batchId:1}; {result:1, rescreenDue:1}.

5.5.24 avtrecords

{ patientId, visitId, ...common, childProfile:{ageAtIdentificationM, ageAtAmplificationM, ageAtImplantationM, devices:[{type, ear, hoursPerDay}], lingSixSound:{distanceM, detection:{m,ah,oo,ee,sh,s}, identification:{...}}, languageAgeM }, measures:[{tool:Enum[CAP,SIR,LITTLEARS,MAIS,IT_MAIS,OTHER], score, date}], goals:[{domain:Enum[DETECTION,DISCRIMINATION,IDENTIFICATION,COMPREHENSION,SPEECH,LANGUAGE,COGNITION,COMMUNICATION], text, status, level}], session:{planSummary, activities[], parentCoaching{topics, parentConfidence0to10}, homePractice}, nextPlan }.

5.5.25 cochlearimplants

{ patientId, ear, manufacturer, model, processor, implantDate, surgeon, centre, activationDate, maps:[{date, clinic, programmer, comments, impedanceMediaId}], aidedThresholds:[{date, hz, db}], speechPerception:[{date, test, scorePct}], status }.

5.5.26 hearingaids

FIELD	TYPE	REQ	NOTES
patientId	→patients	Y	
ear	Enum[RIGHT,LEFT,BILATERAL_PAIR]	Y	
device	{make, model, style:Enum[BTE,RIC,ITE,ITC,CIC,IIC,BODY,BONE_CONDUCTION,CROS,BICROS], techLevel, channels, batterySize, rechargeable, serialNo, earmould{type, material, vent, impressionDate}}	Y	
stockItemId	→stockItems	N	Link to inventory
status	Enum[TRIAL,DISPENSED,RETURNED,UPGRADED,LOST,DAMAGED,UNDER_REPAIR]	Y	
trial	{start, end, outcome, loanDevice:Boolean}	N	
fitting	{date, fittedBy, audiogramEvaluationId, rem:[{hz, targetDb, achievedDb}], aidedThresholds:[{hz, db}],	N	

FIELD	TYPE	REQ	NOTES
	functionalGain:[...], programming:{memories, channelGains[], compression, noiseReduction, feedbackManager, directionality, volumeControl, telecoil}, outcomeMeasures:[{tool, score}], orientationChecklist:{insertion, removal, battery, cleaning, noise, phone, care}}		
reprogramming	[{date, by, summary, evaluationId}]	N	max 100
warranty	{startDate, expiryDate, terms}; lossDamageCover: {expiryDate}	N	
purchase	{invoiceId, pricePaise, date}	N	
services	[{date, type:Enum[CLEANING,TUBING,REPAIR,REPROGRAM,REFIT,LOAN,OTHER], fault, outcome, costPaise, performedBy, nextDue}]	N	max 100
nextServiceDue	Date	N	Drives recalls

Idx: {patientId:1, status:1}; { "device.serialNo":1 } sparse; {nextServiceDue:1}; { "warranty.expiryDate":1 }.

5.5.27 THERAPYPLANS

FIELD	TYPE	NOTES
patientId, authorId	refs	
domain	Enum[ARTICULATION,LANGUAGE,FLUENCY,VOICE,APHASIA,DYSARTHRIA,DYSPHAGIA,AVT,AAC,CAPD,TINNITUS,OTHER]	
impression	String	
frequency	{sessionsPerWeek, durationMin, plannedWeeks}	
startDate, reviewDate, endDate	String date	
status	Enum[ACTIVE,ON_HOLD,ACHIEVED,DISCONTINUED]	
goals	[{goalId, type:Enum[LONG_TERM,SHORT_TERM], text, parentGoalId, baseline:{value, unit, text}, target:{value, unit, text}, mastery:{accuracyPct, consecutiveSessions}, status:Enum[NOT_STARTED,IN_PROGRESS,ACHIEVED,MODIFIED,DISCONTINUED], progress:{latestPct, sessionsAtMastery, achievedOn}, priority}]	max 30
riskOfDiscontinuation	Boolean	From missed sessions
sessionCount	Number	Maintained
dischargeSummary	{date, reason, finalStatus, recommendations, followUpAdvice}	

Idx: {patientId:1, status:1}; {reviewDate:1, status:1}.

5.5.28 SESSIONNOTES

FIELD	TYPE	NOTES
patientId, visitId, therapyPlanId	refs	
sessionNumber	Number	
attendance	{present:Boolean, accompaniedBy, lateByMin}	
durationMin	Number	
soap	{subjective, objective, assessment, plan}	String fields (can be bilingual)
goalData	[{goalId, trials, correct, pct (computed), cueLevel:Enum[INDEPENDENT,VERBAL,VISUAL,MODEL,HAND_OVER_HAND], activity, material, reinforcement, notes}]	
participation	Enum[POOR,FAIR,GOOD,EXCELLENT]	
homePractice	{assigned:[{task, frequency, materials, dueDate}], previousCompliance:Enum[COMPLETED,PARTIAL,NOT_DONE,NA], note}	
caregiverTraining	{topics[], understanding:Enum[POOR,FAIR,GOOD]}	
nextPlan	String	

Idx: {patientId:1, createdAt:-1}; {therapyPlanId:1, sessionNumber:1}; { "goalData.goalId":1 }.

5.5.29 CBTRECORDS (ENCRYPTED)

Because CBT content is highly sensitive, the entire clinical payload is encrypted. Only non-sensitive metadata remains queryable.

FIELD	TYPE	NOTES
patientId, visitId, authorId	refs	
kind	Enum[INTAKE,RISK_ASSESSMENT,SAFETY_PLAN,SCALE,FORMULATION,TREATMENT_PLAN,SESSION,THOUGHT_RECORD,EXPERIMENT,ACTIVITY_SCHEDULE,DISCHARGE]	
sessionNumber	Number	Metadata
recordDate	String date	Metadata
scaleKey	String	For SCALE kind (for example PHQ9) — metadata only, not the score
status, version	common	
secure	{v:1, alg:"AES-256-GCM", kid, wrappedDek, iv, tag, ct}	Encrypted JSON payload (see below)
searchable	**absent**	No searchable fields by design

Decrypted payload shapes (illustrative): - INTAKE: {presentingProblems[], onset, durationMonths, triggers, severity, impact:{work, relationships, sleep, appetite, concentration}, previousTreatment, medicationReported[], substanceUse, medicalConditions[], familyHistory, support, strengthsValues, goalsInOwnWords[]} - RISK_ASSESSMENT: {suicidalIdeation:Enum[NONE,PASSIVE,ACTIVE], plan, intent, means, selfHarm:{recent, history}, harmToOthers, protective[], level:Enum[NONE,LOW,MODERATE,HIGH], actionsTaken[], referral:{to, date}} - SCALE: {instrument, version, items:[0..3 ...], total, severity, item9Flag, interpretation}; PHQ-9: 9 items 0..3; GAD-7: 7 items 0..3. - SESSION: {agenda[], moodBefore0to10, moodAfter0to10, homeworkReview, content, techniques[], insights, newHomework[], rating: {srs|ors}, riskCheck, durationMin, nextPlan} - THOUGHT_RECORD: {situation, automaticThoughts:[{text, beliefPct}], emotions:[{name, intensityPct}], distortions[], evidenceFor, evidenceAgainst, balancedThought, reRating:{beliefPct, intensityPct}, outcome}

Idx: {patientId:1, kind:1, recordDate:-1}; {authorId:1, createdAt:-1}. **Access:** requires cbt.read/cbt.write, step-up authentication, and writes an audit entry on every read.

5.5.30 OUTCOMEMEASURES (FOR TREND CHARTS; NON-CBT)

FIELD	TYPE	NOTES
patientId, visitId	refs	
instrument	String	THI, VHI30, VHI10, APHAB, IOI_HA, OASES, WAB_AQ, FOIS, CAP, SIR, LITTLEARS, PCC, PCT_SS, SSI4, GRBAS_TOTAL, ...
value	Number	
unit	String	

FIELD	TYPE	NOTES
date	String date	
sourceRef	{collection, id}	The document it came from

Idx: {patientId:1, instrument:1, date:1}. CBT scores are **not** copied here.

5.5.31 RECALLS

FIELD	TYPE	NOTES
patientId	→patients	
type	Enum[THERAPY_REVIEW,REEVALUATION,ANNUAL_AUDIOMETRY,HA_SERVICE,WARRANTY_EXPIRY,NEWBORN_RESCREEN,TINNITUS_REVIEW,NO_SHOW_FOLLOWUP,POST_ASSESSMENT,CBT_REVIEW,POST_DISCHARGE,CALIBRATION,CUSTOM]	
dueDate	String date	
priority	Enum[LOW,NORMAL,HIGH]	
status	Enum[PENDING,CONTACTED,SCHEDULED,COMPLETED,DISMISSED,OVERDUE]	
origin	{kind:Enum[RULE,MANUAL], ruleKey, sourceRef:{collection,id}}	
assignedTo	→users	
contactLog	[{at, by, channel, outcome, note}]	
appointmentId	→appointments	On scheduling
snoozeUntil	String date	
closedAt, closedReason		
dedupeKey	String	patientId:type:sourceId:due — unique

Idx: {status:1, dueDate:1}; {patientId:1, status:1}; {assignedTo:1, status:1, dueDate:1}; {dedupeKey:1} unique.

5.5.32 CLINICALREFERRALSOUT

{ patientId, visitId, toType:Enum[ENT,NEUROLOGIST,PSYCHIATRIST,PAEDIATRICIAN,PSYCHOLOGIST,DENTIST,RADIOLOGY,OTHER], toName, toOrg, reason, urgency, documentId, sentAt, responseReceivedAt, outcomeNote }.

5.5.33 INVOICES

FIELD	TYPE	REQ	NOTES
number	String	C	Assigned on issue, for example INV/2026-27/00001; unique
series	String	Y	INV, BOS (bill of supply), CN
financialYear	String	Y	2026-27
status	Enum[DRAFT,ISSUED,PARTIALLY_PAID,PAID,CANCELLED,REFUNDED,CREDIT_NOTED]	Y	
invoiceDate	String date	Y	
patientId	→patients	Y	
billTo	{name, address, mobile, gstin}	Y	Snapshot
visitId, appointmentId	refs	N	
items	[{lineNo, kind:Enum[SERVICE,PRODUCT,PACKAGE,OTHER], refId, description, hsnSac, qty, unitPricePaise, discount:{type:Enum[PCT,AMT], value, reasonKey, approvedBy}, taxableValuePaise, taxRatePct, cgstPaise, sgstPaise, igstPaise, cessPaise, totalPaise, serialNo}]	Y	max 100
totals	{subtotalPaise, discountPaise, taxableValuePaise, cgstPaise, sgstPaise, igstPaise, roundOffPaise, grandTotalPaise}	Y	
paidPaise, balancePaise, advanceAppliedPaise, refundedPaise	Number	Y	Maintained inside transactions
gst	{registered:Boolean, supplierGstin, placeOfSupply, reverseCharge:false}	Y	
notes, terms	String	N	
cancel	{at, by, reason}	C	
relatedInvoiceId	→invoices	C	For credit notes
documentId	→documents	N	PDF
issuedAt, issuedBy		C	Immutable thereafter

Idx: {number:1} unique sparse; {patientId:1, invoiceDate:-1}; {status:1, invoiceDate:-1}; {financialYear:1, series:1, number:1}; {visitId:1}. **Rule:** Mongoose pre('save') blocks edits to monetary fields when status != DRAFT.

5.5.34 PAYMENTS

FIELD	TYPE	NOTES
receiptNo	String	Unique, for example RCT/2026-27/00001
patientId, invoiceId	refs	invoiceId null for advances
kind	Enum[PAYMENT,ADVANCE,REFUND,ADJUSTMENT]	
mode	Enum[CASH,UPI,CARD,BANK_TRANSFER,CHEQUE,OTHER]	
amountPaise	Number (> 0)	Sign carried by kind
reference	{utr, chequeNo, bank, cardLast4, note}	
receivedAt	Date	
receivedBy	→users	
dayCloseId	→dayCloses	
status	Enum[RECORDED,VOID]	
voidReason	String	

Idx: {receiptNo:1} unique; {invoiceId:1}; {patientId:1, receivedAt:-1}; {receivedAt:1, mode:1}.

5.5.35 DAYCLOSES

{ date (String), expected:{cashPaise, upiPaise, cardPaise, bankPaise}, counted:{cashPaise}, variancePaise, note, closedBy, closedAt, status } with unique {date:1}.

5.5.36 PATIENTPACKAGES

{ patientId, packageId, purchasedOn, expiresOn, items:[{serviceId, variantKey, total, used}], invoiceId, status:Enum[ACTIVE,EXHAUSTED,EXPIRED,CANCELLED], uses:[{appointmentId, at}] }. **Idx:** {patientId:1, status:1}.

5.5.37 PRODUCTS

FIELD	TYPE	NOTES
sku, type	String, Enum[HEARING_AID,ACCESSORY,BATTERY,EARMOULD,CONSUMABLE,EQUIPMENT]	
brand, model, description	String	
hsn, taxRatePct		
uom	String	pcs, pack
mrpPaise, sellingPricePaise, purchasePricePaise	Number	
reorderLevel	Number	
serialTracked	Boolean	
warrantyMonths	Number	
photoMediaId, active		

Idx: {sku:1} unique; {type:1, active:1}; {brand:1, model:1}.

5.5.38 STOCKITEMS (SERIALISED UNITS)

{ productId, serialNo, batchNo, status:Enum[IN_STOCK,RESERVED,ON_TRIAL,SOLD,RETURNED,DAMAGED,WITH_VENDOR], purchaseId, purchaseCostPaise, patientId, invoiceId, hearingAidId, trial:{issuedOn, expectedReturn}, location }. **Idx:** {serialNo:1} unique sparse; {productId:1, status:1}.

5.5.39 STOCKMOVEMENTS (IMMUTABLE)

{ productId, stockItemId?, type:Enum[PURCHASE,SALE,TRIAL_OUT,TRIAL_IN,RETURN_TO_VENDOR,ADJUSTMENT,DAMAGED,TRANSFER], qty (signed), unitCostPaise, ref:{kind, id}, reason, at, by }. **Idx:** {productId:1, at:-1}; { "ref.id":1 }. Current quantity = sum of movements (cached on products.stockQty, maintained in transactions).

5.5.40 VENDORS

{ name, contactPerson, mobile, email, gstin, address, paymentTerms, notes, active }. **Idx:** {name:1}.

5.5.41 PURCHASES

{ vendorId, invoiceNo, invoiceDate, items:[{productId, qty, unitCostPaise, taxRatePct, serials[], batchNo}], totals, attachmentMediaIds[], status:Enum[DRAFT,POSTED,CANCELLED], postedAt }. **Idx:** {vendorId:1, invoiceDate:-1}; {vendorId:1, invoiceNo:1} unique.

5.5.42 EQUIPMENT

{ name, type, make, model, serialNo, purchaseDate, warrantyUntil, calibration:{lastOn, dueOn, certificateMediaId, intervalMonths}, serviceHistory:[{date, by, note, costPaise}], location, status }. **Idx:** { "calibration.dueOn":1 }.

5.5.43 EXPENSES

{ date, categoryKey, vendorId?, vendorName, description, amountPaise, taxPaise, paymentMode, reference, attachmentMediaIds[], recurring:{freq, until, parentId}, status:Enum[DRAFT,POSTED,VOID] }. **Idx:** {date:-1}; {categoryKey:1, date:-1}.

5.5.44 NOTIFICATIONS (OUTBOX + IN-APP)

FIELD	TYPE	NOTES
type	String	Section 4.17 types
recipient	{kind:Enum[PATIENT,GUARDIAN,USER,PROSPECT], refId, name, mobile, email, language}	
channels	[Enum[IN_APP,EMAIL,WHATSAPP,SMS]]	Preference order
payload	{templateKey, variables{}, deepLink}	
related	{appointmentId, recallId, invoiceId, documentId}	
scheduledFor	Date	
status	Enum[PENDING,PROCESSING,SENT,DELIVERED,FAILED,CANCELLED,SUPPRESSED]	
suppressedReason	String	
attempts	[{at, channel, ok, providerMessageId, error}]	max 10
lockedBy, lockedAt	String, Date	Dispatcher lease
dedupeKey	String	Unique per appointment+type+channel
readAt	Date	In-app read marker

Idx: {status:1, scheduledFor:1}; {dedupeKey:1} unique; { "recipient.refId":1, createdAt:-1}; { "recipient.kind":1, "recipient.refId":1, readAt:1 }; **TTL** on createdAt 400 days for IN_APP read notifications (partial TTL via separate expiresAt field).

5.5.45 MESSAGETEMPLATES

{ key, channel, language, subject, body, variables[], category:Enum[UTILITY,SERVICE,MARKETING], approval:{status, providerRef, approvedAt}, active }. **Idx:** {key:1, channel:1, language:1} unique.

5.5.46 COMMUNICATIONLOGS

{ patientId, direction:Enum[IN,OUT], channel:Enum[CALL,SMS,WHATSAPP,EMAIL,IN_PERSON,OTHER], summary, outcome, by, at, notificationId? }. **Idx:** {patientId:1, at:-1}.

5.5.47 ENQUIRIES (WEBSITE CONTACT FORM)

{ name, mobile, email, topic, message, language, consent:{privacyVersion, at, ip}, status:Enum[NEW,IN_PROGRESS,RESPONDED,CLOSED,SPAM], assignedTo, notes:[{by, at, text}], respondedAt, patientId?, utm:{source, medium, campaign}, pageUrl, spamScore }. **Idx:** {status:1, createdAt:-1}; {mobile:1}.

5.5.48 REFERRALSIN (DOCTOR REFERRAL FORM)

{ referrer:{name, designation, org, regNo, mobile, email}, patient:{name, age, sex, guardianName, guardianMobile}, reason, serviceRequested, urgency:Enum[ROUTINE,SOON,URGENT], attachmentDocumentIds[], status:Enum[NEW,REVIEWED,CONVERTED,DECLINED], patientId?, appointmentId?, notes, consent, reviewedBy, reviewedAt }. **Idx:** {status:1, createdAt:-1}.

5.5.49 SUBSCRIBERS

{ name, contact:{mobile?, email?}, kind:Enum[WAITLIST,NEWSLETTER,CAMP_ALERTS], language, consent:{at, ip, textVersion}, status:Enum[ACTIVE,UNSUBSCRIBED], unsubscribeTokenHash }. **Idx:** { "contact.email":1 } sparse unique per kind; {status:1}.

5.5.50 PAGES (CMS)

FIELD	TYPE	NOTES
key	String	home, about, legal-privacy, service:AU-PTA ...
slug	{en, mr}	
title	{en, mr}	
blocks	[{id, type:Enum[HERO,TEXT,IMAGE_TEXT,CARDS,FAQ,CTA,GALLERY,TESTIMONIALS,STEPS,MAP,VIDEO,CHECKLIST,SERVICE_GRID,POSTS_GRID], data:{en:{}, mr:{}}, order}]	Validated per block type
seo	{title{en,mr}, description{en,mr}, canonical, ogImageMediaId, noindex}	
status	{en:Enum[DRAFT,IN_REVIEW,SCHEDULED,PUBLISHED,ARCHIVED], mr:Enum[...]}	
publishAt, publishedAt, unpublishedAt	Date	
clinicalContent	Boolean	Needs medical review flag
review	{reviewedBy, lastReviewedAt}	
template	String	Layout variant

FIELD	TYPE	NOTES
currentRevisionId	→contentRevisions	

Idx: {key:1} unique; { "slug.en":1 } unique sparse; { "slug.mr":1 } unique sparse; { "status.en":1, publishedAt:-1 }.

5.5.51 POSTS

{ slug:{en,mr}, title:{en,mr}, excerpt:{en,mr}, body:{en:String(sanitised HTML/JSON), mr:String}, category:String, tags:[String], coverMediaId, authorId, readingMinutes:{en,mr}, status:{en,mr}, publishAt, publishedAt, seo:{...}, related:[postId], clinicalContent:Boolean, review:{reviewedBy, lastReviewedAt}, viewCount }. **Idx:** { "slug.en":1 } unique sparse; { "slug.mr":1 } unique sparse; { "status.en":1, publishedAt:-1 }; { category:1, publishedAt:-1 }; { tags:1 }.

5.5.52 FAQS

{ category, question:{en,mr}, answer:{en,mr}, order, status:{en,mr}, serviceIds[], pageKeys[], review }. **Idx:** {category:1, order:1}.

5.5.53 TESTIMONIALS

{ displayName, relation, text:{en,mr}, rating?, serviceId?, consent:{verified:Boolean, documentId, at, scope}, status, publishedAt, order }. Only consent.verified = true can be published (BR-044).

5.5.54 MEDIA

{ kind:Enum[IMAGE,PDF,VIDEO,AUDIO,OTHER], file:{bucket,key,mime,sizeBytes,sha256,width,height}, variants:[{key,width,format,fileKey}], alt:{en,mr}, caption:{en,mr}, credit, tags[], usage:[{collection,id}], public:Boolean, av:{scanned,result} }. **Idx:** {tags:1}; {kind:1, createdAt:-1}; { "usage.id":1 }.

5.5.55 MENUS, REDIRECTS, EVENTS, CHECKLISTS, CHECKLISTRESPONSES

menus: { key:Enum[HEADER,FOOTER_1,FOOTER_2], items:[{label:{en,mr}, type:Enum[PAGE,POST,SERVICE,URL], refId, url, children[], order}] }. **Idx:** {key:1} unique.
redirects: { from, to, statusCode:Enum[301,302], locale, hits, active }. **Idx:** {from:1} unique.
events: { title:{en,mr}, description:{en,mr}, start, end, venue, registrationUrl, status, coverMediaId, publicNote }. **Idx:** {start:-1}.
checklists: { key, version, title:{en,mr}, intro:{en,mr}, questions:[{id, text:{en,mr}, options:[{value, label:{en,mr}, weight}], redFlag:Boolean}], scoring:{bands:[{min,max,message:{en,mr}}]}, disclaimer:{en,mr}, status }. **Idx:** {key:1, version:-1}.
checklistResponses: { checklistKey, version, answers:[{qid, value}], score, band, language, contactRequested:Boolean, contact?:{...encrypted}, consent?, createdAt }. **Idx:** {checklistKey:1, createdAt:-1}. No personal data unless contactRequested.

5.5.56 CONTENTREVISIONS

{ entity:{collection, id}, version, snapshot:Object, changedBy, changeNote, createdAt }. **Idx:** { "entity.id":1, version:-1 }.

5.5.57 DATAREQUESTS (DPDP RIGHTS)

{ patientId, requesterName, requesterRelation, type:Enum[ACCESS,CORRECTION,ERASURE,GRIEVANCE,NOMINATION,WITHDRAW_CONSENT], receivedAt, identityVerified:{by, at, method}, dueDate, status:Enum[OPEN,IN_PROGRESS,FULFILLED,PARTIALLY_FULFILLED,REJECTED], actions:[{at, by, note}], exportDocumentId, outcomeNote, legalRetentionNote }. **Idx:** {status:1, dueDate:1}.

5.5.58 BREAKGLASSGRANTS

{ userId, patientId?, scope:Enum[PATIENT,CBT_RECORD], reasonKey, justification, grantedAt, expiresAt, notifiedOwnerAt, reviewedBy, reviewedAt, reviewOutcome }. **Idx:** {userId:1, expiresAt:1}; {reviewedAt:1}.

5.5.59 INCIDENTS (SECURITY/PRIVACY)

{ detectedAt, reportedBy, category, description, dataCategories[], estimatedAffected, containment, notifications:[{to, at, method}], status, rootCause, corrective[] }.

5.5.60 DAILYSTATS (PRE-AGGREGATED DASHBOARD DATA)

{ _id: "YYYY-MM-DD:providerId|ALL", date, providerId, appointments:{total, byStatus{}, byService{}, newPatients}, utilisation:{availableMin, bookedMin}, revenue:{invoicedPaise, collectedPaise, byMode{}}, enquiries:{new}, referralsIn:{new}, noShows, cancellations, avgWaitMin }. Rebuilt nightly and incrementally on events. **Idx:** {date:-1}.

5.5.61 APIKEYS

{ name, keyPrefix, keyHash, scopes[], lastUsedAt, expiresAt, revokedAt, createdFor }. **Idx:** {keyPrefix:1} unique.

5.5.62 IMPORTBATCHES

{ kind, filename, uploadedBy, rows, accepted, rejected, errors[{row, message}], status:Enum[DRY_RUN,APPLIED,ROLLED_BACK], appliedAt, rollbackAt }.

5.5.63 JOBS (SCHEDULER COLLECTION)

Managed by the scheduler library (for example Agenda's agendaJobs); documented here for completeness. Jobs: expire-held-appointments (every 5 min), dispatch-notifications (every 1 min), generate-reminders (every 5 min), mark-no-shows (every 10 min), nightly-recalls (02:00 IST), rebuild-daily-stats (02:30 IST), verify-audit-chain (03:00 IST), purge-expired-tokens (hourly), retention-review (monthly), calibration-alerts (daily), low-stock-alerts (daily), recurring-expenses (monthly).

5.6 PATIENT PORTAL READINESS (P3)

patients.portal and a reserved patientAccounts collection ({patientId, mobile, otpHash, lastLoginAt, status}) are specified but **not built** in P1. When built, portal users get a separate auth realm with access limited to their own appointments, invoices, and released documents.

5.7 INDEX SUMMARY AND SIZING

COLLECTION	ESTIMATED DOCS/YEAR	AVG DOC SIZE	NOTES
patients	1,200	3 KB	
appointments	9,000	2 KB	
visits	8,000	1 KB	
audiologyEvaluations	3,000	8 KB	
slAssessments	2,500	6 KB	
sessionNotes	6,000	3 KB	
cbtRecords	1,000	5 KB	encrypted
invoices	8,000	4 KB	
payments	9,000	1 KB	
auditLogs	400,000	0.8 KB	largest; 7-year retention (about 2 GB)
notifications	40,000	1 KB	
media/documents	6,000	1 KB (+ files on S3)	

Total database size after 5 years is estimated at **3 to 6 GB** (excluding files), which fits a single replica set tier with large headroom. File storage (S3) is estimated at 20 to 60 GB in 5 years.

5.8 DATA INTEGRITY RULES (SELECTED)

ID	RULE
DB-100	patients.uhid is unique and never reused, even after deletion or merge.
DB-101	An active appointment cannot overlap another active appointment for the same provider (enforced by the unique multikey index).
DB-102	An appointment with status = COMPLETED must have a visitId.
DB-103	A finalised clinical document cannot be edited in place; amendments create a new version document and mark the previous one AMENDED/SUPERSEDED.
DB-104	An issued invoice's monetary fields are immutable; paidPaise + refundedPaise <= grandTotalPaise + advanceAppliedPaise.
DB-105	totals.grandTotalPaise = Σ item totals + roundOff verified on save.
DB-106	Invoice numbers are gap-free within a series and financial year.
DB-107	Stock cannot go negative unless allowNegativeStock is enabled by the owner.
DB-108	A serial number is unique across stockItems and exists in exactly one active status.
DB-109	A minor patient (age < 18) must have at least one guardian with canConsent = true.

ID	RULE
DB-110	Deleting a patient is not allowed while related clinical or financial records exist inside the retention period; only `ARCHIVED` or the erasure workflow applies.
DB-111	A `recall` has a unique `dedupeKey`.
DB-112	A published `testimonial` requires `consent.verified = true`.
DB-113	`auditLogs` have no update or delete code paths, and the DB role prevents them.
DB-114	CBT documents never contain plaintext sensitive fields in the database.
DB-115	`services.requiresVerifiedCredential = CBT` services cannot be created in `AVAILABLE` status while `cbtQualificationVerified = false`.

5.9 SCHEMA VALIDATION (`$jsonSchema`) POLICY

Critical collections (`patients`, `appointments`, `invoices`, `payments`, `auditLogs`, `cbtRecords`, `consents`) get server-side validators with `validationLevel: "strict"` for new documents and `validationAction: "error"`.
Validators check required fields, types, enums, numeric ranges (for example paise as `long` ≥ 0), and string formats (UHID pattern, E.164 phone pattern).
Validators are created by migrations (Section 17.3) and are versioned.
Example (appointments, abridged) is in Section 6.8.

5.10 ENCRYPTION AND KEY MANAGEMENT (DATA MODEL VIEW)

DATA	PROTECTION
All data at rest	MongoDB Atlas encryption at rest (AES-256) with customer-managed key through AWS KMS (BYOK) where the tier allows
All data in transit	TLS 1.2+ between app and database
CBT payloads, TOTP secrets, PAN, bank account numbers, ABHA numbers, meeting passcodes	**Application-level envelope encryption**: AES-256-GCM data-encryption key (DEK) per record or per patient, wrapped by a KMS key (CMK); stored `{kid, wrappedDek, iv, tag, ct}`; decryption happens only in the API process after authorisation
Key rotation	CMK rotated annually (KMS automatic rotation); DEKs rewrapped lazily; re-encryption script available
Files on S3	SSE-KMS, private bucket, pre-signed URLs valid for 5 minutes
Backups	Encrypted by Atlas; snapshot copies encrypted
Passwords/tokens	Argon2id hashes; token hashes SHA-256; no plaintext secrets stored

Alternative: MongoDB **Client-Side Field Level Encryption (CSFLE)** with automatic encryption can replace the custom envelope encryption if the chosen tier and driver support it; the logical model (which fields are encrypted) remains the same.

5.11 RETENTION AND TTL SUMMARY

DATA	RETENTION	MECHANISM
Refresh tokens	Expiry + 7 days	TTL index
In-app notifications (read)	400 days	TTL on `expiresAt`
Held appointment requests (expired)	Status changes to `EXPIRED`; record kept 2 years	Job
Enquiries (spam)	90 days	Job
Clinical records	Per `clinicSettings.retention` (default 7 years after last visit; minors until age 21 or 7 years, whichever later)	Retention review job, approval workflow
Billing records	8 years	Review job
Audit logs	7 years	Archive to cold storage, then delete after approval
Soft-deleted records	30 days then eligible for purge unless under retention	Job
Checklist responses (anonymous)	3 years	Job

5.12 SAMPLE DOCUMENTS

5.12.1 PATIENT


```
{
  "_id": { "$oid": "6720a1b2c3d4e5f607182930" },
  "uhid": "PGC-27-000042",
  "name": { "salutation": "Master", "first": "Aarav", "last": "Kulkarni", "display": "Aarav Kulkarni", "displayMr": "आरव कुलकर्णी" },
  "sex": "MALE",
  "dob": "2022-03-14",
  "preferredLanguages": ["mr", "en"],
  "homeLanguages": ["mr", "hi"],
  "contact": { "mobile": "+919876543210", "mobileIsShared": true, "preferredChannel": "CALL", "bestTime": "after 5 pm" },
  "address": { "line1": "Plot 12, Sadashiv Peth", "locality": "Sadashiv Peth", "taluka": "Satara", "district": "Satara", "state": "Maharashtra", "pin": "415002", "cou
  "guardians": [
    { "name": "Sneha Kulkarni", "relation": "MOTHER", "mobile": "+919876543210", "isPrimaryContact": true, "isLegalGuardian": true, "canConsent": true, "canPickUp": t
  ],
  "referral": { "sourceKey": "PAEDIATRICIAN", "doctorName": "Dr. R. Joshi", "hospital": "Satara Child Care" },
  "medicalAlerts": [ { "key": "HEARING_AID_USER", "label": "Wears bilateral hearing aids", "severity": "INFO", "addedAt": { "$date": "2027-01-10T05:00:00Z" } } ],
  "tags": ["AVT"],
  "status": "ACTIVE",
  "stats": { "firstVisitAt": { "$date": "2027-01-10T05:00:00Z" }, "lastVisitAt": { "$date": "2027-03-02T06:30:00Z" }, "totalVisits": 9, "noShowCount": 0, "lateCancel(
  "consentSummary": { "treatment": { "status": "GRANTED", "at": { "$date": "2027-01-10T05:05:00Z" } }, "communication": { "status": "GRANTED" } },
  "doNotContact": false,
  "searchKeys": ["aarav", "kulkarni", "आरव", "कुलकर्णी", "9876543210", "pgc-27-000042", "sneha"],
  "createdAt": { "$date": "2027-01-10T05:02:11Z" },
  "updatedAt": { "$date": "2027-03-02T06:31:00Z" },
  "__v": 7
}
```


5.12.2 APPOINTMENT (45-MINUTE THERAPY, 15-MINUTE GRANULARITY, 15-MINUTE BUFFER)


```
{
  "bookingRef": "BK-7K3QH2",
  "patientId": { "$oid": "6720a1b2c3d4e5f607182930" },
  "providerId": { "$oid": "6710aa00bb11cc22dd33ee44" },
  "serviceId": { "$oid": "6711bb00cc11dd22ee33ff44" },
  "variantKey": "THERAPY_45",
  "type": "FOLLOW_UP",
  "mode": "CLINIC",
  "start": { "$date": "2027-03-10T05:30:00Z" },
  "end":   { "$date": "2027-03-10T06:15:00Z" },
  "durationMin": 45,
  "bufferMin": 15,
  "slotKeys": ["2027-03-10T05:30:00Z","2027-03-10T05:45:00Z","2027-03-10T06:00:00Z","2027-03-10T06:15:00Z"],
  "holdsSlots": true,
  "status": "CONFIRMED",
  "statusHistory": [
    { "status": "REQUESTED", "at": { "$date": "2027-03-08T10:12:00Z" }, "by": null, "reason": "online request" },
    { "status": "CONFIRMED", "at": { "$date": "2027-03-08T10:40:00Z" }, "by": { "$oid": "6710aa00bb11cc22dd33ee44" } }
  ],
  "source": "WEBSITE",
  "reason": "Follow-up therapy session",
```

```
  "createdAt": { "$date": "2027-03-08T10:12:00Z" }
}
```

(The appointment is at 11:00 IST on 10 March 2027, stored as 05:30 UTC; IST is always UTC+05:30. `slotKeys` cover the 45-minute session plus the 15-minute buffer.)

5.12.3 AUDIOLOGY EVALUATION (ABRIDGED, PTA)

```
{
  "patientId": { "$oid": "6720a1b2c3d4e5f607182930" },
  "visitId": { "$oid": "6720b000000000000000001" },
  "status": "FINAL", "version": 1,
  "pta": {
    "conditions": { "transducer": "INSERT", "booth": true, "responseMode": "PLAY", "reliability": "GOOD" },
    "right": {
      "air":  [ {"hz":250,"db":40}, {"hz":500,"db":45}, {"hz":1000,"db":50}, {"hz":2000,"db":55}, {"hz":4000,"db":60}, {"hz":8000,"db":65} ],
      "bone": [ {"hz":500,"db":10,"masked":true}, {"hz":1000,"db":15,"masked":true}, {"hz":2000,"db":20,"masked":true}, {"hz":4000,"db":25,"masked":true} ]
    },
    "left": {
      "air":  [ {"hz":250,"db":15}, {"hz":500,"db":15}, {"hz":1000,"db":20}, {"hz":2000,"db":20}, {"hz":4000,"db":25}, {"hz":8000,"db":30} ],
      "bone": []
    },
    "computed": {
      "formulaVersion": "pta-v1",
      "right": { "pta3": 50, "pta4": 55, "degree": "MODERATE", "type": { "suggested": "CONDUCTIVE", "confirmed": "CONDUCTIVE" }, "configuration": "SLOPING", "abGap":
      "left":  { "pta3": 18.3, "pta4": 20, "degree": "SLIGHT", "type": { "suggested": "NORMAL", "confirmed": "NORMAL" }, "configuration": "FLAT" }
    }
  },
  "interpretation": { "en": "Right ear: moderate conductive hearing loss. Left ear: hearing within normal limits to slight loss." },
  "createdAt": { "$date": "2027-01-10T06:20:00Z" }
}
```

5.12.4 INVOICE

```
{
  "number": "INV/2026-27/00123", "series": "INV", "financialYear": "2026-27", "status": "PAID",
  "invoiceDate": "2027-01-10", "patientId": { "$oid": "6720a1b2c3d4e5f607182930" },
  "items": [
    { "lineNo": 1, "kind": "SERVICE", "description": "Pure Tone Audiometry", "hsnSac": "999316", "qty": 1, "unitPricePaise": 50000, "taxableValuePaise": 50000, "taxRa
    { "lineNo": 2, "kind": "SERVICE", "description": "Speech Audiometry", "hsnSac": "999316", "qty": 1, "unitPricePaise": 30000, "discount": { "type": "PCT", "value"
  ],
  "totals": { "subtotalPaise": 80000, "discountPaise": 3000, "taxableValuePaise": 77000, "cgstPaise": 0, "sgstPaise": 0, "igstPaise": 0, "roundOffPaise": 0, "grandTot
  "paidPaise": 77000, "balancePaise": 0, "gst": { "registered": false }
}
```

(HSN/SAC codes and tax treatment are illustrative; actual treatment is configured with the Client's CA.)

5.12.5 NOTIFICATION (REMINDER, OUTBOX)

```
{
  "type": "APPOINTMENT_REMINDER_24H",
  "recipient": { "kind": "GUARDIAN", "refId": { "$oid": "6720a1b2c3d4e5f607182930" }, "name": "Sneha Kulkarni", "mobile": "+919876543210", "language": "mr" },
  "channels": ["WHATSAPP", "IN_APP"],
  "payload": { "templateKey": "appt_reminder_24h", "variables": { "patientName": "Aarav", "date": "10 Mar 2027", "time": "11:00 AM", "clinic": "Palak Speech & Hearing
  "related": { "appointmentId": { "$oid": "6730c000000000000000007" } },
  "scheduledFor": { "$date": "2027-03-09T05:30:00Z" },
  "status": "PENDING",
  "dedupeKey": "6730c000000000000000007:APPOINTMENT_REMINDER_24H"
}
```

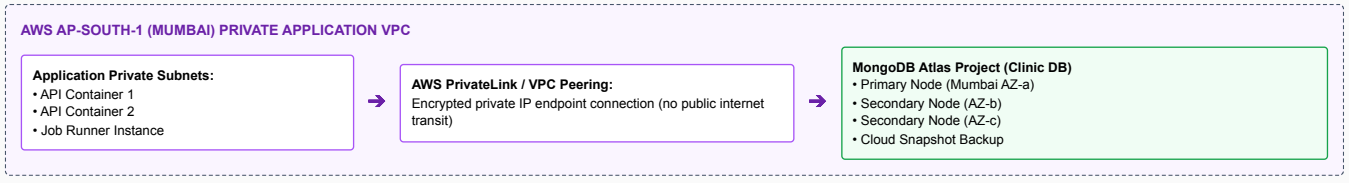
5.12.6 AUDIT LOG ENTRY

```
{
  "ts": { "$date": "2027-03-10T05:31:12Z" }, "seq": 184213,
  "actor": { "userId": { "$oid": "6710aa00bb11cc22dd33ee44" }, "role": "OWNER", "type": "USER", "name": "Palak Gandhi" },
  "action": "clinical.read", "outcome": "SUCCESS",
  "entity": { "type": "audiologyEvaluations", "id": { "$oid": "6720b000000000000000002" }, "patientId": { "$oid": "6720a1b2c3d4e5f607182930" } },
  "context": { "ip": "103.xx.xx.xx", "requestId": "e1c9...", "sessionId": "..." },
  "prevHash": "9f2c...", "hash": "4be1..."
}
```

6. DATABASE CONNECTIVITY AND INFRASTRUCTURE

6.1 TOPOLOGY

FIGURE 6.1: DATABASE CONNECTIVITY & PRIVATE VPC TOPOLOGY



6.2 CLUSTER REQUIREMENTS

ID	REQUIREMENT
NFR-DBC-001	Production shall run on a **MongoDB Atlas dedicated cluster (M10 or higher)**, 3-node replica set across 3 availability zones, region **AWS Mumbai (`ap-south-1`)**, MongoDB 7.0 or later.
NFR-DBC-002	Staging shall run on a separate smaller cluster (M10 or M2/M5 shared where acceptable); development on a local Docker replica set or Atlas free tier. **Production and staging shall never share a cluster, user or credentials.**
NFR-DBC-003	**Cloud Backup** shall be enabled: continuous (point-in-time recovery) with at least **7 days** PITR window; scheduled snapshots: daily (retain 14), weekly (retain 8), monthly (retain 12); a snapshot copy to a second region (Hyderabad `ap-south-2`) shall be enabled, with data residency approved by the Client.
NFR-DBC-004	**Encryption at rest** on; customer-managed key via AWS KMS (BYOK) is recommended.
NFR-DBC-005	**Network isolation**: private connectivity (PrivateLink or VPC peering) preferred; if public endpoints are used, the **IP access list** shall contain only the static egress IPs of the application and the maintainers' VPN. `0.0.0.0/0` is forbidden.
NFR-DBC-006	**TLS** is mandatory; connection strings use `mongodb+srv://` with `tls=true`.
NFR-DBC-007	**Authentication**: SCRAM-SHA-256 database users, or AWS IAM authentication (role-based) as the preferred method; no shared admin user is used by the application.
NFR-DBC-008	**Auto-scaling** of storage enabled; compute auto-scale (M10 to M20) enabled with an upper bound agreed with the Client.
NFR-DBC-009	**Monitoring and alerts** configured in Atlas: replication lag > 10 s, connections > 80%, CPU > 80% for 10 min, disk > 75%, slow queries (> 100 ms) trend, failed logins, backup failure, oplog window < 24 h. Alerts go to the maintenance on-call email and the owner.
NFR-DBC-010	**Maintenance window** set to a low-use time (Sunday 02:00 to 04:00 IST).
NFR-DBC-011	**Database auditing** (Atlas database auditing on tiers that support it) enabled for authentication and administrative operations.

6.3 DATABASE USERS AND PRIVILEGES (LEAST PRIVILEGE)

USER	USED BY	PRIVILEGES
`app_rw`	API processes	`readWrite` on `clinic_prod` **except** `auditLogs` where a custom role grants `insert` and `find` only
`job_rw`	Job runner	Same as `app_rw` (separate credentials for traceability)
`report_ro`	Reporting/exports	`read` on `clinic_prod`, excluding `cbtRecords`, `auditLogs`
`migrator`	CI/CD migration step	`dbAdmin` + `readWrite` on `clinic_prod` (credentials available only to the pipeline during a migration)
`backup_ops`	Restore tests	`read` + restore-capable role, human use only
`owner_break_glass`	Offline, vaulted	Atlas Project Owner access held by the Client with MFA; Black Angler has no standing access

Custom role for audit logs:

```
db.createRole({
  role: "auditAppendOnly",
  privileges: [{ resource: { db: "clinic_prod", collection: "auditLogs" }, actions: ["insert", "find"] }],
  roles: []
})
```

6.4 CONNECTION CONFIGURATION

6.4.1 ENVIRONMENT VARIABLES

VARIABLE	EXAMPLE	NOTES
`NODE_ENV`	`production`	
`MONGODB_URI`	`mongodb+srv://app_rw:<password>@clinic-prod.xxxxx.mongodb.net/clinic_prod?retryWrites=true&w=majority&appName=clinic-api`	From Secrets Manager, never in Git
`MONGODB_DB_NAME`	`clinic_prod`	
`MONGODB_MAX_POOL`	`20`	Per API instance
`MONGODB_MIN_POOL`	`2`	
`MONGODB_SERVER_SELECTION_MS`	`5000`	
`MONGODB_SOCKET_MS`	`45000`	
`MONGODB_AUTO_INDEX`	`false`	`true` only in local/dev
`KMS_KEY_ARN`	`arn:aws:kms:ap-south-1:...`	CBT and secret-field encryption
`JWT_PRIVATE_KEY`, `JWT_PUBLIC_KEY`	PEM	RS256/EdDSA keys
`COOKIE_SECRET`, `PEPPER_SECRET`	random 32+ bytes	
`S3_BUCKET`, `S3_REGION`	`clinic-prod-files`, `ap-south-1`	
`EMAIL_PROVIDER`, `EMAIL_FROM`	`ses`, `no-reply@domain`	
`APP_BASE_URL`, `API_BASE_URL`	URLs	CORS and cookie domain
`LOG_LEVEL`, `SENTRY_DSN`		

Secrets are loaded from **AWS Secrets Manager** (or the host's secret store) at boot; the process fails fast if any required variable is missing (validated by Zod).

6.4.2 CONNECTION MODULE (REFERENCE IMPLEMENTATION)

```
// apps/api/src/config/db.ts
import mongoose from "mongoose";
import { env } from "./env";
import { logger } from "./logger";

mongoose.set("strictQuery", true);
mongoose.set("autoIndex", env.MONGODB_AUTO_INDEX);   // false in production
mongoose.set("sanitizeFilter", true);                // blocks NoSQL operator injection in filters

export async function connectDb(): Promise<typeof mongoose> {
  mongoose.connection.on("connected",    () => logger.info("mongo connected"));
  mongoose.connection.on("disconnected", () => logger.warn("mongo disconnected"));
  mongoose.connection.on("error",       (e) => logger.error({ err: e }, "mongo error"));

  return mongoose.connect(env.MONGODB_URI, {
    dbName: env.MONGODB_DB_NAME,
    maxPoolSize: env.MONGODB_MAX_POOL,          // 20
    minPoolSize: env.MONGODB_MIN_POOL,          // 2
    serverSelectionTimeoutMS: env.MONGODB_SERVER_SELECTION_MS, // 5000
    socketTimeoutMS: env.MONGODB_SOCKET_MS,     // 45000
    connectTimeoutMS: 10000,
    retryWrites: true,
    retryReads: true,
    w: "majority",
    readPreference: "primary",
    compressors: ["zstd", "snappy"],
    appName: "clinic-api",
  });
}

export async function disconnectDb() { await mongoose.disconnect(); }
// apps/api/src/main.ts (excerpt) - startup and graceful shutdown
const app = createApp();
```

```
await connectDb();
const server = app.listen(env.PORT, () => logger.info(`api listening on ${env.PORT}`));

const shutdown = async (signal: string) => {
  logger.info({ signal }, "shutting down");
  server.close(async () => { await disconnectDb(); process.exit(0); });
  setTimeout(() => process.exit(1), 15000).unref();  // hard stop
};
process.on("SIGTERM", () => shutdown("SIGTERM"));
process.on("SIGINT",  () => shutdown("SIGINT"));
```

6.4.3 CONNECTION REQUIREMENTS

ID	REQUIREMENT
NFR-DBC-020	The API shall create **one `MongoClient`/Mongoose connection per process** and reuse it; no per-request connections.
NFR-DBC-021	On startup the API shall verify connectivity with a `ping` and verify that required indexes and validators exist (fail readiness otherwise).
NFR-DBC-022	`/health/live` returns 200 if the process is alive; `/health/ready` returns 200 only if the DB `ping` succeeds within 2 s and migrations are up to date.
NFR-DBC-023	All writes use **write concern `majority`** and journaling; business-critical reads inside transactions use **read concern `snapshot`/`majority`**.
NFR-DBC-024	Dashboard and export queries may use `readPreference: secondaryPreferred` **only** where slightly stale data is acceptable (never for booking, billing, or clinical writes).
NFR-DBC-025	**Retry policy**: transient errors (`TransientTransactionError`, network, `NotWritablePrimary` during failover) are retried with jittered back-off up to 3 attempts for idempotent operations; non-idempotent operations use idempotency keys (Section 7.1).
NFR-DBC-026	Queries shall have **`maxTimeMS`** set (default 5,000; reports 15,000) to protect the database.
NFR-DBC-027	Pagination shall be **keyset-based** for large collections (`_id`/timestamp cursors); `skip` is limited to small offsets (≤ 1,000).
NFR-DBC-028	**Projection discipline**: list endpoints project only required fields; large arrays are never returned in lists.
NFR-DBC-029	**No unbounded queries**: every query must be indexed or explicitly allow-listed; a CI check runs `explain()` on critical queries using a seeded data set and fails on `COLLSCAN` for collections over 1,000 documents.
NFR-DBC-030	**Failover tolerance**: during a primary election (typically < 30 s) the API shall return `503 Retry-After` for writes and recover automatically without restart.

6.5 MONGOOSE MODEL CONVENTIONS

```
// apps/api/src/common/baseSchema.ts
import { Schema, SchemaOptions } from "mongoose";

export const baseOptions: SchemaOptions = {
  timestamps: true,
  optimisticConcurrency: true,       // uses __v for safe concurrent updates
  versionKey: "__v",
  toJSON: { virtuals: true, versionKey: false, transform: (_d, r) => { r.id = r._id; delete r._id; return r; } },
};

export const auditFields = {
  createdBy: { type: Schema.Types.ObjectId, ref: "User" },
  updatedBy: { type: Schema.Types.ObjectId, ref: "User" },
  deletedAt: { type: Date, default: null, index: true },
  deletedBy: { type: Schema.Types.ObjectId, ref: "User" },
  branchId:  { type: Schema.Types.ObjectId, default: null },
};

// soft-delete query helper applied to every schema
export function softDeletePlugin(schema: Schema) {
  schema.pre(/^find/, function () { if (!(this as any).getOptions().withDeleted) (this as any).where({ deletedAt: null }); });
  schema.pre("countDocuments", function () { (this as any).where({ deletedAt: null }); });
}
// apps/api/src/modules/scheduling/appointment.model.ts (abridged)
import { Schema, model } from "mongoose";
import { baseOptions, auditFields, softDeletePlugin } from "../../common/baseSchema";

const ACTIVE = ["REQUESTED", "CONFIRMED", "CHECKED_IN", "IN_PROGRESS"];

const AppointmentSchema = new Schema({
  bookingRef:  { type: String, required: true, unique: true },
  patientId:   { type: Schema.Types.ObjectId, ref: "Patient", index: true },
  prospect:    { type: Object },
  providerId:  { type: Schema.Types.ObjectId, ref: "User", required: true },
  serviceId:   { type: Schema.Types.ObjectId, ref: "Service", required: true },
  variantKey:  { type: String, required: true },
  type:        { type: String, enum: ["NEW","FOLLOW_UP","REVIEW","WALK_IN","SCREENING","CAMP"], required: true },
  mode:        { type: String, enum: ["CLINIC","ONLINE","HOME"], required: true },
  start:       { type: Date, required: true },
  end:         { type: Date, required: true },
  durationMin: { type: Number, required: true, min: 5, max: 480 },
  bufferMin:   { type: Number, default: 0, min: 0, max: 120 },
  slotKeys:    { type: [String], required: true, validate: (v: string[]) => v.length > 0 },
  holdsSlots:  { type: Boolean, required: true },
  status:      { type: String, enum: [...ACTIVE, "COMPLETED","CANCELLED","NO_SHOW","RESCHEDULED","REJECTED","EXPIRED"], required: true },
  statusHistory: [{ status: String, at: Date, by: Schema.Types.ObjectId, reason: String, note: String, _id: false }],
  // ... remaining fields per Section 5.5.12
  ...auditFields,
}, baseOptions);

// DB-enforced no-overlap per provider (ADR-03)
AppointmentSchema.index(
  { providerId: 1, slotKeys: 1 },
  { unique: true, partialFilterExpression: { holdsSlots: true }, name: "uniq_provider_slot_active" }
);
AppointmentSchema.index({ providerId: 1, start: 1 });
AppointmentSchema.index({ patientId: 1, start: -1 });
AppointmentSchema.index({ status: 1, expiresAt: 1 });

AppointmentSchema.pre("validate", function (next) { this.holdsSlots = ACTIVE.includes(this.status); next(); });
AppointmentSchema.plugin(softDeletePlugin);

export const Appointment = model("Appointment", AppointmentSchema);
```

6.6 COUNTERS AND SEQUENCES

```
// apps/api/src/common/counter.ts
export async function nextSeq(id: string, session?: ClientSession): Promise<number> {
  const doc = await Counter.findOneAndUpdate(
    { _id: id },
    { $inc: { seq: 1 }, $set: { updatedAt: new Date() } },
    { upsert: true, new: true, session }
  ).lean();
  return doc!.seq;
```

```
}
// UHID: "PGC-27-000042"  =>  `${prefix}-${yy}-${String(await nextSeq(`uhid:${yy}`)).padStart(6,"0")}`
// Invoice: nextSeq(`invoice:${fy}`, session) inside the issue transaction => gap-free
```

6.7 TRANSACTIONS (CRITICAL FLOWS)

FLOW	OPERATIONS INSIDE ONE TRANSACTION
Create/confirm appointment	insert `appointments` (with `slotKeys`), insert `notifications` (confirmation + reminders), update `patients.stats`, insert `auditLogs` (audit insert is also done in-transaction for strong consistency)
Reschedule	update old appointment (`RESCHEDULED`, `holdsSlots=false`), insert new appointment, cancel pending notifications, insert new ones
Issue invoice	increment invoice counter, set number/date/status, decrement stock (if products), update `patients.stats`, insert audit
Record payment/refund	insert `payments`, update invoice `paidPaise`/`status`, create receipt number, update day-close aggregates
Stock movement	insert `stockMovements`, update `stockItems`, update `products.stockQty`
Merge patients	repoint references across collections, mark merged patient, write audit (batch-limited)
Finalise visit documents	update document statuses, update visit, create recalls, update `outcomeMeasures`

```
// Booking with transaction + duplicate-key mapping
export async function bookAppointment(input: BookInput, actor: Actor) {
  const session = await mongoose.startSession();
  try {
    let result;
    await session.withTransaction(async () => {
      const slotKeys = buildSlotKeys(input.start, input.durationMin + input.bufferMin, GRANULARITY_MIN);
      await assertWithinSchedule(input, session);          // schedule, holidays, blocks, policy
      const bookingRef = await uniqueBookingRef(session);
      const [appt] = await Appointment.create([{
        ...input, bookingRef, slotKeys, status: input.autoConfirm ? "CONFIRMED" : "REQUESTED",
        statusHistory: [{ status: input.autoConfirm ? "CONFIRMED" : "REQUESTED", at: new Date(), by: actor.id }],
        expiresAt: input.autoConfirm ? undefined : addHours(new Date(), HOLD_HOURS),
      }], { session });
      await enqueueConfirmationAndReminders(appt, session);
      await writeAudit("appointment.create", appt, actor, session);
      result = appt;
    }, { readConcern: { level: "snapshot" }, writeConcern: { w: "majority" } });
    return result;
  } catch (e: any) {
    if (e?.code === 11000 && /uniq_provider_slot_active/.test(e.message)) throw new ApiError(409, "SLOT_TAKEN", "This time was just booked. Please choose another slot
    throw e;
  } finally { await session.endSession(); }
}
```

6.8 SERVER-SIDE VALIDATION EXAMPLE (APPOINTMENTS)

```
db.runCommand({
  collMod: "appointments",
  validator: { $jsonSchema: {
    bsonType: "object",
    required: ["bookingRef","providerId","serviceId","variantKey","type","mode","start","end","duration
<truncated 86815 bytes>

NOTE: The output was truncated because it was too long. Use a more targeted query or a smaller range to get the information you need.

# 7. API Specification

## 7.1 Overview & Architectural Principles
The ERP and Website communicate with the backend via a RESTful API operating over HTTPS/TLS 1.3.
- **Base URL**: `https://api.vanisclinic.in/v1`
- **Content Type**: `application/json; charset=utf-8`
- **Authentication**: JWT Bearer token passed in the `Authorization` header (`Authorization: Bearer <token>`).
- **Response Wrapper Standard**:
```json
{
  "success": true,
  "data": { ... },
  "error": null,
  "meta": { "page": 1, "limit": 25, "total": 142, "timestamp": "2026-10-07T14:30:00.000Z" }
}
```
```

7.2 KEY ENDPOINT REGISTRY

MODULE	ENDPOINT	METHOD	ROLE REQUIRED	DESCRIPTION
AUTH	`/v1/auth/login`	POST	Public	Authenticates user, returns JWT and sets httpOnly refresh cookie
AUTH	`/v1/auth/refresh`	POST	Public	Rotates refresh token and issues new short-lived access JWT
AUTH	`/v1/auth/logout`	POST	Authenticated	Revokes current session refresh token
APT	`/v1/appointments/slots`	GET	Public / User	Calculates available slots for provider, service and date range
APT	`/v1/appointments`	POST	Public / User	Creates requested or confirmed appointment
APT	`/v1/appointments/:id/status`	PATCH	Practitioner / Reception	Updates appointment lifecycle state
PAT	`/v1/patients`	GET	ERP Roles	Searches patient master list with filters (UHID, mobile, name)
PAT	`/v1/patients`	POST	Reception / Practitioner	Registers new patient master record
PAT	`/v1/patients/:id`	GET	ERP Roles	Fetches complete 360-degree patient file
CLN	`/v1/clinical/visits`	POST	Practitioner	Finalises session visit and attaches SOAP note
CLN	`/v1/clinical/audiology`	POST	Practitioner (BASLP)	Saves OAE screening & Pure Tone Audiometry evaluation
CLN	`/v1/clinical/speech-assessment`	POST	Practitioner (BASLP)	Saves standardized test batteries (ALD-MM, LPT, CAPC-M, DALI)
CLN	`/v1/clinical/session-report`	POST	Practitioner (BASLP)	Saves therapy session report with target goals, activities & consistency %
CLN	`/v1/clinical/cbt`	POST	Practitioner (CBT)	Saves encrypted CBT session note
DOC-OAE	`/v1/reports/clinical/oae/:evalId`	GET / POST	Practitioner (BASLP)	Generates official OAE Screening PDF (`RP-CLN-OAE`) with ear visuals
DOC-AUD	`/v1/reports/clinical/audiology/:evalId`	GET / POST	Practitioner (BASLP)	Generates Audiological Evaluation PDF (`RP-CLN-AUD`) with dual audiograms
DOC-VOI	`/v1/reports/clinical/voice/:evalId`	GET / POST	Practitioner (BASLP)	Generates Voice Disorders Proforma & Report (`RP-CLN-VOI`)

MODULE	ENDPOINT	METHOD	ROLE REQUIRED	DESCRIPTION
DOC-FLU	`/v1/reports/clinical/fluency/:evalId`	GET / POST	Practitioner (BASLP)	Generates Fluency / Stuttering Evaluation PDF (`RP-CLN-FLU`)
DOC-SLA	`/v1/reports/clinical/speech-language/:evalId`	GET / POST	Practitioner (BASLP)	Generates Comprehensive Speech & Language Report (`RP-CLN-SLA`)
DOC-CBT	`/v1/reports/clinical/cbt/:evalId`	GET / POST	Practitioner (CBT)	Generates Encrypted CBT & Risk Assessment PDF (`RP-CLN-CBT`)
DOC-DYS	`/v1/reports/clinical/dysphagia/:evalId`	GET / POST	Practitioner (BASLP)	Generates Dysphagia & Swallowing Evaluation PDF (`RP-CLN-DYS`)
DOC-HAF	`/v1/reports/clinical/hearing-aid-fitting/:haId`	GET / POST	Practitioner (BASLP)	Generates Hearing Aid Fitting & Warranty Certificate (`RP-CLN-HAF`)
DOC-PRG	`/v1/reports/clinical/therapy-progress/:planId`	GET / POST	Practitioner (BASLP)	Generates Periodic Therapy Progress Summary Report (`RP-CLN-PRG`)
DOC-DSC	`/v1/reports/clinical/discharge-summary/:planId`	GET / POST	Practitioner (BASLP)	Generates Clinical Discharge Summary & Home Plan (`RP-CLN-DSC`)
BIL	`/v1/invoices`	POST	Reception / Practitioner	Generates final tax/clinical invoice
BIL	`/v1/payments`	POST	Reception	Records offline/UPI payment transaction
INV	`/v1/inventory/items`	GET	ERP Roles	Queries stock items and serial numbers
DSH	`/v1/dashboards/summary`	GET	ERP Roles	Returns daily revenue, appointments, and recall statistics

8. EXTERNAL INTERFACE REQUIREMENTS

8.1 USER INTERFACES

Public Website UI: Responsive layout supporting Desktop (1920x1080 to 1280x720), Tablet (768px), and Mobile (375px+). English and Marathi localization.
ERP Clinical Dashboard UI: Single-Page Application (SPA) / Progressive Web App (PWA) optimized for Google Chrome, Apple Safari, and MS Edge. Clean, high-contrast typography, keyboard shortcuts for clinical entry (`Alt+N` for new visit, `Alt+P` for patient lookup).

8.2 HARDWARE INTERFACES

Receipt & Invoice Printers: Standard POS thermal printers (80mm) and A4 laser/inkjet printers via standard browser print dialog.
Barcode & Serial Scanners: USB/Bluetooth barcode scanners operating in keyboard-wedge mode for reading serial numbers on hearing aid packages and consumable stock.

8.3 SOFTWARE & CLOUD INTERFACES

MongoDB Atlas: Connection over TLS 1.3 using Mongoose connection pooling (Mumbai `ap-south-1` region).
Email Gateway (SMTP/API): Transactional email delivery (SendGrid / AWS SES) for appointment confirmations, invoices, and password resets.
WhatsApp Notification Outbox: Provider-neutral outbox queuing system ready to interface with WhatsApp Business API.

9. NON-FUNCTIONAL REQUIREMENTS (NFR)

9.1 PERFORMANCE & RESPONSE TIMES

NFR-PRF-001: Public website Page Load Time (LCP) shall not exceed **1.8 seconds** on standard 4G mobile connections.
NFR-PRF-002: ERP API P95 latency for search and retrieve endpoints shall be **< 200 ms**.
NFR-PRF-003: Clinical record write operations shall complete within **< 350 ms**.
NFR-PRF-004: Automated Clinical PDF report rendering shall complete in **< 1.5 seconds** per report.

9.2 RELIABILITY & AVAILABILITY

NFR-REL-001: System operational availability shall be **99.9%** (excluding scheduled maintenance).
NFR-REL-002: Database Recovery Point Objective (RPO) shall be **< 5 minutes** via continuous point-in-time recovery (PITR).
NFR-REL-003: Database Recovery Time Objective (RTO) shall be **< 1 hour**.

9.3 SECURITY & COMPLIANCE

NFR-SEC-001: All data in transit shall be encrypted using **TLS 1.3 / HTTPS**.
NFR-SEC-002: Sensitive patient clinical data (specifically CBT session notes and risk safety matrices) shall be encrypted at rest using **AES-256 field-level encryption**.
NFR-SEC-003: System shall strictly comply with the **Digital Personal Data Protection (DPDP) Act, 2023 (India)**, maintaining verifiable audit logs for all patient data access.

10. SECURITY ARCHITECTURE

FIGURE 10.1: SECURITY & PRIVACY ARCHITECTURE LAYERS

Layer 1: Network & Edge Security — HTTPS / TLS 1.3 Encryption, Cloudflare WAF, Rate Limiting, Strict CORS Origin Validation

Layer 2: Identity & Session Management — JWT Access Tokens (15 min), httpOnly SameSite Refresh Cookies (7 days), Argon2id Hashing

Layer 3: Authorization (RBAC) — Enforced Role Hierarchy (Owner, Clinician, Reception, Accountant) with granular action permissions

Layer 4: Data Protection & Field Encryption — AES-256 Field-Level Encryption for restricted CBT session notes

Layer 5: DPDP Act 2023 Compliance & Audit — Immutable append-only audit logs recording every clinical file access

11. USER INTERFACE REQUIREMENTS & DESIGN SYSTEM

Color System: Primary Deep Slate (`#1E293B`), Medical Teal (`#0D9488`), Neutral Gray (`#F8FAFC`), Accent Amber (`#D97706`).
Typography: Primary Body: `Calibri` / `Segoe UI` / `Arial`; Headings: `Georgia` serif font for legal/formal views and `Segoe UI` for ERP web views.
Accessibility: Compliance with **WCAG 2.1 Level AA** standards for color contrast, screen reader compatibility, and focus management.

12. BUSINESS RULES CATALOGUE

RULE ID	NAME	DESCRIPTION
BR-APT-001	Double-Booking Prevention	The database shall prevent two active overlapping appointments for the same provider via unique partial indexes (`uniq_provider_slot_active`).
BR-PAT-002	UHID Generation Standard	Every patient shall be assigned a unique immutable identifier in format `PGC-YY-XXXXXX`

RULE ID	NAME	DESCRIPTION
		(e.g. `PGC-26-000104`).
BR-BIL-003	Invoice Immutability	Once an invoice status transitions to `ISSUED` or `PAID`, its line items and prices cannot be modified; amendments require a Credit Note.
BR-INV-004	Serialized Stock Tracking	Hearing aids and major equipment must have unique serial numbers recorded upon purchase and verified upon billing.
BR-DOC-005	Diagnostic Report Finalization	Once an assessment report transitions to `FINAL`, clinical parameters become immutable; corrections create an explicit `AMENDED` version with reason.
BR-DOC-006	Practitioner Stamped Signature	All generated diagnostic reports must embed the RCI Registration Number and official stamped digital signature of the authoring practitioner.

13. VALIDATION RULES CATALOGUE

RULE ID	TARGET FIELD	VALIDATION CRITERIA	ERROR MESSAGE / ACTION
VR-PAT-001	Mobile Number	10-digit numeric string matching Indian mobile numbers (`^[6-9]\d{9}$`)	"Please enter a valid 10-digit Indian mobile number."
VR-APT-002	Duration	Integer between 5 and 480 minutes	"Appointment duration must be between 5 and 480 minutes."
VR-BIL-003	Line Item Amount	Non-negative numerical value (amount >= 0)	"Item total cannot be negative."
VR-CLN-004	Audiometry dB Range	Multiples of 5 within range -10 dB HL to 120 dB HL	"Threshold must be in 5 dB steps between -10 and 120 dB HL."
VR-CLN-005	OAE SNR Threshold	Numeric value between -20.0 dB and +35.0 dB	"SNR must be a valid numeric measurement in dB."
VR-CLN-006	SSI-4 %SS Field	Percentage between 0.0% and 100.0%	"Percentage of syllables stuttered must be between 0 and 100%."

14. REPORTS CATALOGUE

14.1 MASTER CLINICAL ASSESSMENT & DIAGNOSTIC REPORTS MATRIX

The software features a built-in automated clinical reporting engine that generates **10 specialized clinical assessment and diagnostic reports**, each rendered as a standardized, print-ready A4 PDF document with clinic branding, bilingual metadata, auto-computed formulas, and official RCI clinician sign-off:

REPORT ID	REPORT NAME	CLINICAL ASSESSMENT BATTERY	TARGET AUDIENCE	KEY PARAMETERS & AUTO-CALCULATIONS	VISUAL LAYOUT STANDARD	FREQUENCY
RP-CLN-OAE	**Otoacoustic Emission (OAE) Screening Report**	TEOAE & DPOAE Screening	Pediatricians, ENT Doctors, Parents	Frequencies 1k to 4k Hz, SNR (dB), Reproducibility %, Pass/Refer logic (SNR ≥ 6 dB across ≥ 3 bands)	Single-page A4, prominent Red Right & Blue Left ear graphics with binary checkboxes, impression & recommendations	Per screening encounter
RP-CLN-AUD	**Audiological Evaluation Report**	Pure Tone & Speech Audiometry, Tympanometry, Tuning Fork	ENT Specialists, Neurologists, Audiologists	AC/BC thresholds 125-8000 Hz, PTA1, PTA2, SDT, SRT, WRS %, MCL, UCL, Rinne, Weber, Tympanogram Type, Compliance, ECV	Single-page A4, dual side-by-side audiogram graphs (Red Right circles / Blue Left crosses), full-width parameters & immittance tables	Per evaluation
RP-CLN-VOI	**Voice Disorders Assessment Proforma & Report**	Perceptual Voice (CAPE-V, GRBAS), Acoustic & Aerodynamics, VHI	ENT / Laryngologists, Voice Therapists, Patients	CAPE-V VAS (0-100 mm), GRBAS (0-3), MPT (sec for /a/, /i/, /u/), s/z ratio (cutoff > 1.4), HNR (dB), VHI-30/10 severity	Multi-page structured proforma, visual severity meters, acoustic benchmark comparison grid	Baseline & re-evaluation
RP-CLN-FLU	**Fluency Disorders Assessment Report**	SSI-4 (Stuttering Severity Instrument 4th Ed), Speech Rate, OASES	Speech-Language Pathologists, Schools, Parents	%SS (reading & conversation), 3 longest block durations (sec), physical concomitants (0-20), total SSI-4, percentile, SPM/WPM	Standardized proforma, disfluency pie chart, severity band thermometer, avoidance checklist	Intake & periodic review
RP-CLN-SLA	**Comprehensive Speech & Language Assessment**	ALD-MM, LPT, CAPC-M, DALI, PCC, AAC, AVT	Developmental Paediatricians, Schools,	Receptive/Expressive Language Age (RLA/ELA),	Comprehensive multi-page report, developmental age	Comprehensive evaluation

REPORT ID	REPORT NAME	CLINICAL ASSESSMENT BATTERY	TARGET AUDIENCE	KEY PARAMETERS & AUTO-CALCULATIONS	VISUAL LAYOUT STANDARD	FREQUENCY
	Report		Parents	Developmental Quotient (DQ %), PCC %, Pragmatics rating, Erber hierarchy	gap charts, phonological error breakdown, SMART goal tables	
RP-CLN-CBT	**Psychological & CBT Clinical Assessment Report**	PHQ-9, GAD-7, PSS-10, Suicide Risk Matrix, Beck's ABC Model	Psychiatrists, Clinical Psychologists, Therapist	PHQ-9 (0-27; Item #9 suicidal ideation alert), GAD-7 (0-21), PSS-10 (0-40), Risk stratification (Low/Mod/High), 5-Step Safety Plan	Secure encrypted report, psychometric trend gauges, ABC cognitive formulation matrix, crisis plan banner	Intake & review sessions
RP-CLN-DYS	**Swallowing & Dysphagia Evaluation Report**	Bedside Clinical Swallow Exam, FOIS, IDDSI Framework	Neurologists, Gastroenterologists, Caregivers	FOIS Level (1-7), IDDSI Liquid (0-4) & Food (3-7) levels, water swallow test trials (3ml/90ml), cough/wet-voice signs	Bedside proforma, IDDSI color-coded diet pyramid, compensatory posture diagrams, aspiration precautions	Post-stroke / dysphagia visit
RP-CLN-HAF	**Hearing Aid Fitting & Verification Certificate**	Amplification Dispensing, REM Target Matching, Sound Field	Patients, Hearing Aid Manufacturers, ENT	Device Make/Model/Serial, REM/Coupler target curve fit, Aided vs Unaided sound field thresholds, functional gain	Formal stamped certificate, serial barcode/QR, warranty dates, trial agreement terms	Dispensing / fitting visit
RP-CLN-PRG	**Periodic Therapy Progress Summary Report**	Longitudinal SOAP Synthesis, Goal Mastery Tracking	Parents, Sponsoring Bodies, School Counsellors	Baseline vs Current Goal Accuracy %, Ling 6-sound detection distance, session attendance rate %, parent coaching score	Multi-page progress report, longitudinal goal mastery progress bars, attendance summary	Every 12 sessions / quarterly
RP-CLN-DSC	**Clinical Discharge Summary & Home Plan**	Episode-of-Care Discharge Protocol	Referring Doctors, Parents, Patient File	Treatment duration, total sessions attended, mastered SMART goals summary, discharge criteria validation	Formal clinical discharge certificate, home maintenance regimen, annual review recall schedule	Discharge from therapy

14.2 DETAILED SPECIFICATIONS FOR CORE DIAGNOSTIC REPORTS

14.2.1 RP-CLN-OAE: OTOACOUSTIC EMISSION (OAE) SCREENING REPORT

Clinical Scope: TEOAE (Transient Evoked) & DPOAE (Distortion Product) infant and pediatric hearing screening.
Reference Standard: Joint Committee on Infant Hearing (JCIH) & Rehabilitation Council of India (RCI) newborn screening protocol.
Header Elements: Clinic identity (*Vani's Speech & Hearing Clinic, Satara*), Consultant Audiologist credentials (*Ms. Palak Paresh Gandhi, BASLP, RCI Reg. No.*), contact numbers, and official logo.
Patient Demographics: Patient Name, Date of Examination, Age / Gender, OAE No., Date of Birth (DOB), UHID / Reg. No., Referring Pediatrician / Hospital.
Visual Ear Containers:
Right Ear Section: Prominent Red ear outline graphic with binary checkboxes: `Pass [ ]` / `Refer [ ]`.
Left Ear Section: Prominent Blue ear outline graphic with binary checkboxes: `Pass [ ]` / `Refer [ ]`.
Measurement Data Table: Frequencies tested (1000 Hz, 1500 Hz, 2000 Hz, 3000 Hz, 4000 Hz), Signal-to-Noise Ratio (`SNR in dB`), Noise Floor (dB SPL), and Reproducibility (%).
Automated Pass Criteria:
Frequency band PASS: `SNR ≥ 6 dB` AND `Reproducibility > 70%`.
Overall Ear PASS: Minimum 3 out of 5 frequencies pass.
Clinical Impression & Recommendations:
`IMPRESSION`: Formatted clinical conclusion (e.g. *"Bilateral Pass - Normal Outer Hair Cell Function"* or *"Right Ear Refer, Left Ear Pass - Outer Hair Cell Dysfunction Noted"*).
`RECOMMENDATION`: Clear next steps (e.g. *"Diagnostic BERA / AABR within 4 weeks"*, *"ENT review for middle ear effusion"*, *"Behavioral hearing review at 6 months"*).
Authentication: Official RCI-registered Audiologist signature block and clinic seal.

14.2.2 RP-CLN-AUD: COMPREHENSIVE AUDIOLOGICAL EVALUATION REPORT

Clinical Scope: Diagnostic Pure Tone Audiometry (Air & Bone Conduction), Masking, Speech Audiometry, Tympanometry, and Tuning Fork tests.
Patient Metadata: Date, Registration Number / UHID, PTA Number, Patient Name, Age / Gender, Audiometer Make/Model (e.g. *Proton SX5 / Interacoustics*), Reliability / Response Consistency (`Consistent` / `Fairly Consistent` / `Inconsistent`).
Dual Visual Audiogram Graphs (Side-by-Side):
Right Ear Audiogram (Red): Frequency axis (125, 250, 500, 750, 1000, 1500, 2000, 3000, 4000, 6000, 8000 Hz) vs Sound Pressure Level (-10 dB HL to 120 dB HL). Standard Red Circles `○` for unmasked AC, triangles `△` for masked AC, `<` for unmasked BC, `[` for masked BC, down arrows for No Response.
Left Ear Audiogram (Blue): Matching frequency/intensity grid with Blue Crosses `X` for unmasked AC, squares □ for masked AC, `>` for unmasked BC, `]` for masked BC, down arrows for No Response.
Standard Symbol Legend: Complete visual key illustrating Right vs Left ear symbols.
Full-Width Audiometric Parameters Table:
`Ear | PTA1 (500, 1k, 2k Hz) | PTA2 (500, 1k, 2k, 4k Hz) | SDT (dB HL) | SRT (dB HL) | WRS % at dB HL | MCL (dB HL) | UCL (dB HL) | Rinne (512 Hz) | Weber (512 Hz)`
Direct automated formula computation: $$ ext{PTA1} = rac{ ext{Threshold}*{500} + ext{Threshold}*{1000} + ext{Threshold}*{2000}}{3}$$ $$ ext{PTA2} = rac{ ext{Threshold}*{500} + ext{Threshold}*{1000} + ext{Threshold}*{2000} + ext{Threshold}_{4000}}{4}$$
Tympanograms & Middle Ear Immittance Grid:
Parameters: Tympanogram Type (Type A, As, Ad, B, C), Peak Pressure (daPa), Static Compliance (ml), Ear Canal Volume (ml), and Clinical Normative ranges.
Clinical Interpretation & Recommendations: Structured diagnostic finding (Degree, Type, Configuration of hearing loss) and actionable management recommendations.
Footer: Official RCI Audiologist stamp, registration number, and clinic address.

14.2.3 RP-CLN-VOI: CLINICAL PROFORMA & EVALUATION REPORT FOR VOICE DISORDERS

Clinical Scope: Multidimensional voice evaluation for dysphonia, vocal nodules, muscle tension dysphonia, and professional voice users.
Intake & Demographics: Vocal loading history (hours/day), vocal abuse/misuse habits (throat clearing, shouting, hard attacks), hydration, reflux symptoms, ENT laryngoscopy findings.
Perceptual Voice Evaluation:
CAPE-V: Visual-analogue ratings (0-100 mm scale) across Overall Severity, Roughness, Breathiness, Strain, Pitch deviation, Loudness deviation; flagged as Consistent (`C`) or Intermittent (`I`).
GRBAS Scale: Grade (`G`), Roughness (`R`), Breathiness (`B`), Asthenia (`A`), Strain (`S`), each rated from 0 (Normal) to 3 (Severe). Auto-computed `GRBAS Total`.
Acoustic & Aerodynamic Measures:
Maximum Phonation Time (**MPT** in seconds): Sustained /a/, /i/, /u/ across 3 trials; best trial recorded and compared against normative benchmark (15-20 sec).
s/z Ratio: Ratio of sustained /s/ to /z/; ratio > 1.4 automatically flagged as indicative of laryngeal pathology / glottic insufficiency.
Fundamental Frequency (F_0 in Hz), Jitter (%), Shimmer (%), and Harmonics-to-Noise Ratio (**HNR** in dB).
Patient-Reported Outcome: **VHI-30 / VHI-10** (Voice Handicap Index) with Functional, Physical, Emotional subscale scores, total score (0-120), and severity classification (Mild, Moderate, Severe).
Therapy Plan: Prescribed vocal hygiene protocols, resonant voice therapy (RVT), vocal function exercises (VFE), and follow-up timeline.

14.2.4 RP-CLN-FLU: FLUENCY DISORDERS ASSESSMENT REPORT & STUTTERING PROFORMA

Clinical Scope: Diagnostic evaluation of stuttering and cluttering in children and adults.
Standardized Diagnostic Battery — SSI-4:
%SS Calculation: Real-time computation of percentage of syllables stuttered across conversational speech and oral reading samples: $$\% ext{SS} = \left(rac{ ext{Stuttered Syllables}}{ ext{Total Syllables}} ight) imes 100$$
Disfluency Typology Breakdown: Part-word repetitions, whole-word repetitions, prolongations, silent postural fixations (blocks), interjections, revisions.
Duration Measure: Average duration of the 3 longest stuttering blocks in seconds, converted to scaled SSI-4 duration points.
Physical Concomitants: Scaled evaluation (0-20 points) across secondary behaviors (facial grimaces, head turns, limb tension, distracting noises).
Total SSI-4 Score & Severity: Summed total score, mapped percentile rank, and severity classification (Very Mild: 10-17, Mild: 18-24, Moderate: 25-31, Severe: 32-36, Very Severe: 37+).
Speech Rate Metrics: Syllables Per Minute (SPM) and Words Per Minute (WPM) compared against developmental norms.
Covert & Emotional Factors: Avoidance behaviors, situational anxiety rating, and OASES (Overall Assessment of Speaker's Experience of Stuttering) score.
Recommendations: Fluency shaping vs stuttering modification goals, parent guidance, and session frequency.

14.2.5 RP-CLN-SLA: COMPREHENSIVE SPEECH & LANGUAGE ASSESSMENT REPORT

Clinical Scope: Evaluation of delayed speech and language, developmental language disorder (DLD), speech sound disorders (SSD), and neurodevelopmental communication profiles.
Oral Peripheral Mechanism Examination (OPME): Structural and functional examination of lips, tongue, jaw, hard/soft palate, dentition, and Diadochokinetic (DDK) rates (/pa/, /ta/, /ka/, /pataka/).
Standardized Assessment Batteries:
ALD-MM (Assessment of Language Development - Mukesh Mohan): Receptive Language Age (**RLA**), Expressive Language Age (**ELA**), and Developmental Quotient: $$ ext{DQ} = \left(rac{ ext{Language Age}}{ ext{Chronological Age}} ight) imes 100$$
LPT (Linguistic Profile Test): Mastery percentages across phonology, semantics, syntax, morphology, and pragmatics.
CAPC-M (Clinical Assessment of Pragmatics & Communication): Eye contact, joint attention, communicative intent (requesting, commenting, protesting), turn-taking, topic maintenance.
DALI (Dyslexia Assessment for Indian Children): Phonological awareness, rapid automatized naming (RAN), decoding, reading comprehension.

PCC (Percentage of Consonants Correct): Phonology error analysis (Substitutions, Omissions, Distortions, Additions); computed as: $$ ext{PCC} = \left(rac{ ext{Number of Correct Consonants}}{ ext{Total Consonants}} ight) imes 100$$ Severity rating: Mild (>85%), Mild-Moderate (65-85%), Moderate-Severe (50-65%), Severe (<50%).
AAC Evaluation: Candidacy, motor access, symbol recognition, and communication board / PECS phase.
AVT Tracking: Erber's listening hierarchy (Detection, Discrimination, Identification, Comprehension), Ling 6-Sound test results.
Roadmap: Prioritized SMART therapy goals, prognostic statement, and home stimulation plan.

14.2.6 RP-CLN-CBT: PSYCHOLOGICAL & COGNITIVE BEHAVIOURAL THERAPY (CBT) REPORT

Security & Privacy Architecture: Encrypted at rest using **AES-256 field-level encryption**; accessible only when `cbtQualificationVerified = true` with `cbt.read` permission; excluded from general search and public exports.
Standardized Psychometric Scales (Item-Level & Auto-Summed):
PHQ-9 (Patient Health Questionnaire - 9): Total Score (0-27) with severity bands (Minimal: 0-4, Mild: 5-9, Moderate: 10-14, Moderately Severe: 15-19, Severe: 20-27). **Clinical Red Flag**: Any positive response on Item #9 ("Thoughts that you would be better off dead") triggers an automated high-priority clinical risk alert banner.
GAD-7 (Generalized Anxiety Disorder - 7): Total Score (0-21) with severity bands (Minimal: 0-4, Mild: 5-9, Moderate: 10-14, Severe: 15-21).
PSS-10 (Perceived Stress Scale): Total Score (0-40) with stress appraisal categories (Low: 0-13, Moderate: 14-26, High: 27-40).
Clinical Risk Assessment & Safety Plan Matrix:
Evaluation of suicidal ideation, intent, plan, accessibility of lethal means, historical attempts, and protective factors.
Clinical Risk Level assignment: `LOW`, `MODERATE`, `HIGH`.
5-Step Stanley-Brown Safety Plan: Warning signs, internal coping strategies, social contacts for distraction, trusted family contacts, crisis hotlines & psychiatric referral contacts.
Cognitive Case Formulation (Beck's Model):
Core beliefs, intermediate rules/assumptions, automatic thoughts.
The ABC Model: `A` (Activating Event), `B` (Cognitive Beliefs / Automatic Thoughts), `C` (Emotional & Behavioral Consequences).
Cognitive distortion tagging (Catastrophizing, All-or-Nothing, Mind Reading, Emotional Reasoning, Overgeneralization).
CBT Treatment Plan: Behavioral activation, cognitive restructuring targets, exposure hierarchies, psychoeducation.

14.2.7 RP-CLN-DYS: SWALLOWING & DYSPHAGIA CLINICAL EVALUATION REPORT

Clinical Bedside Swallowing Exam (CSE): Cranial nerve examination, oral hygiene, lip seal, mastication, bolus propulsion, residue.
Water Swallow Trials: 3 ml, 5 ml, and 90 ml continuous swallow testing with pre/post SpO2 monitoring, cough reflex, throat clearing, and post-swallow wet vocal quality.
Standardized Staging & Diet Mapping:
FOIS (Functional Oral Intake Scale): Levels 1 to 7 (Level 1: No oral intake / tube fed; Level 7: Total oral intake with no restrictions).
IDDSI Framework: Color-coded diet and fluid levels: Drinks (Level 0 Thin to Level 4 Extremely Thick) and Foods (Level 3 Liquidised to Level 7 Regular).
Compensatory Maneuvers & Guidelines: Postural adjustments (chin tuck, head turn), swallowing maneuvers (Mendelsohn, effortful swallow, supraglottic swallow), pacing, aspiration precautions.

14.2.8 RP-CLN-HAF: HEARING AID FITTING, TRIAL & VERIFICATION CERTIFICATE

Device Details: Ear (Right, Left, Bilateral), Manufacturer, Model, Style (RIC, BTE, CIC, ITE), Serial Numbers, Earmould / Dome type, Acoustic Matrix.
Fitting Verification & REM: Real-Ear Measurement (REM) / Coupler Gain target matching (NAL-NL2 / DSL v5), Aided Sound Field thresholds vs Unaided thresholds, Functional Gain.
Dispensing & Warranty Terms: Fitting date, trial period outcome, warranty duration (years), loss-and-damage terms, battery care instructions, cleaning guidance, and service recall schedule.

14.2.9 RP-CLN-PRG: PERIODIC CLINICAL THERAPY PROGRESS SUMMARY REPORT

Therapy Review Synthesis: Synthesized from longitudinal SOAP session notes and goal tracking data over a specified review block (e.g. 12 sessions / 3 months).
Core Progression Metrics: Baseline vs Current Goal Accuracy (%), Ling 6-sound detection distance, MLU increase, %SS reduction.
Compliance Metrics: Attendance rate (% sessions attended vs scheduled), home practice adherence score, parent coaching milestones, and updated SMART objectives for the next treatment phase.

14.2.10 RP-CLN-DSC: CLINICAL DISCHARGE SUMMARY & HOME MAINTENANCE PLAN

Discharge Evaluation: Initial baseline vs discharge performance benchmarks, total therapy sessions attended, criteria met for formal discharge.
Home Maintenance & Annual Recall: Structured home maintenance plan, speech/auditory stimulation exercises, scheduled annual audiological / speech re-evaluation recall date.

14.3 OPERATIONAL & ADMINISTRATIVE REPORTS CATALOGUE

REPORT ID	NAME	FREQUENCY	TARGET AUDIENCE	KEY METRICS INCLUDED
RP-DSH-001	Daily Practice Summary	Daily	Practitioner / Admin	Appointments total, completed vs no-show, daily cash collected, pending recalls
RP-BIL-002	Monthly Revenue & GST Register	Monthly	Accountant / Admin	Gross revenue, tax collected breakdown, payment mode breakdown (Cash, UPI, Card)
RP-CLN-003	Speech & Audiology Outcome Report	Quarterly	Practitioner	Pre/post treatment score trends, hearing aid fitting success rates

15. TESTING AND ACCEPTANCE

Unit & Integration Testing: Automated test coverage exceeding **80%** using Jest / Vitest for backend services and React Testing Library for UI components.
Clinical Report Verification Testing: Automated regression testing verifying that all 10 clinical assessment PDF reports compile with valid geometry, zero text-wrapping, crisp SVG graphics, and exact mathematical accuracy on all formulas (PTA1, PTA2, WRS, SNR, %SS, SSI-4, CAPE-V, GRBAS, MPT, s/z, ALD-MM DQ, PHQ-9, GAD-7, FOIS, IDDSI).
User Acceptance Testing (UAT): Execution of formal UAT scripts by Ms. Palak Paresh Gandhi covering booking, patient registration, clinical recording, billing, and report generation.

16. DEVOPS, DEPLOYMENT AND OPERATIONS

CI/CD Pipeline: GitHub Actions automated pipeline performing linting, type-checking, automated unit testing, container build, and deployment.
Hosting Architecture: Frontend hosted on Vercel CDN; Node.js API hosted on containerized App Service (Render / AWS ECS); Database on MongoDB Atlas Replica Set (Mumbai).

17. DATA MIGRATION AND SEEDING

CSV Seeding Pipeline: Automated migration script for importing legacy patient Excel files into `patients` and `caseHistories` collections.
System Seeding: Pre-populating default lookup lists (audiometry frequencies, speech therapy goal banks, ICD-10 speech/hearing diagnosis codes, default clinic roles).

18. ASSUMPTIONS, DEPENDENCIES AND RISK REGISTER

RISK ID	RISK DESCRIPTION	IMPACT	PROBABILITY	MITIGATION STRATEGY
RSK-001	Internet connectivity disruption at clinic	High	Medium	PWA offline capability for draft note entry and local caching of active daily schedule
RSK-002	Third-party WhatsApp API service outage	Medium	Low	Outbox queue holds messages and automatically retries with fallback SMS/Email notification
RSK-003	Complex SVG rendering latency in PDF generation	Medium	Low	Server-side pre-compiled headless Chromium rendering with in-memory caching

19. REQUIREMENTS TRACEABILITY MATRIX

SCOPE BASELINE ID	REQUIREMENT ID	MONGODB COLLECTION	REPORT ID	TEST CASE ID
WS-01 (Website)	FR-WEB-001	`pages`, `posts`	—	TC-WEB-001
DR-01 (Appointments)	FR-APT-012	`appointments`, `providerSchedules`	—	TC-APT-012
CB-01 (Clinical Records)	FR-CLN-004	`visits`, `slAssessments`, `audiologyEvaluations`	—	TC-CLN-004
CB-01 (Clinical Records)	FR-DOC-002A	`audiologyEvaluations.oae`	**RP-CLN-OAE**	TC-DOC-002A
CB-01 (Clinical Records)	FR-DOC-002B	`audiologyEvaluations.pta`, `immittance`	**RP-CLN-AUD**	TC-DOC-002B
CB-01 (Clinical Records)	FR-DOC-002C	`slAssessments.VOICE`	**RP-CLN-VOI**	TC-DOC-002C
CB-01 (Clinical Records)	FR-DOC-002D	`slAssessments.FLUENCY`	**RP-CLN-FLU**	TC-DOC-002D
CB-01 (Clinical Records)	FR-DOC-002E	`slAssessments.LANGUAGE`, `ARTICULATION`	**RP-CLN-SLA**	TC-DOC-002E

SCOPE BASELINE ID	REQUIREMENT ID	MONGODB COLLECTION	REPORT ID	TEST CASE ID
CB-01 (Clinical Records)	FR-DOC-002F	`cbtRecords` (encrypted)	**RP-CLN-CBT**	TC-DOC-002F
CB-01 (Clinical Records)	FR-DOC-002G	`slAssessments.DYSPHAGIA`	**RP-CLN-DYS**	TC-DOC-002G
CB-01 (Clinical Records)	FR-DOC-002H	`hearingAids`	**RP-CLN-HAF**	TC-DOC-002H
CB-01 (Clinical Records)	FR-DOC-002I	`therapyPlans`, `sessionNotes`	**RP-CLN-PRG**	TC-DOC-002I
CB-01 (Clinical Records)	FR-DOC-002J	`therapyPlans.dischargeSummary`	**RP-CLN-DSC**	TC-DOC-002J
CB-02 (Billing & Stock)	FR-BIL-008	`invoices`, `payments`, `stockItems`	**RP-BIL-002**	TC-BIL-008

20. APPENDICES & EXECUTION SIGN-OFF

20.1 APPENDIX A: STANDARDS COMPLIANCE CROSS-REFERENCE

- **ISO/IEC/IEEE 29148:2018**: Systems and software engineering — Life cycle processes — Requirements engineering structure.
- **Digital Personal Data Protection (DPDP) Act, 2023**: Compliance rules governing data fiduciary notice, consent tracking, and patient data audit logs.
- **Information Technology Act, 2000 (Section 43A / Section 10A)**: Reasonable security practices & electronic contract validity.

20.2 APPENDIX B: DOCUMENT REVISION INDEX

- Baseline v1.0 complete technical specification archived under Git commit SHA tagging `v1.0-SRS-FINAL` in Black Angler repository.

20.7 BILATERAL EXECUTION & ACCEPTANCE SIGNATURE BLOCK

IN WITNESS WHEREOF, the Parties hereto have caused this Software Requirements Specification (Doc ID: **BA-SRS-WEB-ERP-001**) to be executed by their duly authorized representatives as of the Effective Date written above.

By signing below, both Black Angler Pvt Ltd and Vani's Speech & Hearing Clinic formally ratify and approve this SRS document as the authoritative specification governing the engineering, design, data architecture, security compliance, and testing criteria for the Public Website, Content Management System, and Clinic Practice Management System (ERP).

FOR BLACK ANGLER PVT LTD:	FOR VANI'S CLINIC (CLIENT):
_____ **Signature** **Name:** Shreeharsh M. Shivpuje **Title:** Founder / Lead Solutions Architect **Date:** October 07, 2026 *(Authorized Signatory & Technical Representative)*	_____ **Signature** **Name:** Ms. Palak Paresh Gandhi **Title:** Product Owner & Principal Practitioner **Date:** October 07, 2026 *(Client Representative & Business Owner)*

BLACK ANGLER EXECUTIVE DOCUMENTATION • SOFTWARE REQUIREMENTS SPECIFICATION • STAMPED BILATERAL RECORD